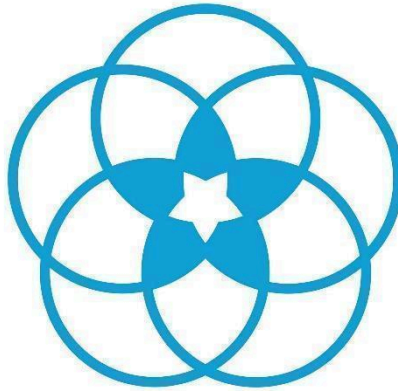




Health Information System Program Middle East and North Africa (HISP MENA)

User Manual: Middleware

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1

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HISP MENA

Table 1: Version control

Version	Date	Description of Changes
1.0	2025-09-05	Initial creation of the document.
2.0	2025-10-06	Update Document Added New Modules

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1. Introduction

This User Manual provides detailed guidance on using the Middleware System. It is designed for system users, administrators. The manual covers all modules within the system and includes step-by-step instructions with screenshots for performing various operations such as search, add, edit, delete, and export.

1.1 Login Page - User Authentication

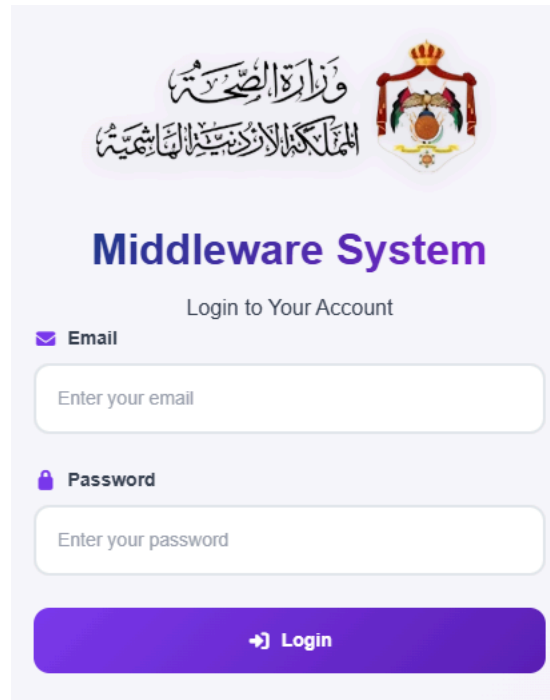


Figure 1 - Login Page

Access to the Middleware System is restricted to authorized users. Each user must log in with a valid email and password to reach the dashboard and system modules.

When navigating to the system URL, the login screen appears:

1. Email: enter your registered email address.
2. Password: enter your account password.
3. Login: press to authenticate.

If the email or password is incorrect, the system will display an error message:

1. Passwords are case-sensitive.
2. Accounts may be locked after repeated failed attempts, depending on admin policy.
3. Only users assigned roles (e.g., Admin, Operator) can access modules after login.

Security Reminder

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1. Do not share your password.
2. Log out after finishing your session.
3. Use strong passwords and change them regularly.

2. Modules Overview

2.1 Dashboard

2.1.1 Overview

The Dashboard provides a real-time, read-only snapshot of system activity across transactions, scheduled jobs, and user events. It is the default landing page after sign-in and offers shortcuts to the most used modules.

2.1.2 Screen Areas

1. Daily Execution Summary (today)

Cards show same-day counts sourced from Observability:

- Number of Transactions: total API transactions processed today (all statuses).
- Number of Failed Transactions: transactions with runtime failure today.
- Number of Succeeded Transactions: transactions completed with today.
- Number of User Events (Audit Logs): audit events recorded today (logins, CRUD actions, API calls).
- Number of Executed Scheduled Jobs: all job runs today (success + failed).

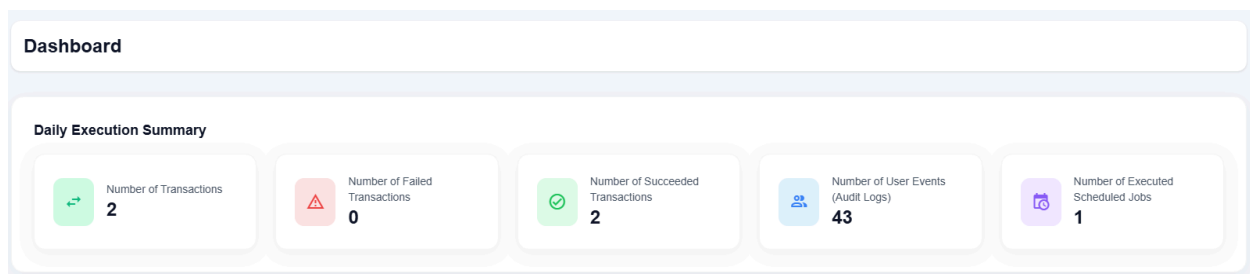


Figure 2 - Dashboard / Daily Execution Summary

2. Quick Access

One-click navigation to:

- Transaction List (API Transaction Logs)
- Dynamic Routes (manage/view routes)

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- Users (manage accounts & tokens)
- Scheduled Jobs (create/edit/pause jobs)

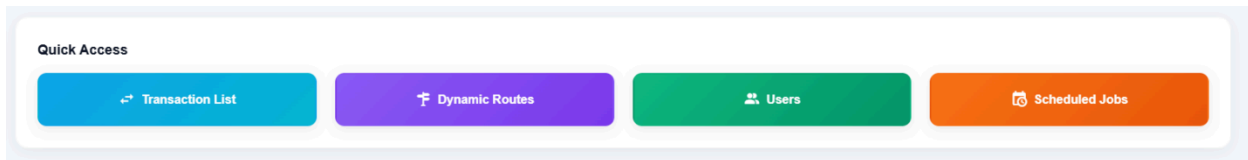


Figure 3 - Quick Access [Transaction List,Dynamic Routes,Users,Scheduled Jobs]

3. Transactions Overview (time series)

A timeline chart of daily totals for the recent period:

- Total Transactions, Failed, Succeeded, In Progress

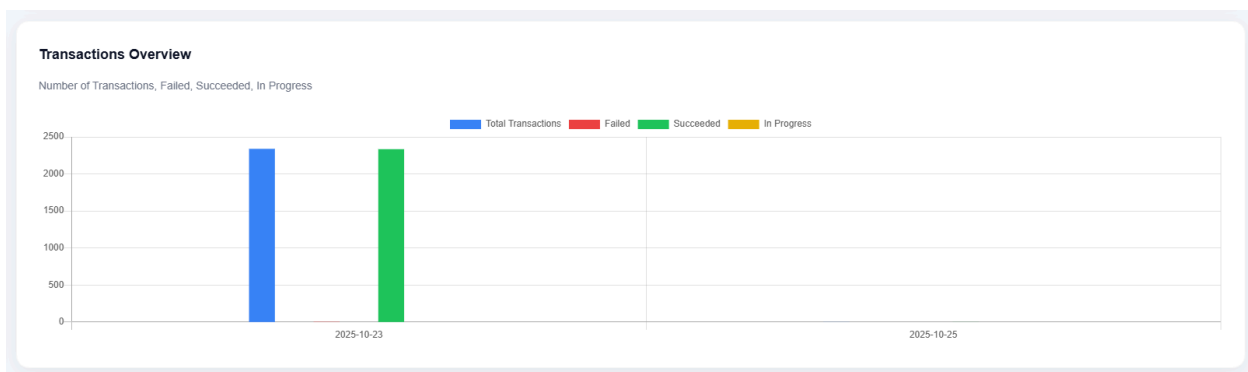


Figure 4 - Total Transactions

4. Scheduled Jobs Overview (time series)

- Daily counts of Succeeded and Failed scheduled job runs to monitor automation health.

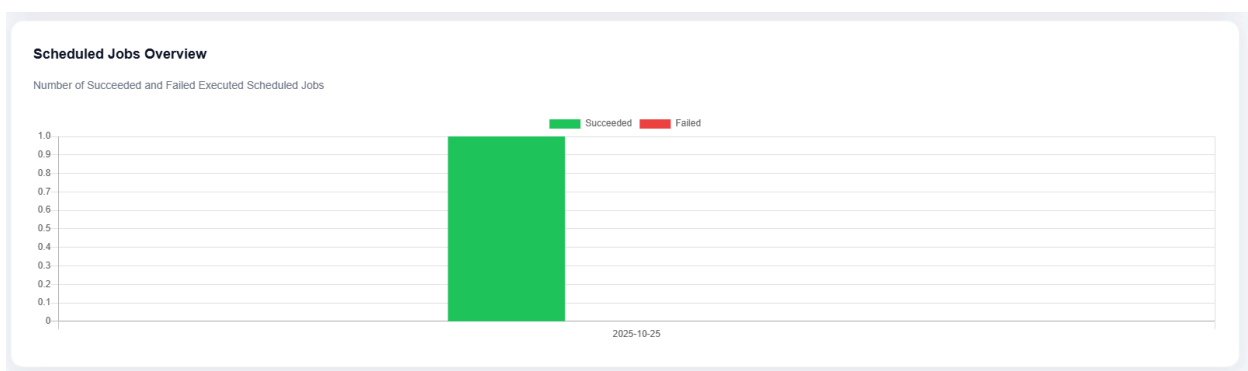


Figure 5 - Scheduled Jobs Overview

5. Last 10 Failed Transactions

- Table of the most recent failed API transactions with columns: ID, Endpoint, Status, Timestamp, Duration, Action.
- Click View All to open Observability : API Transaction Logs pre-filtered to failed items.

6. Last 10 Executed Scheduled Jobs

- Latest job executions with ID, Name, Status, Execution Time, Duration, Action.
- View All opens Observability : Scheduled Jobs Logs.

The screenshot displays two side-by-side panels. The left panel, titled 'Last 10 Failed Transactions', shows a table with columns: ID, Endpoint, Status, Timestamp, Duration, and Action. It lists two failed transactions. The right panel, titled 'Last 10 Executed Scheduled Jobs', shows a table with columns: ID, Name, Status, Execution Time, Duration, and Action. It lists one successful job execution.

ID	Endpoint	Status	Timestamp	Duration	Action
4e938f24-607d-4b04-b2c9-f3cb5c02f440	servlet/register?httpMethodRestrict=POST	Failed	23 Oct 2025, 13:17	0.1s	View All
7efe91c5-559c-48c4-a550	servlet/register?httpMethodRestrict=POST	Failed	23 Oct 2025,	0.1s	View All

ID	Name	Status	Execution Time	Duration	Action
1	Monthly Job	Succeeded	25 Oct 2025, 22:00	0min	View All

Figure 6 - Last 10 Transactions / Scheduled Jobs

7. Expired and About to Expire Tokens

- Lists API tokens that are expired or nearing expiry with Token ID, User, Expiration Date, Status, Action.
- View All opens Access Control : Users for token generation/revocation.

The screenshot shows a table titled 'Expired and About to Expire Tokens' with columns: Token ID, User, Expiration Date, Status, and Action. It contains one row for an expired token.

TOKEN ID	USER	EXPIRATION DATE	STATUS	ACTION
032	healthmapuser1	24 Oct 2025, 00:00	Expired	View All

Figure 7 - Expired & About To Expired Tokens

2.2 Configuration

2.2.1 Overview

The Configuration module allows administrators to manage various system settings and mappings, ensuring the middleware operates efficiently and according to defined rules. This includes error handling, external system integration, and other foundational settings.

2.2.2 Error Mapping / Configuration

Purpose

The Error Mapping module standardizes error handling across the middleware by mapping raw system errors from external APIs into predefined codes and user-friendly messages.

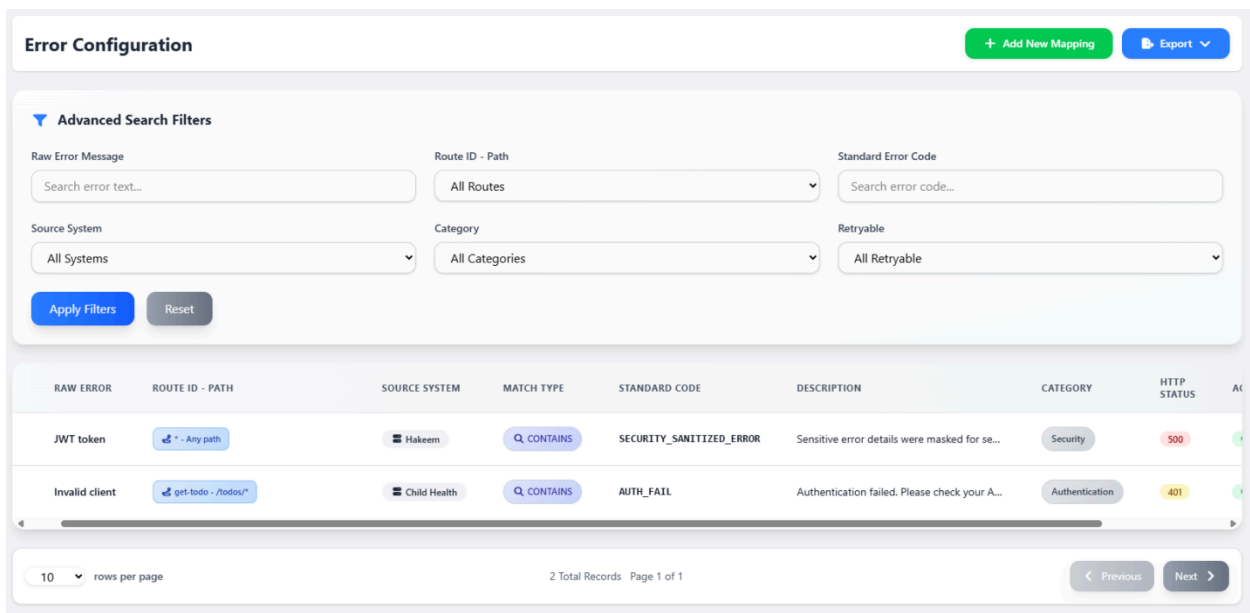
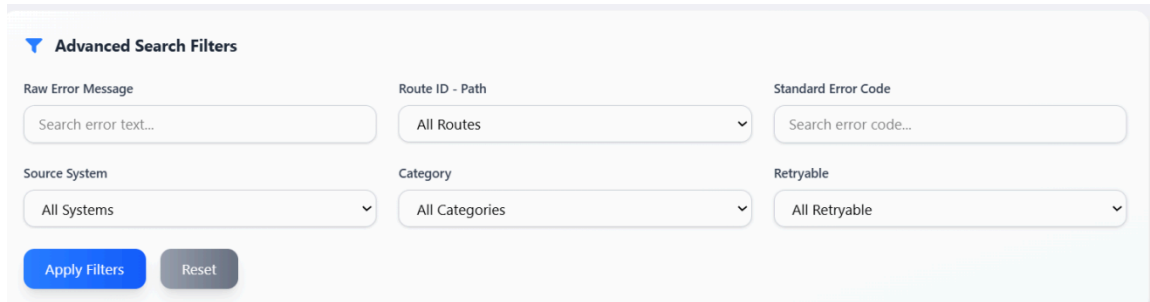


Figure 8 - Error Configuration

Key Features

Search & Filters: Find error mappings by system, code, category, or status.

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Advanced Search Filters

Raw Error Message:

Route ID - Path:

Standard Error Code:

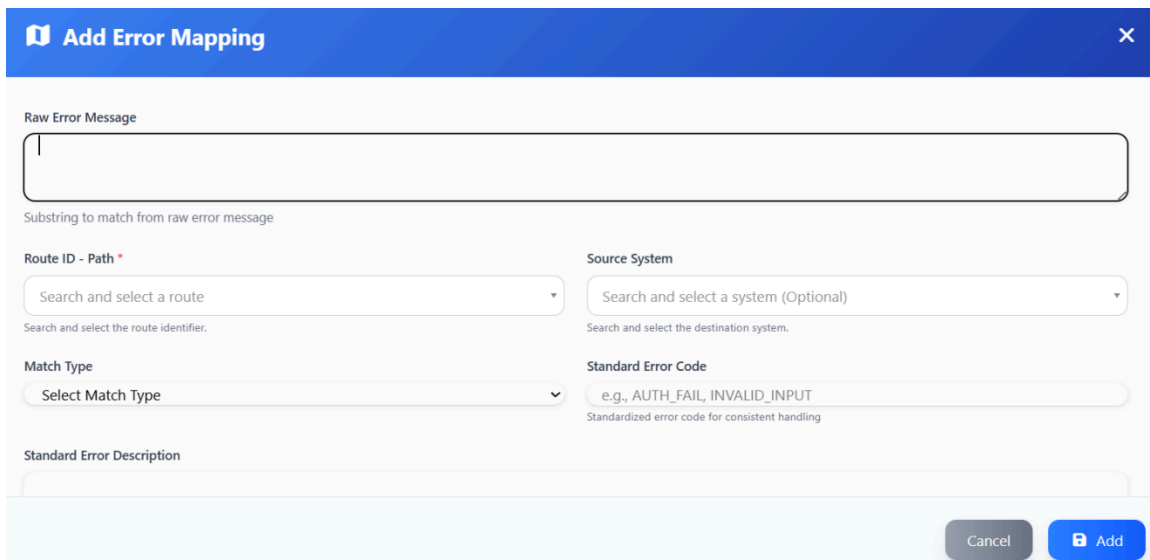
Source System:

Category:

Retryable:

Figure 9 - Search Error Configuration

Add Error Mapping: Define new mappings by system, raw error substring, match type (Contains/Equals/Regex), mapped error code, friendly message, category, and HTTP status code.



Add Error Mapping ×

Raw Error Message:

Substring to match from raw error message

Route ID - Path *

Source System

Match Type

Standard Error Code

Standard Error Description:

Figure 10 - Add Error Mapping / Configuration

Edit: Update existing mapping details. **Delete:** Remove obsolete mappings.

Figure 11 - Edit Error Mapping / Configuration

Activate/Deactivate: Toggle whether the mapping is applied in real-time processing from the edit error mapping screen.

Figure 12 - Activate/Deactivate Error Mapping

Export: Export mappings to CSV/Excel for review or backup.

Figure 13 - Export Error Mapping

Process Flow :

1. Middleware detects an external error.
2. Look up mapping in the error_mapping table (by system + code/message).
3. If matched, returns the unified error code + friendly message.
4. Logs the mapped error in Transaction Logs and displays it on the dashboard.

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2.2.3 Error Category

Purpose

The Error Category module allows for the definition and management of categories for various errors within the system. This standardization aids in consistent error handling, reporting, and analysis across the middleware.

ID	NAME	DESCRIPTION	STATUS	CREATED	ACTIONS
4	Security	-	Active	10-09-2025, 08:54:58 AM	⏸️ ✎️
3	Business	Business Error Category	Active	10-09-2025, 08:47:54 AM	⏸️ ✎️
2	Validation	Validation Error Category	Active	09-09-2025, 04:51:17 PM	⏸️ ✎️
1	Authentication	Authentication Error Category	Active	09-09-2025, 04:51:10 PM	⏸️ ✎️

10 Rows per page 4 Total Records Page 1 of 1 < Previous Next >

Figure 14 - Error Category

Search Error Category

This section allows users to search and filter Error Categories using the following fields:

1. Search Error Categories: Search by name.
2. Description: Search by description.
3. Status: Choose status (Active / Inactive / All).
4. Date Filter: Select a date range.
5. Apply Filters: Apply the filters.

6. Reset: Clear filters and return to default state.

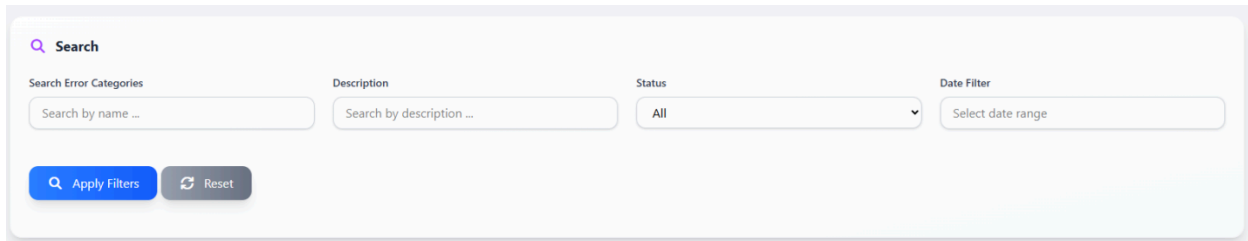
The image shows a search interface for error categories. At the top left is a 'Search' icon and label. Below it are four input fields: 'Search Error Categories' with a placeholder 'Search by name ...', 'Description' with a placeholder 'Search by description ...', 'Status' with a dropdown menu currently showing 'All', and 'Date Filter' with a placeholder 'Select date range'. At the bottom left are two buttons: a blue 'Apply Filters' button with a magnifying glass icon and a grey 'Reset' button with a circular arrow icon.

Figure 15 - Error Category Search Form

Add Error Category

By clicking the + Add Error Category button, a new window opens to add a new error category. Users can enter the required details (Name, Description, Status) and save the record .

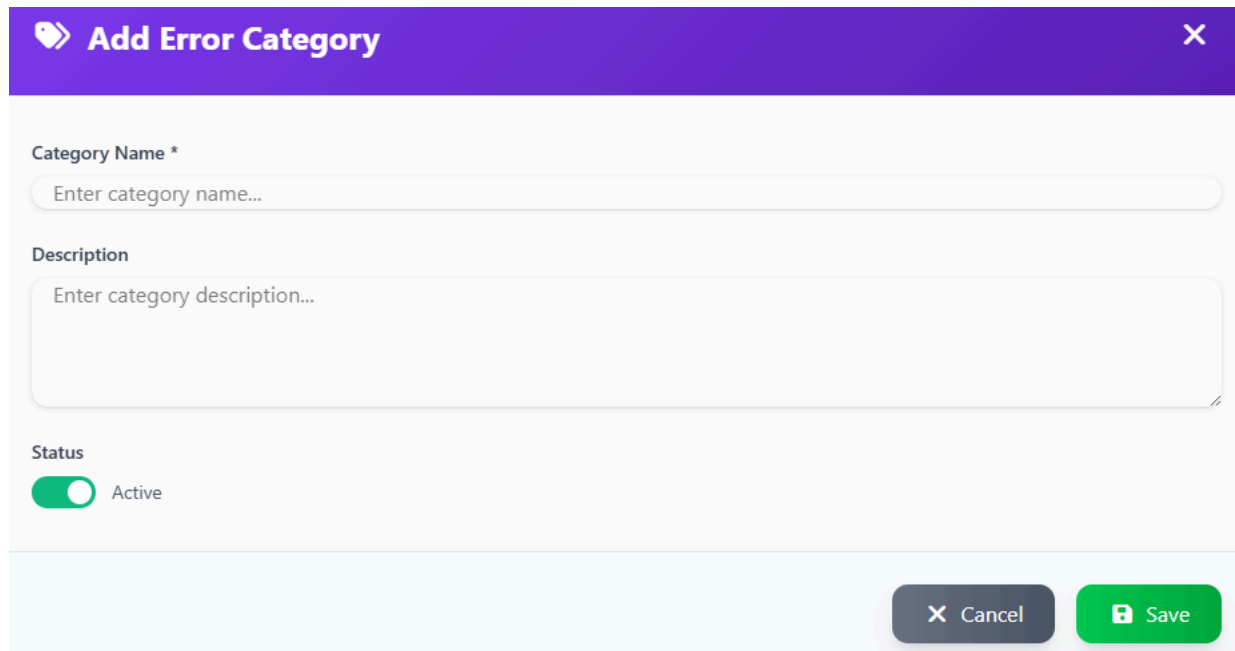
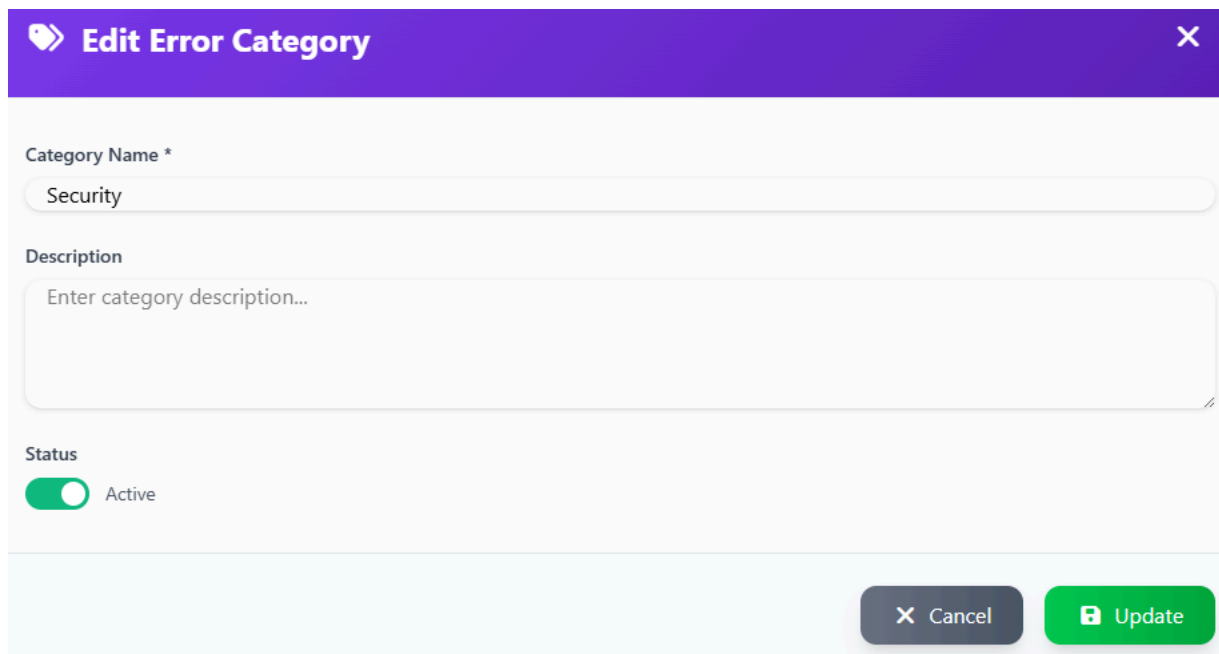
The image shows a modal window titled 'Add Error Category' with a close button (X) in the top right corner. The form contains three main sections: 'Category Name *' with a text input field containing the placeholder 'Enter category name...'; 'Description' with a larger text area containing the placeholder 'Enter category description...'; and 'Status' with a toggle switch currently turned on, labeled 'Active'. At the bottom right of the modal are two buttons: a grey 'Cancel' button with an X icon and a green 'Save' button with a save icon.

Figure 16 - Add Error Category

Edit Error Category

By clicking the blue edit icon, a window opens to modify the selected category details. After editing, the user can save the changes.



The image shows a modal window titled "Edit Error Category" with a close button (X) in the top right corner. The form contains three main sections: "Category Name *" with a text input field containing "Security"; "Description" with a larger text area containing the placeholder "Enter category description..."; and "Status" with a toggle switch currently turned on, labeled "Active". At the bottom right of the modal are two buttons: "Cancel" (with an X icon) and "Update" (with a document icon).

Figure 17 - Edit Error Category

Export Error Categories

The list of Error Categories can be exported into Excel or CSV format. By clicking the Export button, users can select the desired format (Excel / CSV).

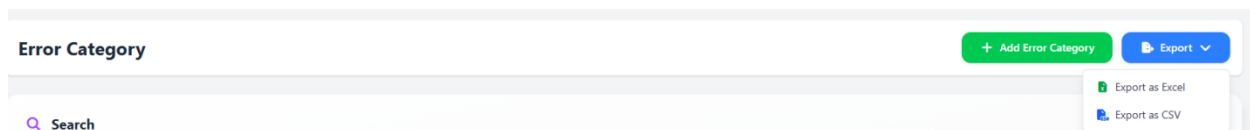


Figure 18 - Export Error Categories

Activate / Deactivate Status Error Categories

Each Error Category can be either Active or Inactive (Deactivated) depending on its usage status in the system.



Active

- The category is enabled and available for use across the middleware system.
- Users and services can assign or reference this error category in logs, mappings, and reports.
- Appears in filtering/search results as “Active.”

Inactive / Deactivated

- The category is disabled and not available for future assignments.
- Existing records may still display the category for historical purposes, but it cannot be newly assigned.
- Useful when a category becomes obsolete or no longer needed.

ID	NAME	DESCRIPTION	STATUS	CREATED	ACTIONS
4	Security	-	Active	10-09-2025, 08:54:58 AM	 
3	Business	Business Error Category	Inactive	10-09-2025, 08:47:54 AM	 

Figure 19 - Error Category Activation & Deactivation

2.2.4 External Source Systems

Overview

The Source System module manages all external systems integrated with the middleware. It allows administrators to add, update, activate/deactivate, delete, and search source systems.

External Source System List

Each record displays:

1. ID: auto-generated unique identifier.
2. Name: the system's name.
3. Description: optional details about the system.
4. Status: shows whether the source system is Active (green) or Inactive (red).
5. Created: the date and time of creation.
6. Actions: available options:

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- A. Activate/Deactivate: toggle system status.
- B. Edit: update record details.

ID	NAME	DESCRIPTION	STATUS	CREATED	ACTIONS
4	HEALTHMAP	HEALTHMAP	Inactive	10-09-2025, 08:57:16 AM	
3	Child Health	Child Health Source System	Active	09-09-2025, 04:50:56 PM	
2	Email sys	Email Source System	Active	09-09-2025, 04:50:40 PM	
1	Hakeem	Hakeem Source System	Active	09-09-2025, 04:50:34 PM	

10 Rows per page 4 Total Records Page 1 of 1 < Previous Next >

Figure 20 - External Source System List

Search External Source System

Users can search and filter source systems by:

1. Name: search by system name.
2. Description: search by description keyword.
3. Status: filter by All, Active, or Inactive.
4. Date Filter: filter by creation date.

Search

Search Source Systems Description Status Date Filter

Search by name Search by description ... All Select date range

Apply Filters Reset

Figure 21 - Search External Source System

Add External Source System

The Add Source System button allows creation of a new source system record. Required fields include Name, Description, and Status.

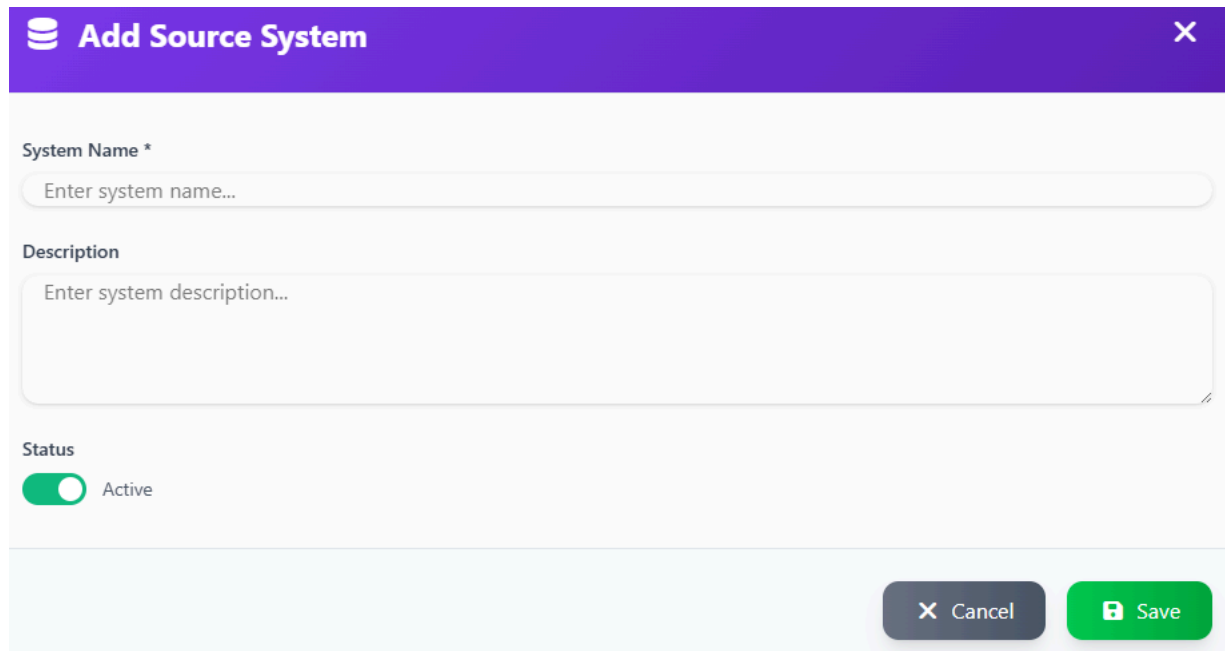
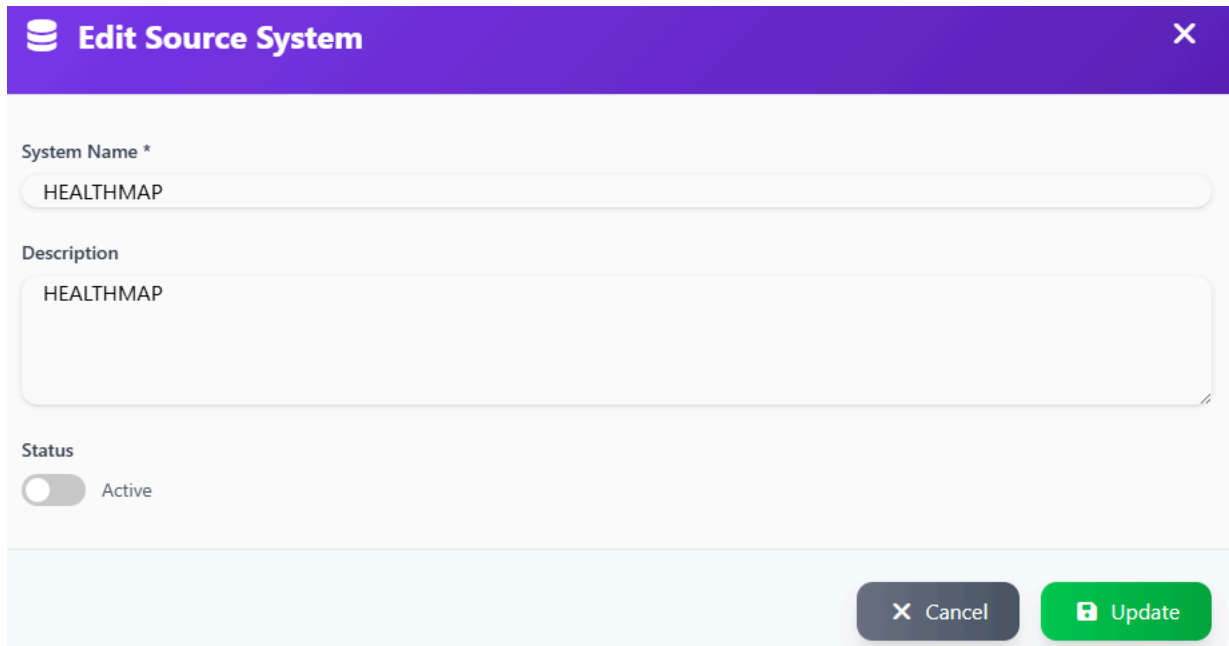
A modal dialog box titled "Add Source System" with a purple header bar containing a database icon and a close button (X). The dialog has a light gray background. It contains three main sections: "System Name *" with a text input field containing the placeholder "Enter system name..."; "Description" with a larger text area containing the placeholder "Enter system description..."; and "Status" with a green toggle switch labeled "Active". At the bottom right, there are two buttons: a gray "Cancel" button with an X icon and a green "Save" button with a document icon.

Figure 22 - Add Source External System

Edit External Source System

By clicking the blue edit icon, a window opens to modify the selected external system details. After editing, the user can save the changes.



The image shows a modal window titled "Edit Source System" with a purple header bar containing a database icon and a close button. The form has three main sections: "System Name *" with a text input containing "HEALTHMAP"; "Description" with a larger text area also containing "HEALTHMAP"; and "Status" with a toggle switch currently set to "Active". At the bottom right, there are two buttons: a grey "Cancel" button and a green "Update" button with a save icon.

Figure 23 - Edit External Source System

Export External Source Systems

The list of external systems can be exported into Excel or CSV format. By clicking the Export button, users can select the desired format (Excel / CSV).

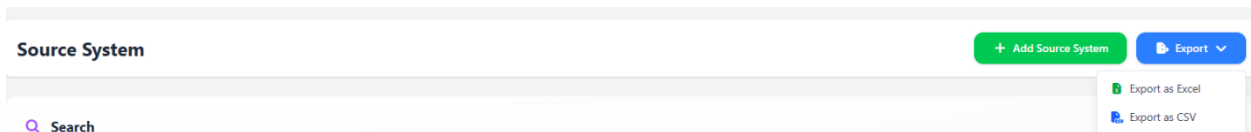


Figure 24 - Export External Source System

Activate / Deactivate Status External Source Systems

Each external system can be either Active or Inactive (Deactivated) depending on its usage status in the system.



Active

- The external system is enabled and available for use across the middleware system.
- Users and services can assign or reference this error category in logs, mappings, and reports.
- Appears in filtering/search results as “Active.”

Inactive / Deactivated

- The external system is disabled and not available for future assignments.
- Existing records may still display the external system for historical purposes, but it cannot be newly assigned.
- Useful when an external system becomes obsolete or no longer needed.

ID	NAME	DESCRIPTION	STATUS	CREATED	ACTIONS
4	HEALTHMAP	HEALTHMAP		10-09-2025, 08:57:16 AM	 
3	Child Health	Child Health Source System		09-09-2025, 04:50:56 PM	 

Figure 25 - Activate/Deactivate External Source System

2.2.5 Throttling & Rate Limit

Purpose

Protects the platform from overload and abuse by limiting how many requests can be processed in a given time window. Admins can set:

- A System-wide limit (applies to all APIs).
- A Camel APIs limit (applies to routes under /camel/**).
- Specific route limits (granular exceptions or tighter rules for certain endpoints).

Changes take effect immediately for new requests.

Screen Areas

1. System Rate Limit Configuration

- Request Limit: Maximum number of requests allowed within the time window.
- Time Window (Seconds): The measurement window.
- Update Rate Limit: Opens the edit dialog.

2. Camel Rate Limit Configuration

- Overall /camel/** Request Limit: Global cap for all Camel routes.
- Overall /camel/** Time Window (Seconds): Window for the cap above.
- Update /camel/** Rate Limit: Opens the edit dialog.
- Add Specific Route: Create an override for an individual Camel route.

Throttling & Rate Limit

System Rate Limit Configuration

Update Rate Limit

Request Limit

100

Time Window (Seconds)

60

Camel Rate Limit Configuration

Update /camel/** Rate Limit

Overall /camel/** Request Limit

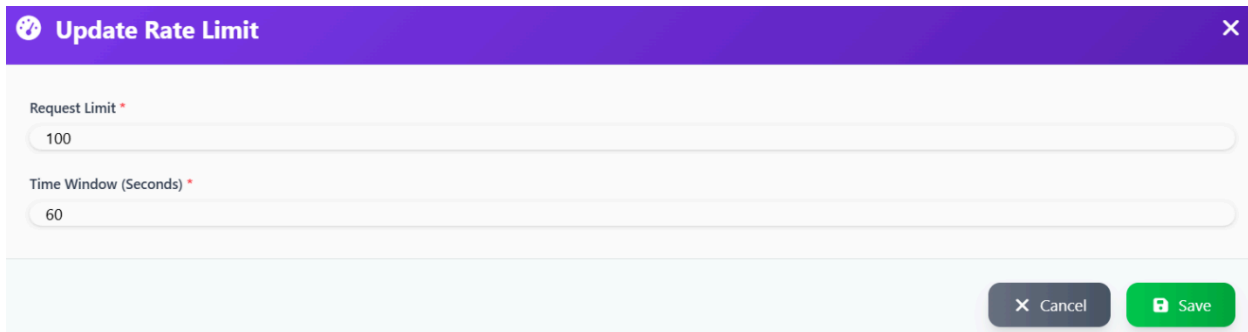
100

Overall /camel/** Time Window (Seconds)

60

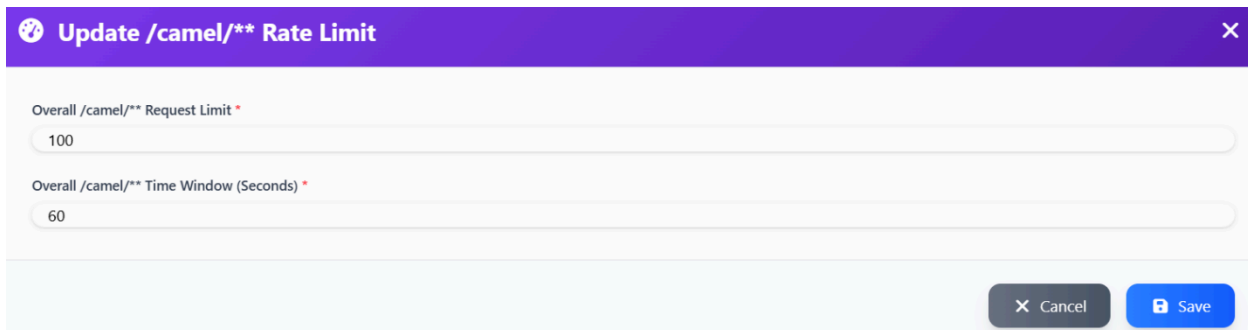
+ Add Specific Route

Figure 26 - System / Camel Rate Limit - Throttling



The dialog box has a purple header with a gear icon and the text "Update Rate Limit" and a close button (X). It contains two input fields: "Request Limit" with a value of 100 and "Time Window (Seconds)" with a value of 60. At the bottom right are "Cancel" and "Save" buttons.

Figure 27 - Update Rate Limit

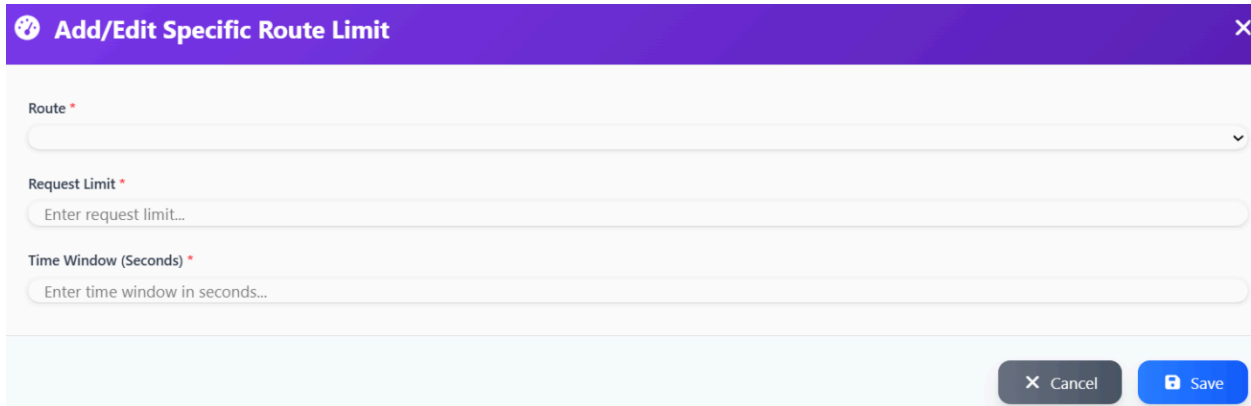


The dialog box has a purple header with a gear icon and the text "Update /camel/** Rate Limit" and a close button (X). It contains two input fields: "Overall /camel/** Request Limit" with a value of 100 and "Overall /camel/** Time Window (Seconds)" with a value of 60. At the bottom right are "Cancel" and "Save" buttons.

Figure 28 - Update /camel/** Rate Limit

3. Add/Edit Specific Route Limit (Dialog)

- Route: Select the target route (e.g., /camel/dhis2/push).
- Request Limit: Max requests in the defined window.
- Time Window (Seconds): Window for the limit above.
- Save: Creates/updates the rule.



Add/Edit Specific Route Limit

Route *

Request Limit *

Enter request limit...

Time Window (Seconds) *

Enter time window in seconds...

Cancel Save

Figure 29 - Add Specific Route Limit

2.2.6 SYSTEM Settings

Purpose

System Settings is a centralized registry of key/value configuration items read by the middleware.

1. Search & Filters

- **Key:** Search by setting name (supports partial match).
- **Value:** Search by current value (supports partial text).
- **Type:** Filter by data type: All / String / Number.
- **Creation Date:** Filter by when a key was created.
- **Apply Filters / Reset:** Run or clear filters.

2. Configuration Grid

- **KEY:** Unique setting name (recommended style: UPPER_SNAKE_CASE, e.g., API_TIME_OUT).
- **VALUE:** Current value (text or numeric).
- **TYPE:** Data type (STRING or NUMBER).
- **ACTIONS:** View (eye), Edit (pencil).
- **Pagination:** Navigate between pages when there are many keys.

Configuration Management

+ Add New

Search & Filters

Key

Search by key...

Value

Search by value...

Type

All

Creation Date

Select creation date

Apply Filters

Reset

KEY	VALUE	TYPE	DESCRIPTION	CREATED BY	CREATED AT	ACTIONS
API_TIME_OUT	3000	NUMBER	TIMEOUT FOR AN API	System	20/10/2025 09:26:10	

10

Records per page

Total Records: 1Page 1 of 1

< Prev

Next >

Figure 30 - Search/Filter System Settings

3. Add New (Configuration)
- Key: Unique identifier (duplicates are not allowed).
 - Type: STRING or NUMBER.
 - Value: The setting value (numeric validation applies when Number).
 - Description: Optional explanation.
 - Save: Creates the configuration entry.

Add Configuration

Key *

Enter unique key...

Type *

String

Value *

Enter value...

Description

Enter description (optional)...

Cancel

Save

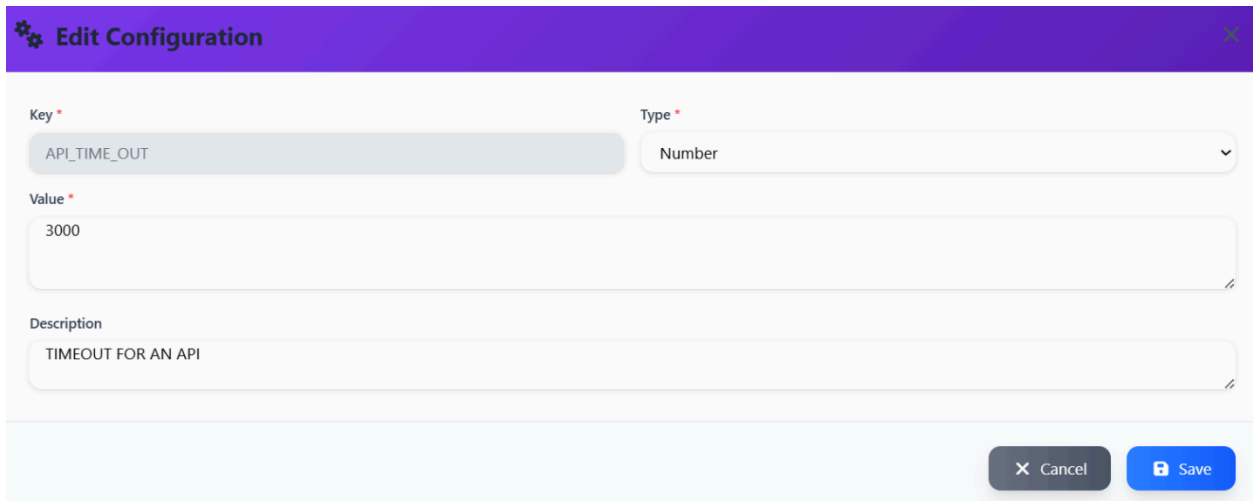
Figure 31 - Add System Settings

4. Edit (Configuration)

From the list, click Edit:

- **Key:** Is locked (cannot be changed after creation).
- You can change Type, Value, and Description.

Click Save to apply changes.



The screenshot shows a web interface for editing system settings. The title bar is purple with a gear icon and the text "Edit Configuration". Below the title bar, there are four input fields arranged in a form. The first field is labeled "Key" and contains the text "API_TIME_OUT". The second field is labeled "Type" and is a dropdown menu currently showing "Number". The third field is labeled "Value" and contains the number "3000". The fourth field is labeled "Description" and contains the text "TIMEOUT FOR AN API". At the bottom right of the form, there are two buttons: a grey "Cancel" button and a blue "Save" button.

Figure 32 - Edit System Settings

5. Validation & Behavior

- The key is required and must be unique.
- Type determines how Value is validated.
- Updates are logged in Audit Logs.

6. Example

- KEY: API_TIME_OUT
- TYPE: NUMBER
- VALUE: 3000
- DESCRIPTION: TIMEOUT FOR AN API (milliseconds)

2.2.7 Integrated Systems

Purpose

Integrated Systems stores connection metadata for external systems (e.g., DHIS2 instances) that the middleware calls. For each system you define protocol, host/port, authentication, and optional connection options. Records drive routing and observability and are auditable.

Screen Areas

1. Search & Filters

Top filter panel fields:

- Code: system code (unique).
- Host: host-name or base URL.
- Description: free-text search.
- Protocol: All / HTTP / HTTPS.
- Creation Date: filter by created-at date.
- Actions: Apply Filters, Reset.

2. Grid Columns

- CODE: unique identifier (e.g., DHIS2.1).
- HOST: target host/base path.
- PROTOCOL: HTTP or HTTPS.
- AUTHENTICATION TYPE: None, Basic, or JWT.
- DESCRIPTION: purpose or environment note.
- CREATED BY / CREATED AT: audit metadata.
- ACTIONS: View and Edit.

Integrated Systems Management + Add New

Search & Filters

Code:

Host:

Description:

Protocol:

Creation Date:

Apply Filters Reset

CODE	HOST	PROTOCOL	AUTHENTICATION TYPE	DESCRIPTION	CREATED BY	CREATED AT	ACTIONS
DHIS2.1	play.im.dhis2.org/stable-2-41-5	HTTPS	BASIC	DHIS2 Instance one	SYSTEM	10/20/2025, 9:26:10 AM	🟢 🔗
DHIS2.2	hmis-dev.moh.gov.jo/dwh	HTTPS	BASIC	DHIS2 Instance two	SYSTEM	10/20/2025, 9:26:10 AM	🟢 🔗

Figure 33 - Search/Filter Integrated Systems

3. Add Integrated System

Click Add New to open Add Integrated System:

- Protocol: HTTP or HTTPS.
- Host: host-name or base path (e.g., play.im.dhis2.org/stable-2-41-5).
- Port: optional; default per protocol if empty (80/443).
- Authentication Type: one of:
 - A. None: no auth headers.
 - B. Basic: provide Username and Password (stored securely; UI masks secrets).
 - C. JWT: provide Token (bearer token; UI masks the value).
- Code: unique system code (cannot be changed later).
- Additional Key 1 / Additional Value 1: optional pair for connection options (e.g., timeout :30000).
- Additional Key 2 / Additional Value 2: optional second pair (e.g., connect-timeout : 10000).
- Description: optional note.

Click **Save**.

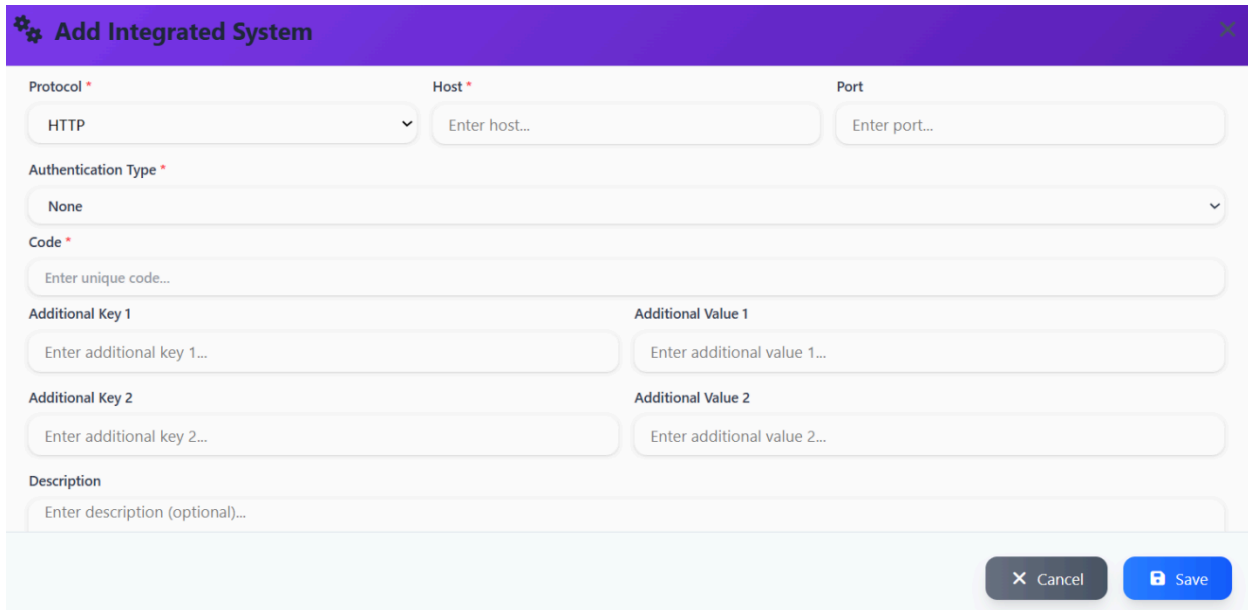
The image shows a web form titled "Add Integrated System" with a purple header bar. The form contains several input fields: "Protocol" (a dropdown menu with "HTTP" selected), "Host" (a text input field with placeholder "Enter host..."), "Port" (a text input field with placeholder "Enter port..."), "Authentication Type" (a dropdown menu with "None" selected), "Code" (a text input field with placeholder "Enter unique code..."), "Additional Key 1" (a text input field with placeholder "Enter additional key 1..."), "Additional Value 1" (a text input field with placeholder "Enter additional value 1..."), "Additional Key 2" (a text input field with placeholder "Enter additional key 2..."), "Additional Value 2" (a text input field with placeholder "Enter additional value 2..."), and "Description" (a text input field with placeholder "Enter description (optional)..."). At the bottom right of the form are two buttons: "Cancel" (a grey button with a close icon) and "Save" (a blue button with a save icon).

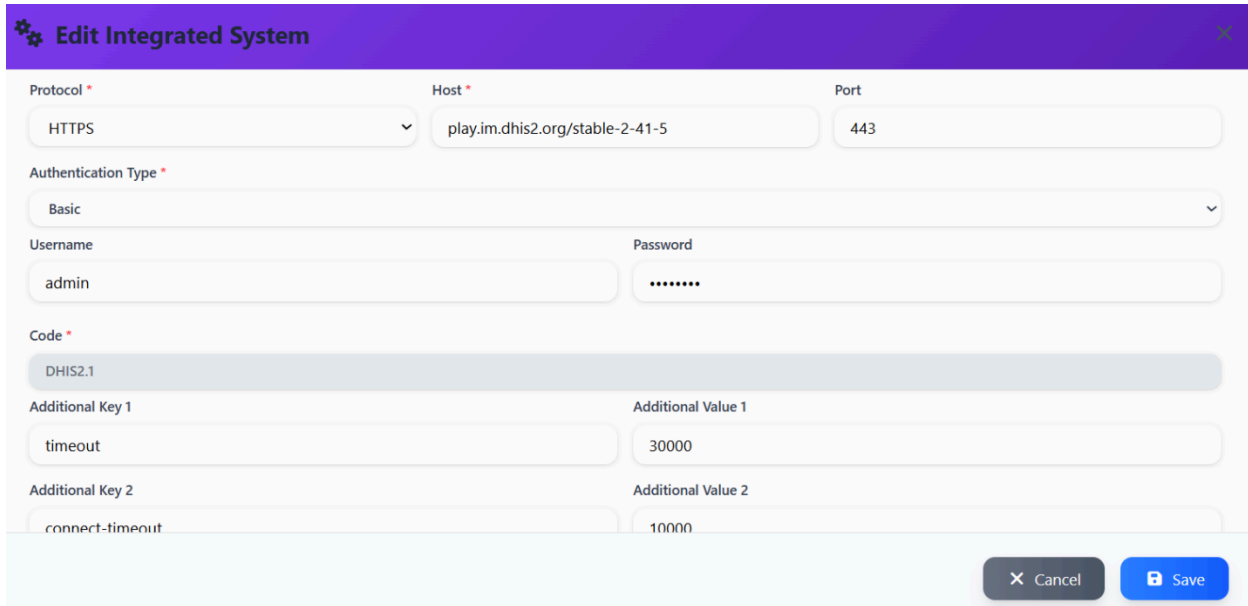
Figure 34 - Add New Integrated System

4. Edit Integrated System

From the list, click Edit:

- You may adjust Protocol, Host, Port, Authentication Type/credentials, optional Additional Key/Value pairs, and Description.
- Code is read-only after creation.

Click **Save**.

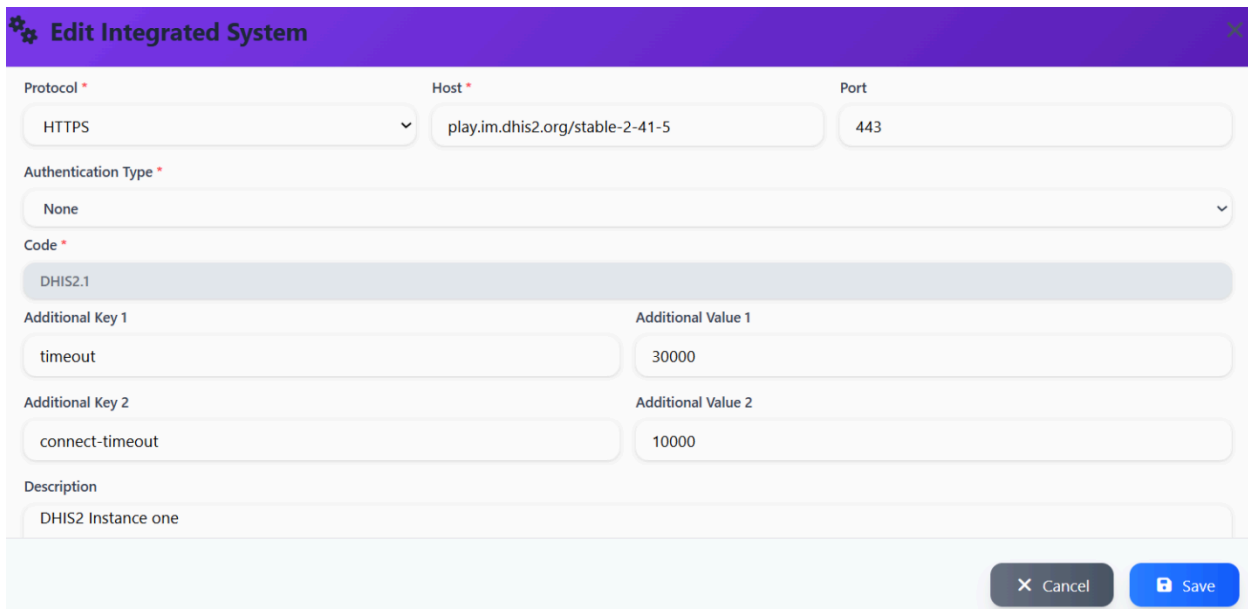


The screenshot shows the 'Edit Integrated System' dialog box with the following fields and values:

- Protocol:** HTTPS
- Host:** play.im.dhis2.org/stable-2-41-5
- Port:** 443
- Authentication Type:** Basic
- Username:** admin
- Password:** (masked with dots)
- Code:** DHIS2.1
- Additional Key 1:** timeout
- Additional Value 1:** 30000
- Additional Key 2:** connect-timeout
- Additional Value 2:** 10000

Buttons at the bottom right: Cancel, Save.

Figure 35 - Edit Integrated System / Basic Auth Type



The screenshot shows the 'Edit Integrated System' dialog box with the following fields and values:

- Protocol:** HTTPS
- Host:** play.im.dhis2.org/stable-2-41-5
- Port:** 443
- Authentication Type:** None
- Code:** DHIS2.1
- Additional Key 1:** timeout
- Additional Value 1:** 30000
- Additional Key 2:** connect-timeout
- Additional Value 2:** 10000
- Description:** DHIS2 Instance one

Buttons at the bottom right: Cancel, Save.

Figure 36 - Edit Integrated System / None Auth Type

Edit Integrated System

Protocol * Host * Port

Authentication Type *

Token

Code *

Additional Key 1 Additional Value 1

Additional Key 2 Additional Value 2

Figure 37 - Edit Integrated System / JWT Auth Type

5. Example

- CODE: DHIS2.1
- PROTOCOL/HOST/PORT: HTTPS, play.im.dhis2.org/stable-2-41-5, 443
- AUTH TYPE: Basic (username: admin) or JWT (token provided)
- OPTIONS: timeout=30000, connect-timeout=10000
- DESCRIPTION: DHIS2 Instance one

2.2.8 Integrated API Management

The Integrated API Management module allows system administrators to define reusable, parameterized API configurations that the Middleware can call dynamically, without hard-coding URLs in Java code.

Each Integrated API acts as a template for calling a DHIS2 endpoint (either Analytics or Metadata).

The template stores:

- query structure (dimensions such as dx, ou, pe, attributes, etc.),
- connection target (which DHIS2 environment / Integrated System),

- and activation state.

This lets the platform request DHIS2 data in a controlled, consistent, auditable way.

Purpose

The listing page provides:

- a filter/search panel, and
- a table (grid) of saved Integrated APIs.

From this page you can:

- search for existing APIs,
- create new APIs,
- edit an API's configuration,
- export all APIs (CSV / Excel),
- activate or deactivate an API.

This module is crucial for:

1. Building dynamic DHIS2 Analytics queries (e.g. data elements, indicators, org units, periods, attribute breakdowns).
2. Binding Metadata endpoints to specific business routes.

Search & Filters

Use the filter panel at the top of the page to quickly locate a specific API configuration.

Available filters:

- API Code Exact or partial match on the unique technical identifier.
- API Name Search by the human-readable display name.

- API Type Filter by type (for example: All Types, Analytics, Metadata).
- Integrated System Limit results to a specific target system configuration (for example: DHIS2.tracker, DHIS2.DWH).
- Status Show only Active APIs, only Inactive APIs, or All Status.
- Description Contains Free-text search within the API description field.
- Bound API Code Filter by a related or parent API code.
(Mainly used for Metadata APIs that are “bound” to a business API / route.)
- API URL Contains Search by part of the DHIS2 URL or query string.

Actions:

- Click Search to apply the selected filters.
- Click Reset to clear all filters and reload the full list.

Integrated API Management + Create API Export

Search & Filter APIs

API Code: API Name: API Type: Integrated System:

Status: Description Contains: Bound API Code: API URL Contains:

Search Reset

API CODE	API NAME	API TYPE	INTEGRATED SYSTEM	BOUND API	API URL	STATUS	CREATED
HealthCenter	HealthCenter	Analytics	DHIS2.2	-	/api/analytics.json?dimension=dx:gsconUle30EY;uL...	ACTIVE	10/19/2025, 1:40:52 PM
HealthCenter Metadata	HealthCenter	Metadata	DHIS2.2	HealthCenter-mapping	/api/metadata.json?organisationUnits=true&field...	ACTIVE	10/15/2025, 10:03:33 AM

Figure 38 - Search/Filter Integrated API

Create a New Integrated API

The Create New Integrated API form allows system administrators to register a new reusable API definition that the Middleware can call dynamically when interacting with DHIS2 or other

integrated systems.

Instead of writing full API URLs in code, this interface defines all necessary parameters (type, target system, status, and purpose) in a structured and user-friendly way.

Once saved, the API becomes visible in the Integrated API Management list and can be referenced by other modules such as Schedulers, Metadata bindings, or Integration Routes.

Main Fields Description

The top section of the dialog includes the core configuration fields, these must be completed before specifying any Analytics or Metadata details.

Main Fields

- API Code: A unique technical identifier for this API.
 - Spaces are replaced with underscores `_`.
 - Example: HealthCenter_Metadata, OPD_Analytics.
- API Name : A clear, descriptive name shown to users in the list.
 - Example: “Health Center Analytics”.
- API Type : Defines the purpose of the API.
 - Analytics: used to fetch data values and indicators.
 - Metadata: used to fetch descriptive info (e.g., organisation units).
 - Selecting a type shows its specific configuration section.
- Integrated System: Select which DHIS2 environment this API belongs to (e.g., DHIS2.DWH).
 - This controls which server and credentials are used.
- Status : Marks whether the API is active and available for use.
- Description : Optional text explaining what the API does.

Once all fields are filled, click Save API.

After saving:

- If API Type = Analytics, you'll configure dx, ou, pe, and attributes.
- If API Type = Metadata, you'll specify the metadata URL and bound API.

+ Create New Integrated API ×

API Code *
Unique technical identifier
 Spaces will be automatically replaced with underscores ()

API Name *
Descriptive name for this API

API Type *
The type of DHIS2 API

Integrated System *
The target system identifier (e.g., HMIS-DWH, DHIS2-Play)

Status ☒ Active
Only active APIs will be available for use

Description
Optional description of the API's purpose and functionality

× Cancel Save API

Figure 39 - Create New Integrated API

API Type Configuration

After filling the basic fields (API Code, API Name, API Type, Integrated System, Status, and Description), the form will expand based on the selected API Type.

There are two supported types:

1. Analytics ,used to fetch data values or indicators from DHIS2.
2. Metadata ,used to fetch descriptive or structural information (organisation units)

A. Analytics API Configuration

When API Type = Analytics, the system displays the Analytics API Configuration section as shown below:

Edit Integrated API
✕

API Code *

HealthCenter

Unique technical identifier

Spaces will be automatically replaced with underscores ()

API Name *

HealthCenter

Descriptive name for this API

API Type *

Analytics

The type of DHIS2 API

Integrated System *

DHIS2.2

The target system identifier (e.g., HMIS-DWH, DHIS2-Play)

Status

☒ Active
 Only active APIs will be available for use

Analytics API Configuration

Define the DHIS2 Analytics API parameters. The system will automatically generate the complete URL based on these inputs.

Data Dimensions (dx) *

gscmTue30EY;wAyxe0razbM;nQpou1fu9H9;NXTYyp5QrLH;1e7t3f0LLGo;cIdwvoAgZLV;v4K1pDjcywB;zEAeW2HEVOB;KUvaaMBYSN1;k5A4Y81dCwK;eBbqiatcnKn;Xc5MqYLZ3Cd;ASVFI0KGKji;vnmveK2PZ54;11hdh9E0KwE;wU1QKpjSwrM;Km0qVQob9MY;Msm0TSTRN9k;mCKqPcYw7t;Nce6S25UISw;qGMXZnCezpm;szrz2G03B65;M0bZGvks8Wc;Z1n0D0FTQ2h;0z94ja0edw0;SvV7nmwMLI2

Organization Units (ou)

fQa9PpK9pwS

Use from Request ☐

Periods (pe)

Use from Request ☒

Period value will be taken from API request parameter (?pe=...)

Attribute Dimensions (Optional)

+ Add Attribute Dimension

Add custom attribute dimensions (e.g., category option combos, attributes)

Other Parameters (Optional)

+ Add Other Parameter

Add other query parameters (e.g., outputScheme=NAME, skipRounding=true, completedOnly=true)

Generated URL Preview

/api/analytics.json?dimension=dx:gscmTue30EY;wAyxe0razbM;nQpou1fu9H9;NXTYyp5QrLH;1e7t3f0LLGo;cIdwvoAgZLV;v4K1pDjcywB;zEAeW2HEVOB;KUvaaMBYSN1;k5A4Y81dCwK;eBbqiatcnKn;Xc5MqYLZ3Cd;ASVFI0KGKji;vnmveK2PZ54;11hdh9E0KwE;wU1QKpjSwrM;Km0qVQob9MY;Msm0TSTRN9k;mCKqPcYw7t;Nce6S25UISw;qGMXZnCezpm;szrz2G03B65;M0bZGvks8Wc;Z1n0D0FTQ2h;0z94ja0edw0;SvV7nmwMLI2&ou=fQa9PpK9pwS&pe=

✕ Cancel

Save API

This section defines the structure of your DHIS2 Analytics request.

The Middleware automatically builds the final URL (`/api/analytics.json?...)` based on the parameters you provide.

1 Data Dimensions (dx)

- Defines which data elements or indicators will be fetched from DHIS2.
- You can enter multiple UIDs separated by semicolons ;.

Example:

abc123;def456;ghi789

This will generate a query part like:

dimension=dx:abc123;def456;ghi789

- This field is mandatory ,without it, the API cannot generate valid Analytics data.

2 Organization Units (ou)

- Defines which organisation units (facilities, districts, etc.) to include.
- Enter UIDs separated by semicolons ;.

Example:

fQg3PxK9pkS;ab123cdEFgH

- Use from Request (checkbox):
 - Checked : The system will accept the OU dynamically from the caller (?ou= . . .).
 - Unchecked : The OU list here will be fixed in the configuration.
- This allows reuse of the same Analytics API for multiple organisational levels by passing different OU parameters at runtime.

3 Periods (pe)

- Defines the time periods for which data should be retrieved.
- You can use:

- Explicit periods: 202401 ; 202402
- Dynamic keywords: LAST_MONTH, THIS_YEAR, etc.
- Use from Request (checkbox):
 - Checked : The period will be read from the API request.
 - Unchecked : The period defined here will always be used.

Example final URL:

```
/api/analytics.json?dimension=dx:abc123&dimension=ou:D1930&dimension=pe:202505
```

4 Attribute Dimensions (Optional)

- Used to break down results by an additional dimension such as gender, age group, or category option combo.
- Each Attribute entry contains:
 - Attribute UID (the dimension key).
 - Values (allowed options, separated by ;).
- Useful when DHIS2 indicators are categorized by demographic or program-specific attributes.

5 Other Parameters (Optional)

- Add custom query parameters for advanced use cases.
- Common examples:
 - `outputIdScheme = NAME`
 - `skipRounding = true`
- These parameters are appended to the generated URL automatically.

6 Generated URL Preview

- Displays a **live preview** of the final DHIS2 Analytics URL.
- It updates automatically as you edit fields above, helping verify correctness before saving.

B. Metadata API Configuration

When API Type = Metadata, the form changes to show the Metadata API Binding section as below:

The screenshot shows the 'Edit Integrated API' form with the following fields and sections:

- API Code ***: Text input containing 'HealthCenter_Metadata'. Below it, a note states: 'Unique technical identifier' and 'Spaces will be automatically replaced with underscores ()'.
- API Name ***: Text input containing 'HealthCenter'. Below it, a note states: 'Descriptive name for this API'.
- API Type ***: Dropdown menu set to 'Metadata'. Below it, a note states: 'The type of DHIS2 API' and 'METADATA APIs require a Bound API Code to link with business APIs'.
- Metadata API Binding**: A section with a blue header and a descriptive paragraph: 'For METADATA APIs, you need to specify which business API this metadata is bound to. This allows the system to automatically fetch the correct metadata configuration.'
 - Bound API Code ***: Text input containing 'HealthCenter-mapping - /HealthCenter'. Below it, a note states: 'Select the dynamic route (Route ID - Path) that this metadata API supports'.
- Integrated System ***: Dropdown menu set to 'DHIS2.2'. Below it, a note states: 'The target system identifier (e.g., HMIS-DWH, DHIS2-Play)'.
- Status**: A checkbox labeled 'Active' is checked. Below it, a note states: 'Only active APIs will be available for use'.
- API URL ***: Text area containing the following JSON:


```
/api/metadata.json?
organisationUnits=true&fields=id,code,name,displayName,shortName,description,openingDate,closedDate,level,phoneNumber,email,address,url,parent[
id,code,name,displayName,level],translations,organisationUnitGroups[id,code,name],lastUpdated,lastUpdatedBy[id,displayName,username],attributeV
alues[value,attribute[id,code,name]]
```

At the bottom right, there are two buttons: 'Cancel' and 'Save API'.

Metadata APIs are used to fetch non-numeric, descriptive information ,such as names, levels, codes, and hierarchies of organisation units or data elements.

These APIs usually support Analytics routes by enriching their data.

1 Bound API Code

- Links this Metadata API to a specific Analytics API that it supports.
- This allows the system to automatically fetch metadata relevant to that business route.

Example:

HealthCenter-mapping : /HealthCenter

- When the HealthCenter Analytics API runs, it can automatically use this Metadata API to map UUIDs to readable names (e.g., “D1930” : “Irbid Health Directorate”).

2 Integrated System

- Selects which DHIS2 environment the Metadata API belongs to (same as in the main form).
- Ensures that the correct host and credentials are used during calls.

3 API URL

- Defines the actual DHIS2 endpoint for metadata retrieval.
- Usually starts with /api/metadata.json?

Example:

```
/api/metadata.json?organisationUnits=true&fields=id,code,name,displayName,level,parent[id,name]
```

Explanation:

When you use this endpoint, the system retrieves metadata details for all organisation units available in DHIS2.

It returns, for each organisation unit, the specified fields ,such as `id`, `code`, `name`, `displayName`, `level`, and its parent’s `id` and `name`.

You can control exactly which information to include by editing the `fields` parameter.

For example, if you add `email`, `address`, `phoneNumber`, the API will also return those details for every organisation unit.

4 Status & Description

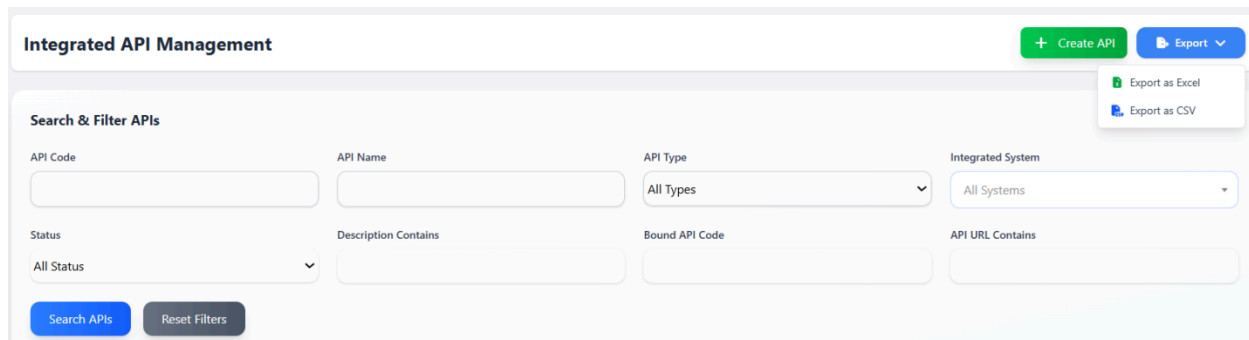
- Status: Active means the API is available for use.
- Description: Optional short text to explain the metadata purpose (e.g., “Fetch organisation unit structure with contact info”).

View Integrated API Details

By clicking the eye icon in the Integrated API Management table, the system opens a pop-up displaying the complete API configuration details in read-only mode for review and verification.

Export

Use Export : Export as Excel/CSV to download the current list for review, sharing, or backup.



The screenshot displays the 'Integrated API Management' interface. At the top right, there are two buttons: a green '+ Create API' button and a blue 'Export' button with a dropdown arrow. Below the 'Export' button, a dropdown menu is open, showing two options: 'Export as Excel' (with a green Excel icon) and 'Export as CSV' (with a blue CSV icon). The main area of the interface is titled 'Search & Filter APIs' and contains several input fields and dropdowns for filtering APIs. These include: 'API Code' (text input), 'API Name' (text input), 'API Type' (dropdown menu showing 'All Types'), 'Integrated System' (dropdown menu showing 'All Systems'), 'Status' (dropdown menu showing 'All Status'), 'Description Contains' (text input), 'Bound API Code' (text input), and 'API URL Contains' (text input). At the bottom left of the search area, there are two buttons: a blue 'Search APIs' button and a grey 'Reset Filters' button.

Figure 42 - Export Integrated APIs as Excel/CSV

Activation

Only Active APIs are selectable for execution. Toggle the status in the edit dialog to enable/disable availability.

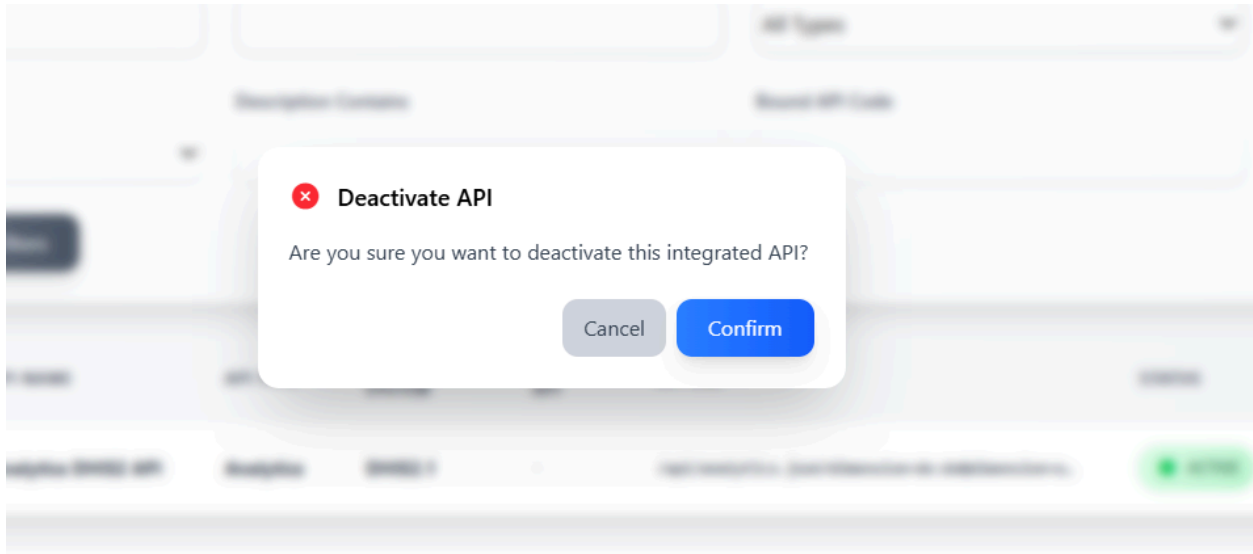


Figure 43 - Activation Integrated API

Example: DHIS2 Analytics

- Integrated System: DHIS2.DWH
- API Type: Analytics
- dx: gscmTUE3OEY;WAYxe0razbM;mQpoulfu9H9;
- ou: fQa9PpK9pWS(or Use from Request)
- pe:202505
- Other parameters: outputIdScheme=NAME; completedOnly=true

At runtime the middleware assembles the full analytics URL and executes it on the selected DHIS2 instance; results are traceable in API Transaction Logs.

Notes & Validation

- Required fields are marked with *.
- Semicolon (;) separates multiple UUIDs/values in dx, ou, pe, and attribute values.
- “Use from Request” lets callers override the field via API input; if not provided, the saved default is used.
- Edits are tracked in Audit Logs; executions are tracked in API Transaction Logs.

2.2.9 Integration Mapping

Overview

The Integration Mapping Management module allows administrators to define how data values from DHIS2 (or other Integrated APIs) are mapped to external system keys used in Middleware business routes.

Each Integration Mapping links an Integrated API (such as a DHIS2 Analytics API) to a specific Route ID, defining how individual data elements, indicators, or attributes correspond to their external representations.

Mappings serve as the translation layer between internal DHIS2 UUIDs (e.g., `xvHA0fLRcyX`) and external codes or field names used in integrated systems (e.g., `gp_general_visits`).

This ensures consistency, readability, and reusability across integrations and scheduled data exchange processes.

Purpose

The Integration Mapping module is primarily used to:

- Define how data elements or indicators are identified in Middleware routes.
- Associate each DHIS2 UUID (data element, indicator, or attribute) with a business key used by external systems.
- Manage mappings per Integration Route and Integrated API.
- Control which mappings are active and available for data transformation during runtime.

In short, this module acts as a bridge between DHIS2 analytics data and the business layer, allowing the Middleware to automatically translate DHIS2 field identifiers into meaningful keys that external consumers understand.

Search & Filters

The top section of the page provides a filter panel to quickly locate specific mappings.

Each filter refines the results displayed in the table below.

Available filters:

- Route ID - Path : Filter by the integration route to which mappings belong (e.g., `integration-mapping-api`).
- Integrated API : Select the DHIS2 API configuration linked to the mapping (e.g., `Health Analytics API (HEALTH_ANALYTICS)`).
- Mapping Type : Filter by the mapping category such as:
 - Data Element
 - Indicator
 - Data Element with Disaggregation
- Status : Show only *Active*, *Inactive*, or *All* mappings.
- External Key : Search by the external business key (e.g., `gp_general_visits`, `icu_patient_days`).
- Data (UID) : Search by the internal DHIS2 UID of the data element or indicator.
- Notes Contains : Search for mappings containing a specific note or description.
- Date Range : Filter by the creation or update date of mappings.

Actions:

- Click Search to apply filters.
- Click Reset to clear all filters and reload the full mapping list.

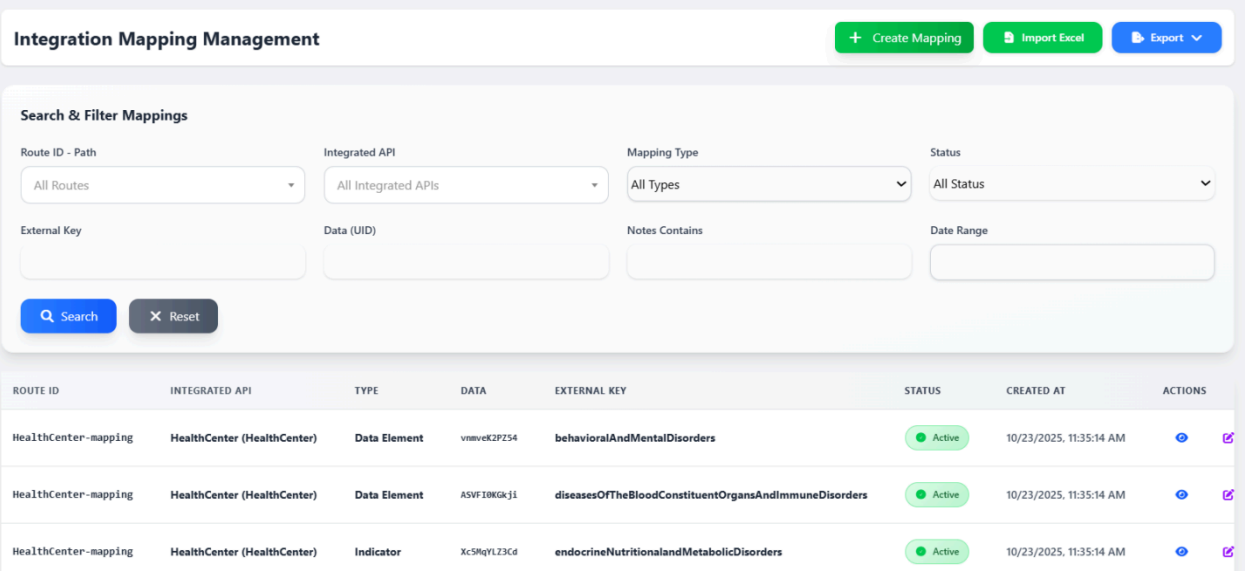


Figure 44 - Search Integration Mapping

Create a New Integration Mapping

The Create New Integration Mapping form allows administrators to define how individual DHIS2 data elements, indicators, or attributes are mapped to business-friendly field names used by Middleware APIs.

Each mapping connects a specific DHIS2 UID or expression to an external output key, under a selected route and API configuration

Mappings created here are used whenever the Middleware executes the corresponding route, ensuring consistent data translation between DHIS2 and external systems.

1 Route ID - Path

- The dynamic route this mapping belongs to.

- Select from predefined routes configured in the Integration Routes module.
- Determines where this mapping will be applied when data is fetched or transformed.

2 Data (UID / Expression)

- Enter the DHIS2 UID or expression depending on the mapping type.
- Example inputs:
 - For Data Element : xvHA0fLRcyX
 - For Indicator : TrtkNBSAdS
 - For Disaggregation : xvHA0fLRcyX.M0TzDNQdeju

The system uses this UID to fetch the value from DHIS2 Analytics results.

3 Integrated API

- Select the DHIS2 API configuration (created under *Integrated API Management*) that this mapping should extract data from.
- This field links the mapping directly to an Analytics or Metadata API, ensuring the correct DHIS2 source is used at runtime.

4 Mapping Type

- Defines how the system will interpret and process the DHIS2 data.
- Available types include:
 - Data Element : Maps a single data element UID to an external key.
 - Data Element with Disaggregation : Combines multiple category option combos under one mapping.
 - Indicator : Maps a DHIS2 indicator UID to an output field.
- The selected type determines the input format required in the “Data (UID/Expression)” field.

5 External Key (Output Field)

- The friendly name that will appear in the Middleware API response.
- This acts as the “output field name” for consumers.
- Example:
 - gp_general_visits
 - icu_patient_days
- Helps make the API output more descriptive instead of showing raw UIDs.

6 Status

- Determines if the mapping is active and should be used in API calls.
 - Active : included during data extraction.
 - Inactive : ignored at runtime but kept for record or later use.

7 Notes & Comments

- Optional text area for documentation or usage notes.
- Can include:
 - Context for the mapping (e.g., “Used in monthly facility report”).
 - Business logic references or testing notes.

+ Create New Integration Mapping

X

Route ID - Path

Select Route

The dynamic route that this mapping belongs to

Data (UID/Expression)

Select mapping type first

Please select a mapping type to see format requirements

Integrated API

Select Integrated API Configuration

Choose the DHIS2 API configuration to extract data from

External Key (Output Field)

The field name that will appear in the API response

Mapping Type

Select Mapping Type

Choose how to extract and process the data from DHIS2

Status

☒ Active

Only active mappings will be processed in API calls

Notes & Comments

Optional: Add context, usage notes, or any special considerations

Figure 45 - Create New Integration Mapping

Edit an Integration Mapping

From the list, click Edit on a row to modify fields. You can change the route path, integrated api, mapping type, Data(UID/Expression) , External Key , notes , and Status . Save to apply changes.

 **Edit Integration Mapping** 

Route ID - Path

HealthCenter-mapping - /HealthCenter 

The dynamic route that this mapping belongs to

Integrated API

HealthCenter (HealthCenter) 

Choose the DHIS2 API configuration to extract data from

Mapping Type

Data Element 

Choose how to extract and process the data from DHIS2

Notes & Comments
Optional: Add context, usage notes, or any special considerations

Data (UID/Expression)

vnmveK2PZ54

Select a mapping type above to see format examples

External Key (Output Field)

behavioralAndMentalDisorders

The field name that will appear in the API response

Status

☒ Active

Only active mappings will be processed in API calls

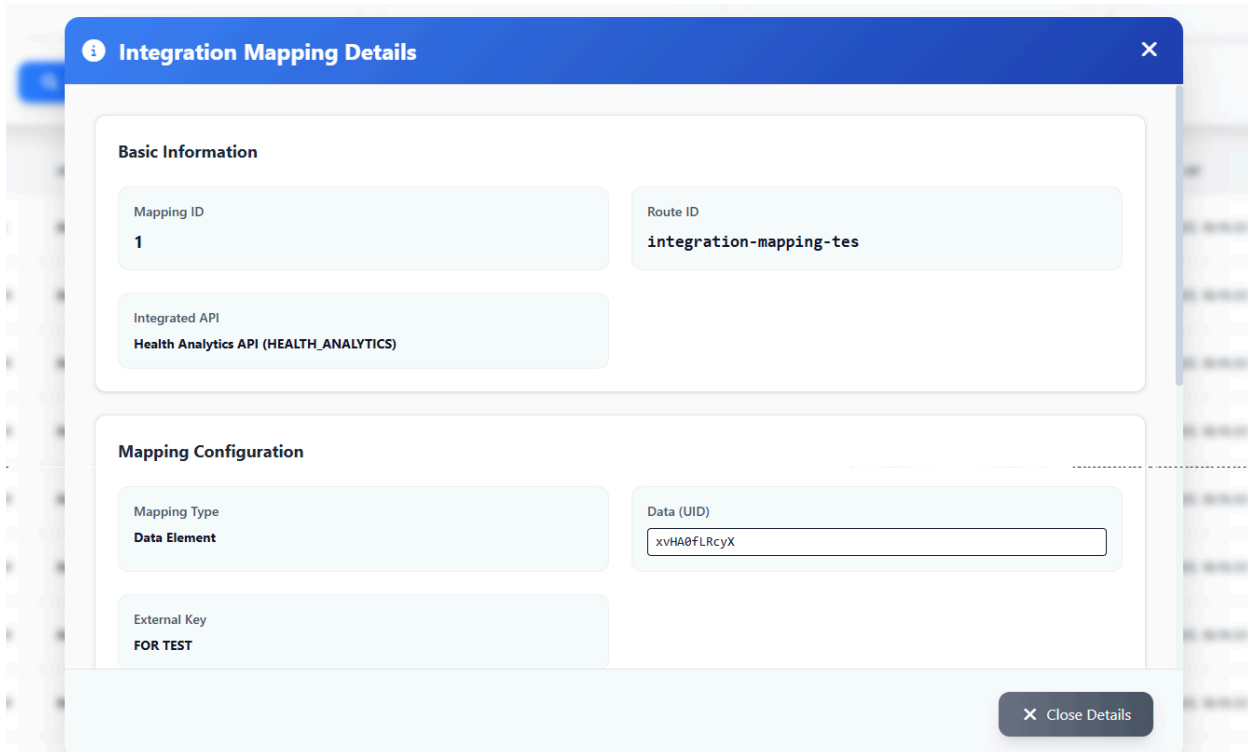
 Cancel

 Save Mapping

Figure 46 - Edit Integration Mapping

View Integration Mapping Details

By clicking the eye icon in the Integration Mapping table, the system displays a pop-up showing the full mapping details, including all configuration fields for review in read-only mode.



The image shows a modal window titled "Integration Mapping Details" with a close button in the top right corner. It is divided into two main sections: "Basic Information" and "Mapping Configuration".

Basic Information

- Mapping ID: 1
- Route ID: integration-mapping-tes
- Integrated API: Health Analytics API (HEALTH_ANALYTICS)

Mapping Configuration

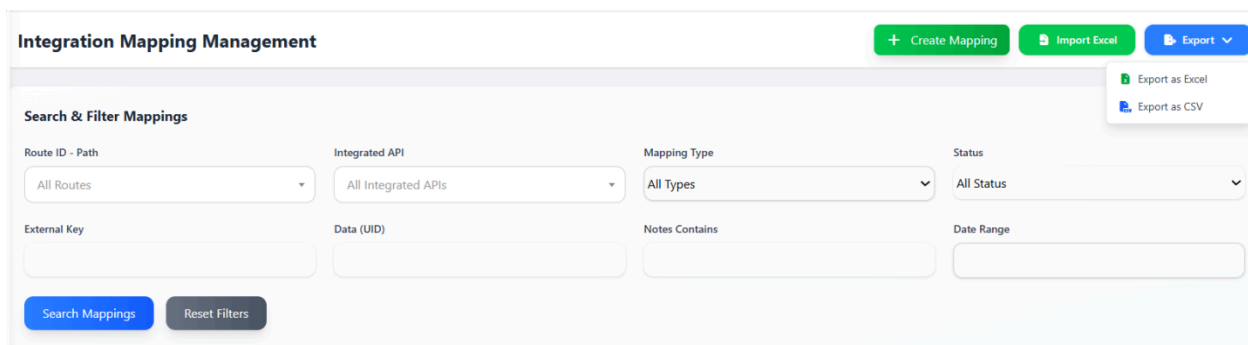
- Mapping Type: Data Element
- Data (UID): xvhA0fLRcyX
- External Key: FOR TEST

A "Close Details" button is located at the bottom right of the modal.

Export & Import

Use Export : Export as Excel/CSV to download the current list for review, sharing, or backup.

Use Import Excel to bulk : create/update mappings from a template. Invalid rows are rejected with detailed messages; valid rows are processed.



The image shows the "Integration Mapping Management" interface. At the top, there are three buttons: "+ Create Mapping", "Import Excel", and "Export" (with a dropdown arrow). Below these is a "Search & Filter Mappings" section with several filters:

- Route ID - Path: All Routes (dropdown)
- Integrated API: All Integrated APIs (dropdown)
- Mapping Type: All Types (dropdown)
- Status: All Status (dropdown)
- External Key: (text input)
- Data (UID): (text input)
- Notes Contains: (text input)
- Date Range: (text input)

At the bottom of the search section are two buttons: "Search Mappings" and "Reset Filters". A dropdown menu for the "Export" button is visible, showing options for "Export as Excel" and "Export as CSV".

Figure 47 - Export / Import Integration Mapping

Activation

Unchecked the Status box to deactivate; check again to reactivate. Deactivated mappings never affect runtime transformations or API responses.

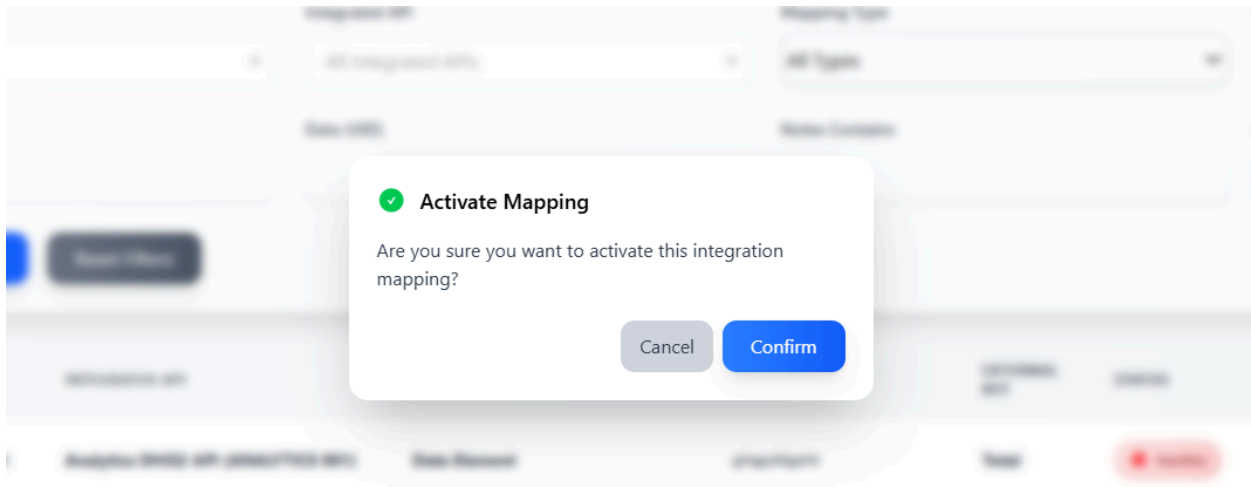


Figure 48 - Activate / Deactivate Mapping

2.3 Dynamic Routes

2.3.1 Overview

This module lets administrators manage the HTTP routes that the middleware (Apache Camel + Spring Boot) exposes and executes. From here you can create, version, activate/deactivate, and export route definitions. All routes are stored centrally and can be updated without re-deploying the application.

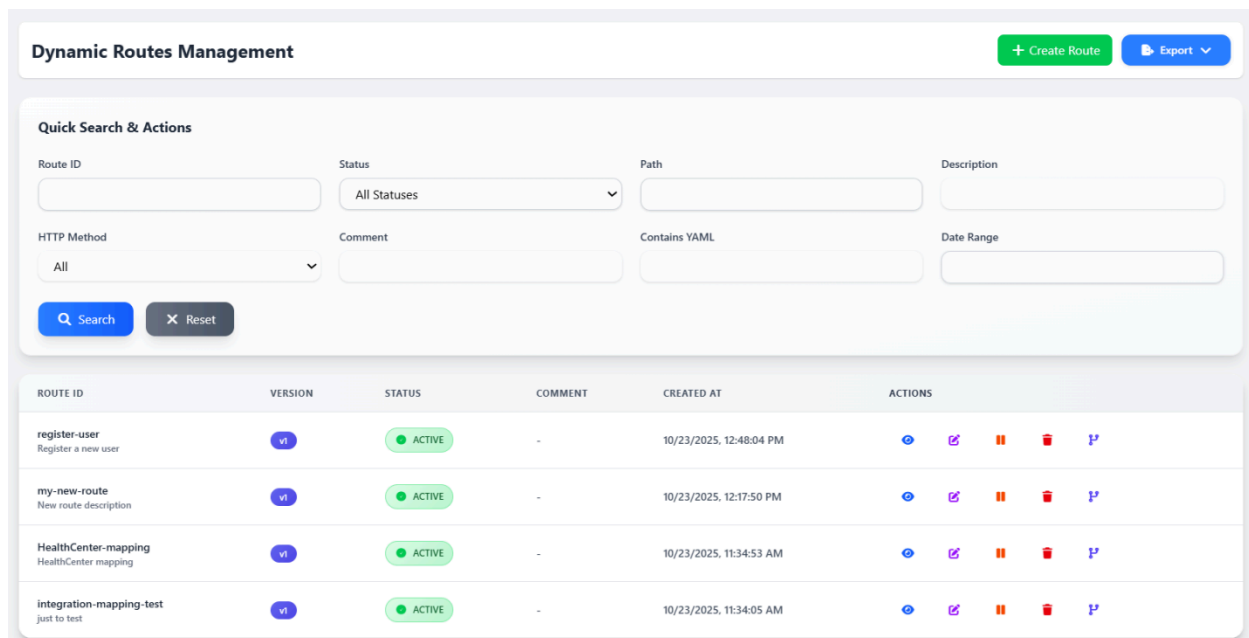


Figure 49 - Dynamic Routes

2.3.2 Features and Actions

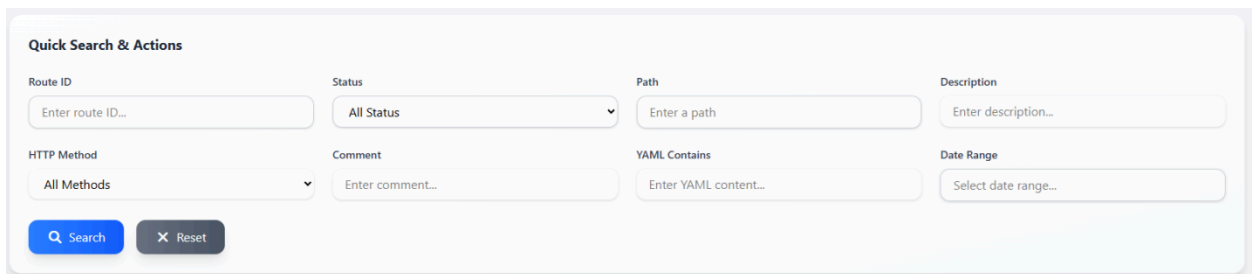
Features

- **Create Route:** define a new route in YAML/DSL, including method, path, and processing steps.
- **Quick Search & Filters:** instantly find routes by ID, status, HTTP method, path, comment, or YAML content.
- **Versioning:** each route can have multiple versions. One version can be marked as the default active.
- **Activation:** toggle routes ON/OFF without deleting them.
- **Export:** download selected routes for backup or migration.

- Actions per route: preview, edit, activate/deactivate, delete, and more (based on permissions).

Quick Search & Filters

- Route ID: Enter a full or partial route identifier.
- Status: All, Active, or Inactive.
- HTTP Method: Filter by GET, POST, PUT, DELETE, etc.
- Path: Search the exposed HTTP path (e.g., /external/integrate).
- Comment: Matches routes that include this text in the comment field.
- YAML Contains: Search inside the YAML body of the route.
- Date Range: Return routes created or updated within the selected time window.
- Search / Reset: Apply filters or clear them to return to the full list.



The screenshot shows a 'Quick Search & Actions' panel with a light blue background. It contains several search filters arranged in two rows. The first row includes 'Route ID' (text input), 'Status' (dropdown menu showing 'All Status'), 'Path' (text input), and 'Description' (text input). The second row includes 'HTTP Method' (dropdown menu showing 'All Methods'), 'Comment' (text input), 'YAML Contains' (text input), and 'Date Range' (text input). At the bottom left of the panel are two buttons: a blue 'Search' button with a magnifying glass icon and a grey 'Reset' button with an 'X' icon.

Figure 50 - Quick Search Dynamic Routes

Routes table columns

1. Route ID: Unique identifier of the route. The subtitle shows the route description.
2. Version: The current version label (e.g., v1).
3. Status: Active/Inactive badge indicating whether the route is enabled.
4. Comment: Optional note saved with the route version.
5. Created At: Timestamp of when this version was created.
6. Actions: Context actions available for the route (see below).

ROUTE ID	VERSION	STATUS	COMMENT	CREATED AT	ACTIONS
register-user Register a new user	v1	ACTIVE	-	10/23/2025, 12:48:04 PM	    
my-new-route New route description	v1	ACTIVE	-	10/23/2025, 12:17:50 PM	    
HealthCenter-mapping HealthCenter mapping	v1	ACTIVE	-	10/23/2025, 11:34:53 AM	    
integration-mapping-test just to test	v1	ACTIVE	-	10/23/2025, 11:34:05 AM	    

Figure 51 - Route Table Columns

Row actions

- 1. Preview: Open the route to view its YAML/definition in read-only mode.
- 2. Edit: Modify the route metadata or YAML and save as a new version if required.
- 3. Activate / Deactivate: Toggle the route availability. Only active routes will receive traffic.
- 4. Delete: Remove the selected version (permission-restricted).
- 5. More actions: Additional options (e.g., clone/duplicate, version history) may appear based on your role.

STATUS	COMMENT	CREATED AT	ACTIONS
 ACTIVE		2025-09-09T13:05:38.69989	    

Figure 52 - Row Actions / Dynamic Routes

Create a new route

- 1. Click Create Route.
- 2. Fill in basic metadata (Route ID, description/comment).
- 3. Paste or write the route definition (YAML/DSL).
- 4. Validate and save. A new version (e.g., v1) is created.
- 5. Activate the route to start receiving requests.

+ Create New Route

Comment (Optional):
Describe changes or notes...

YAML Content:

```

- route:
  id: "my-new-route"
  description: "New route description"
  from:
    uri: "rest:post:/api/hello"
    steps:
      log:
        message: "received: ${body}"
      set-body:
        constant: "hello from new route!"
      to: "log:output"

```

Cancel Validate Test Route Save Route

Figure 53 - Create New Dynamic Route

Edit an Existing Route (Using the API Designer)

This section explains how to modify a route's logic, structure, and metadata using the API Designer, either through the Visual Design Editor or the YAML Source Code Editor.

Each update is automatically versioned , the previous version remains available for rollback.

Open the Editor

1. From the Dynamic Routes / APIs : Routes table, locate the route you want to edit.
Click Edit under the Actions column.
2. The selected route opens in the API Designer with the current version pre-selected.

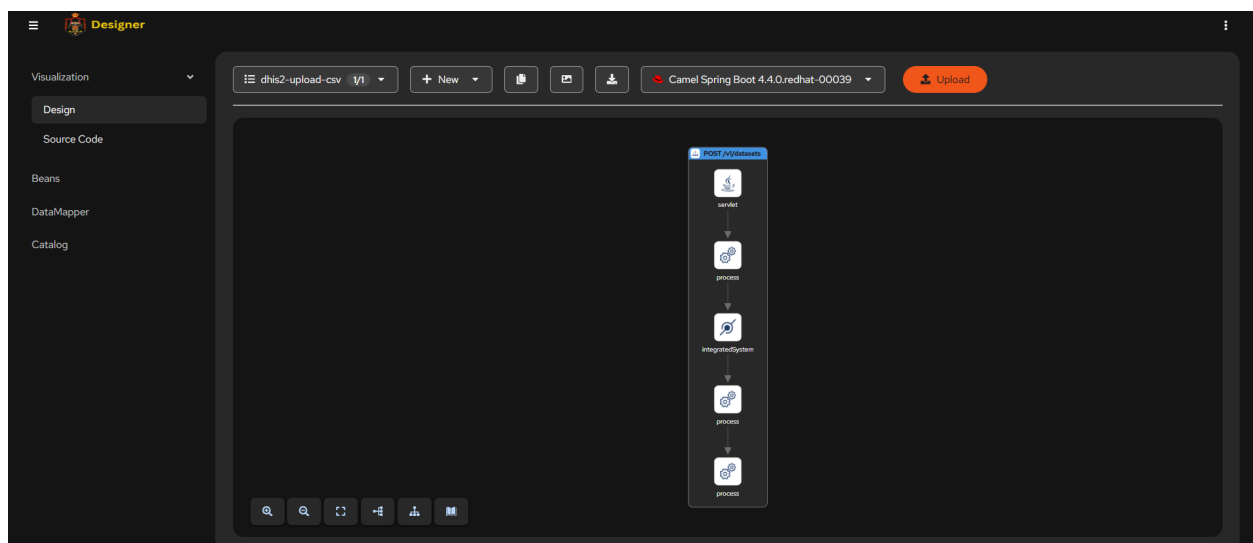


Figure 54 - Edit Route (Visual Designer)

Editing with the Visual Designer

Use this mode when you prefer a graphical, block-based view of the flow.

The Visual Designer enables you to modify routes interactively through drag-and-drop actions.

1. **Add / Remove Steps** : drag a new process or component from the Catalog and drop it between existing steps, or select an existing one and press Delete.
2. **Reorder Steps** : drag and move blocks vertically to change their execution order.
3. **Configure a Step** : click a block to open its Properties Panel on the right.
 - Edit parameters such as bean ref, expression, options, or endpoint URI.
4. **HTTP Method & Path** : click the from: servlet:/... node to modify endpoint details like httpMethodRestrict or URI pattern.
5. **Add Comments** :in the Properties Panel, fill in the Description or Comment field to document the step's function.

Comments appear inside the YAML as annotations, for example:

```
# Validates uploaded CSV structure
```

```
- process: { ref: csvGuard }
```

6. **Zoom & Layout Tools** – use the bottom toolbar to zoom in/out, reset view, or switch between horizontal and vertical layout modes.

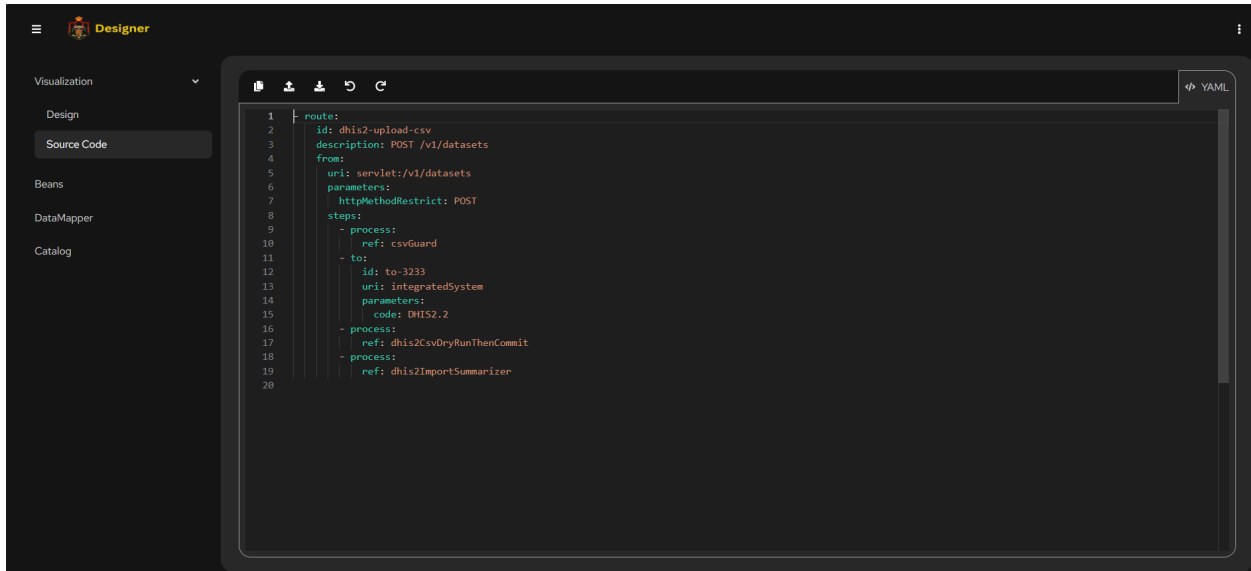


Figure 55 - Edit Route (Source Code / YAML)

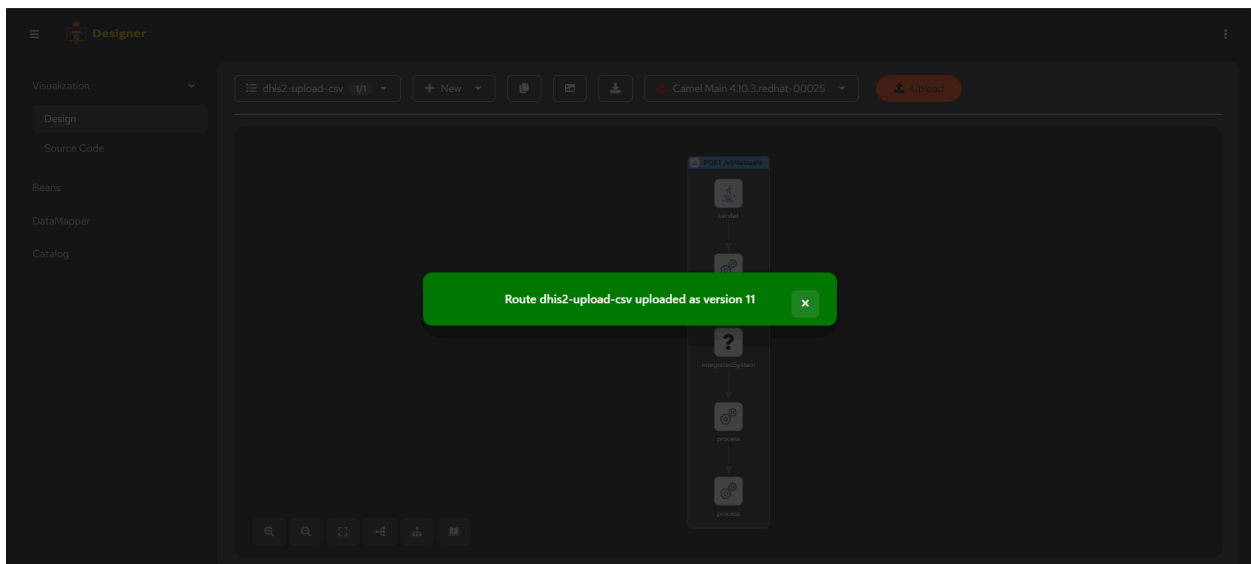


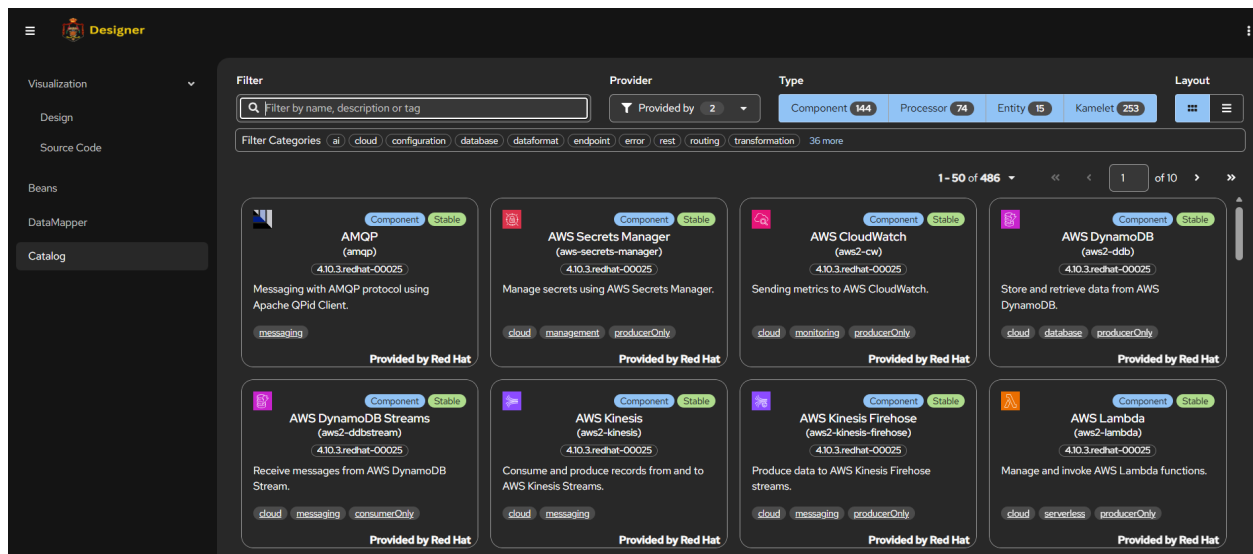
Figure 56 - Upload as Activated New Version

Using the Catalog

The Catalog contains all Camel components, processors, and Kamelets available in the system.

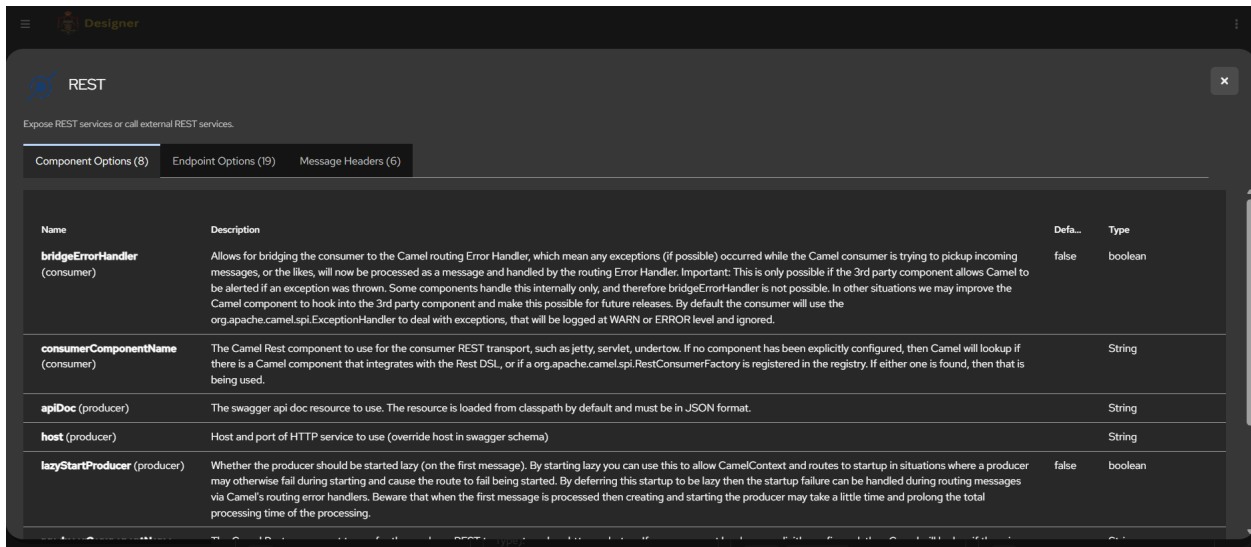
To add a new integration step:

1. Click Catalog from the left navigation menu.



2. Use the search bar to find a specific component (for example, `rest`, `http`, `setBody`, `bean`, etc.).
3. Click the desired component card to open its detail view.
4. Review its Component Options, Endpoint Options, and Message Headers tabs to understand configuration fields.
5. Drag the component onto your canvas or click Add to Route to insert it directly.

Tip: the Catalog also displays documentation for each parameter, its data type, default value, and usage scope (consumer / producer).



Editing in Source Code (YAML)

Switch to the Source Code tab for precise text-based editing.

Changes here are reflected instantly in the visual view.

1. Click Source Code in the left navigation.
2. Update the YAML directly.

Example:

```

1  - route:
2    id: dhis2-upload-csv
3    description: POST /v1/datasets
4    from:
5      uri: servlet:/v1/datasets
6      parameters:
7        httpMethodRestrict: POST
8      steps:
9        - process:
10          ref: csvGuard
11        - to:
12          id: to-3233
13          uri: integratedSystem
14          parameters:
15            code: DHIS2.2
16        - process:
17          ref: dhis2CsvDryRunThenCommit
18        - process:
19          ref: dhis2ImportSummarizer
20
```

3. Click Upload to save your edits.

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Versioning and Activation

After editing, click Upload to edit route version.

A confirmation message appears (for example:

“Route dhis2-upload-csv uploaded as version 2”).

- Activate / Deactivate a Route:
 - From the Routes table, click **Activate** to enable or **Deactivate** to disable.
 - The status badge updates accordingly.

Exporting and Documentation

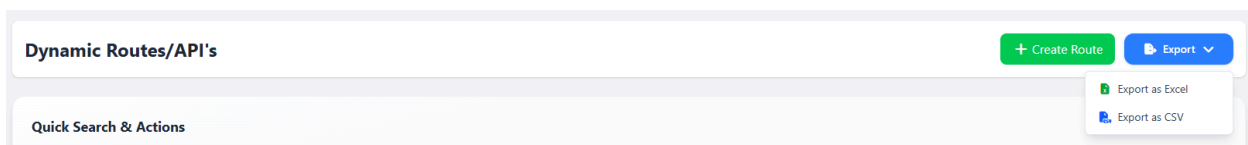


Figure 57 - Export Dynamic Routes

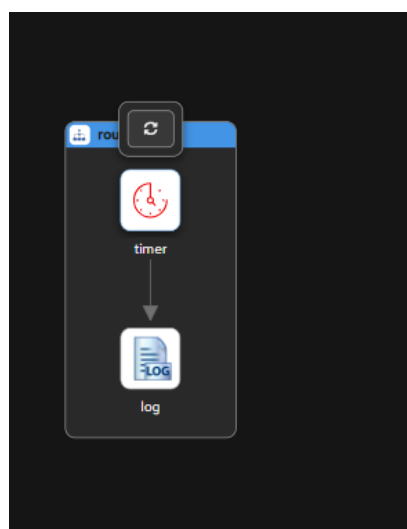
Practical Example – dhis2-upload-csv Route

This practical example demonstrates how to build and configure an integration route visually in the API Designer , starting from a basic flow, adding components, editing their parameters, and automatically generating the corresponding YAML.

Step 1: Start from the Default Template

When a new route is created, the Designer displays a default timer-to-log flow .

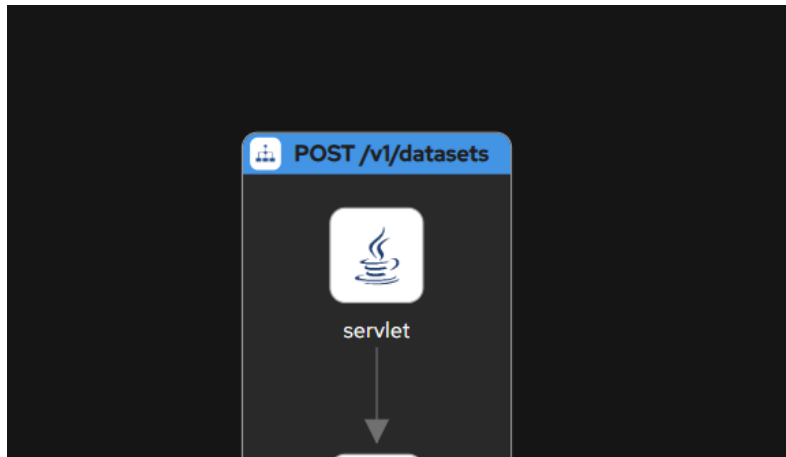
This serves as a basic starting point to visualize how the route structure works.



Step 2: Replace the Timer with a Servlet Endpoint

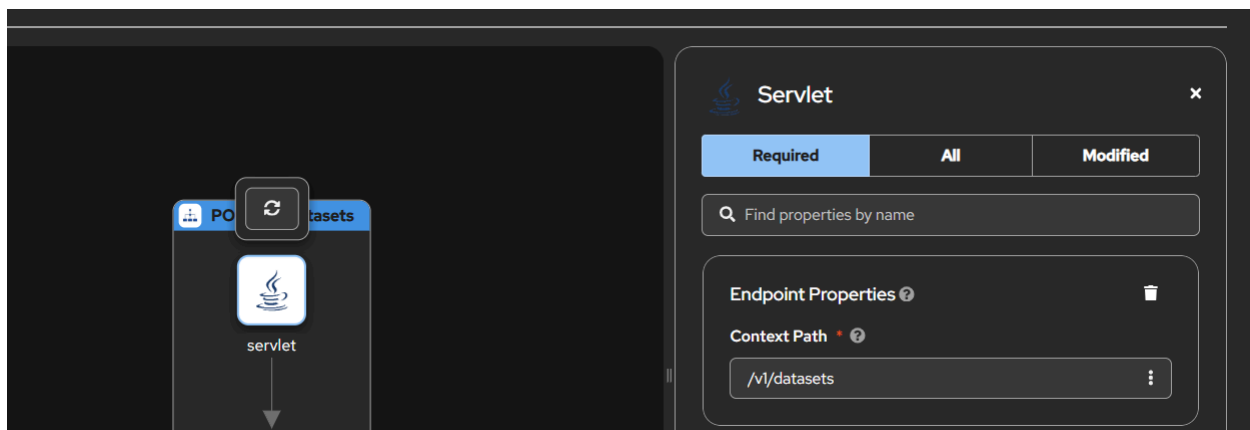
1. Click the timer block.
2. A small toolbar appears above the block click the Replace icon (↻) to open the Catalog.
3. In the Catalog search bar, type “servlet”.
4. Select the Servlet component and confirm replacement.

The timer block is now replaced with a servlet endpoint.

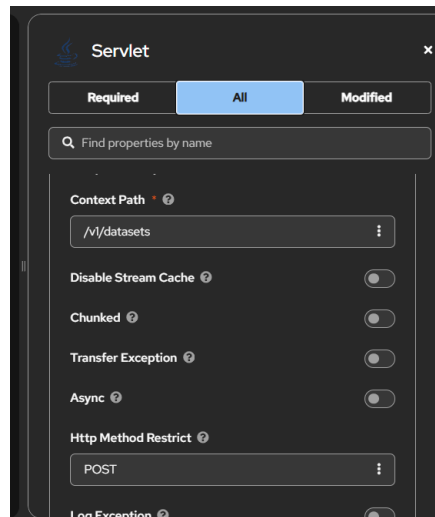
**Step 3: Configure the Servlet Endpoint**

With the servlet block selected, a Properties Panel appears on the right.

This panel defines how incoming requests are handled.

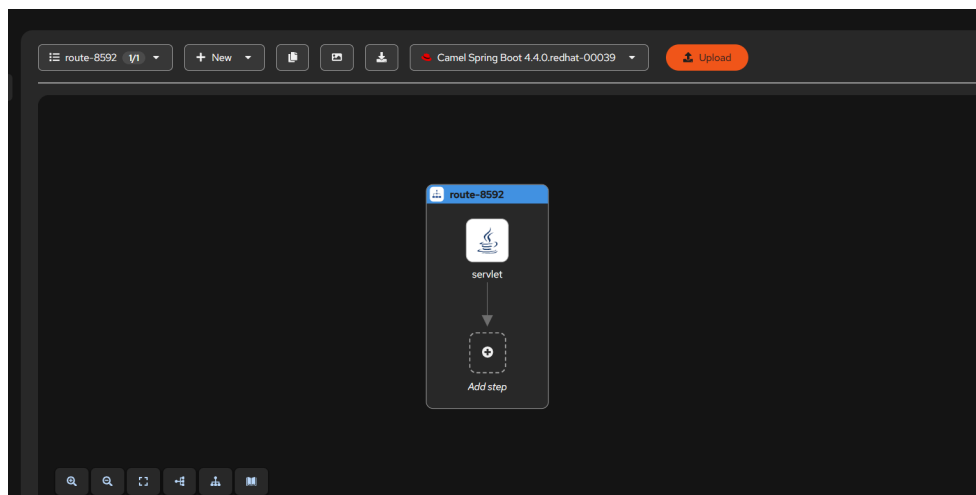


Property	Value	Description
Context Path	/v1/datasets	Defines the exposed HTTP endpoint path
Http Method Restrict	POST	Restricts requests to POST only

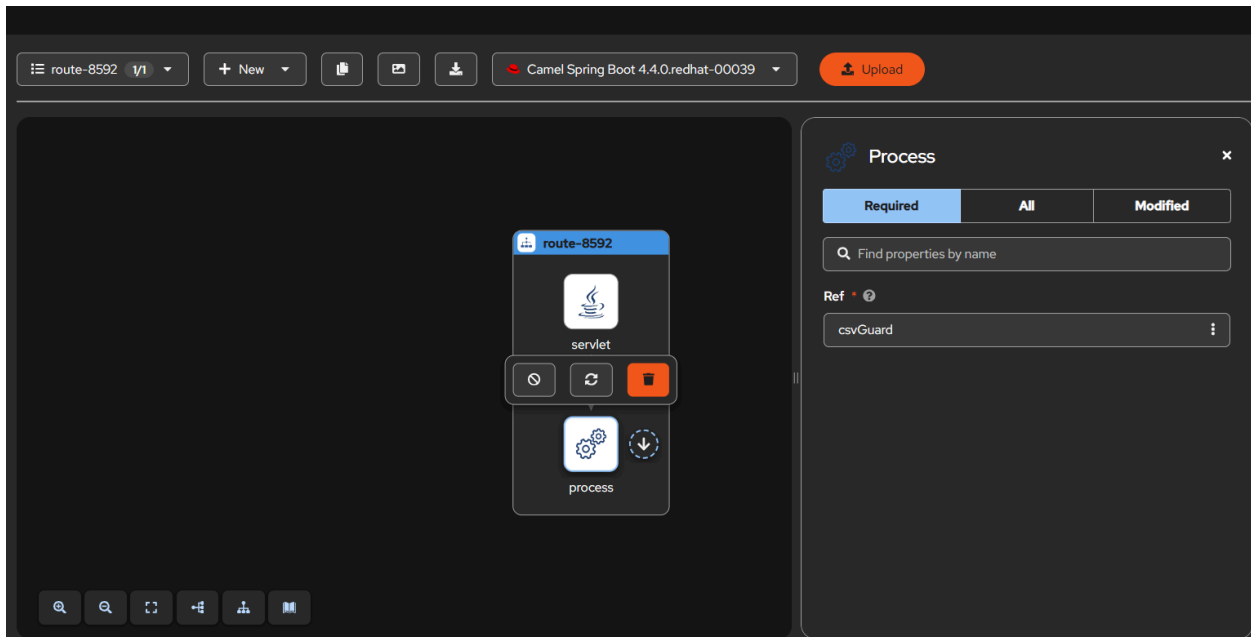


Step 4: Add a Process Step (csvGuard)

1. Click the Catalog button in the left panel.
2. In the Filter field, type **process**.
3. Drag the Process component and drop it below the servlet node.
4. Click the newly added Process block to open its Properties Panel.
5. Set the following:



Property	Value
Ref	csvGuard
Description	(optional – add a short note such as “validates uploaded CSV”)



Step 5: Add an Integrated System Step

Next, we connect the route to an external system (for example, DHIS2.TRACKER).

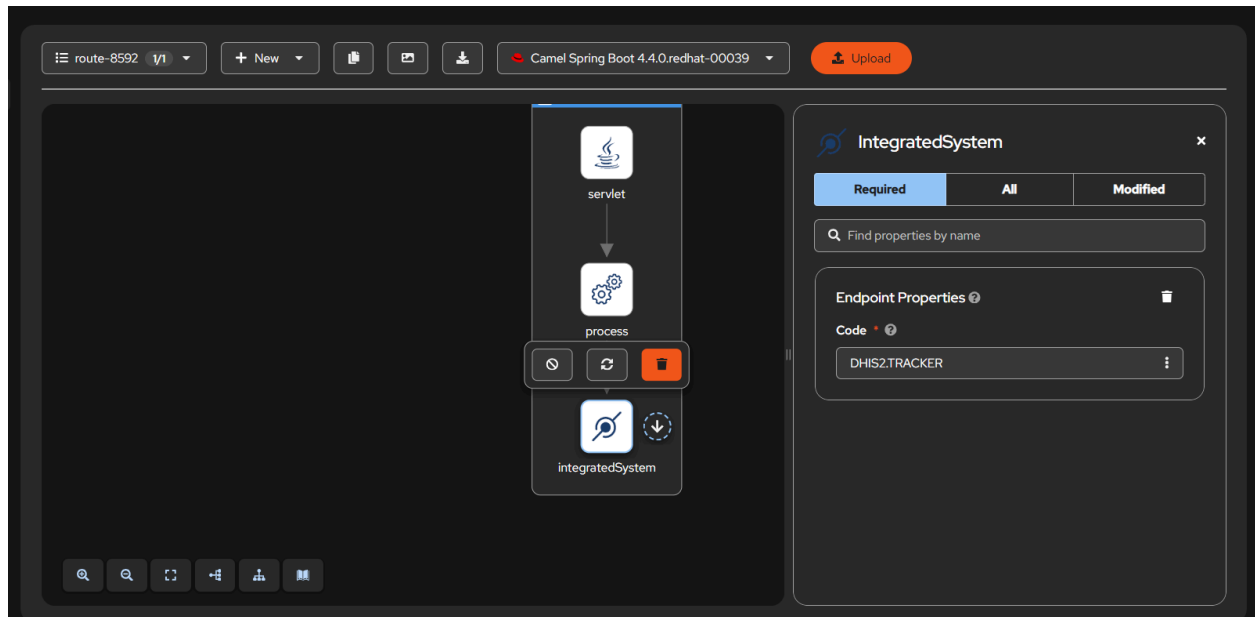
1. From the Catalog, drag the IntegratedSystem component and drop it below the previous process step.
2. Click the IntegratedSystem block to open its properties on the right panel.
3. Fill in the following details:

Property	Value
Code	The integration code that identifies the target external system (e.g., DHIS2.TRACKER).
Description	Notes about the connection or data exchange.

Note:

The Code value (e.g., DHIS2 . TRACKER) must already be defined in the Integrated Systems Management page within the Middleware system.

If the integration code is not registered there, the route will not be able to connect to the target external system.

**Step 6: Add Additional Processing Steps**

After the external call, add additional process blocks to complete your flow:

1. Add a Process component for `dhis2CsvDryRunThenCommit`.
2. Add another Process for `dhis2ImportSummarizer`.

Each process can include a custom `ref`.

Step 7: Review and Generate YAML Automatically

The Designer automatically converts your visual route into its YAML definition.

Click Source Code in the left panel to view the generated YAML:

```
1  - route:
2    id: dhis2-upload-csv
3    description: POST /v1/datasets
4    from:
5      uri: servlet:/v1/datasets
6      parameters:
7        httpMethodRestrict: POST
8    steps:
9      - process:
10        ref: csvGuard
11      - to:
12        id: to-3233
13        uri: integratedSystem
14        parameters:
15          code: DHIS2.2
16      - process:
17        ref: dhis2CsvDryRunThenCommit
18      - process:
19        ref: dhis2ImportSummarizer
20
```

Step 8: Save and Upload the Route

1. Click the Upload button on the top bar.
2. The system will save your new route version and display a confirmation message.

Result:

This case demonstrates how to create and configure an API Route visually through the API Designer.

The same steps can be followed to build a Hakeem API Route or any other integration route. After completing these steps, and ensuring the integration code (e.g., DHIS2.TRACKER) is properly registered under Integrated Systems Management the route becomes fully operational and ready for activation within the Middleware environment.

2.4 Scheduled Jobs

2.4.1 Overview

The Scheduled Jobs module in the Middleware System provides a robust mechanism for automating tasks and processes that need to run at specific intervals or times. This feature is for maintaining system efficiency, performing routine data synchronization, generating reports, or executing any API calls or workflows on a predefined schedule.

2.4.2 Features and Actions

Scheduled Jobs List

The main page lists all jobs with:

- Method: HTTP method.
- Target: Endpoint.
- Schedule: Frequency and time.
- Status: Active/Inactive.
- Created At: Creation timestamp.
- Last Updated At: Last modification timestamp.
- Actions: Icons for details, edit, activate/deactivate, delete.

Scheduled Jobs								
+ Create Job Export								
Filters Job Name API Endpoint Method Status Search by name... Search by endpoint... All All Apply Filters Reset								
Job ID	Job Name	Method	Target	Schedule	Status	Created At	Last Updated At	Actions
1	get User	GET	https://jsonplaceholder.typicode.com/p...	Every 10 seconds	Active	09-09-2025, 05:56:55 PM	10-09-2025, 04:24:39 PM	Details Edit Activate Delete
3	Get Information	GET	https://api.agify.io/?name=neelad	Every 4 seconds	Inactive	09-09-2025, 07:37:56 PM	09-09-2025, 07:38:19 PM	Details Edit Activate Delete
2	Create User	POST	https://jsonplaceholder.typicode.com/p...	Every 5 seconds	Inactive	09-09-2025, 06:03:42 PM	09-09-2025, 06:05:37 PM	Details Edit Activate Delete
10 Records per page 3 Total records Page 1 of 1 Previous Next								

Figure 58 - Scheduled Jobs List

Creating a Scheduled Job

To create a new scheduled job, navigate to the 'Scheduled Jobs' section within the Middleware interface and click on the 'Create Job' button. This action will open a dialog box titled 'Create Scheduled Job'.

Users are required to provide the following details:

- **Job Name:** A unique and descriptive name for the scheduled task. This name helps in identifying the job within the system.
- **Description:** An optional field to provide a more detailed explanation of the job's purpose or functionality.
- **Schedule Type:** This field defines the frequency and timing of the job's execution. Upon clicking the 'Select Schedule' dropdown, a list of predefined scheduling options appears, including:
 1. Every N Seconds
 2. Every N Minutes
 3. Hourly
 4. Daily
 5. Weekly
 6. Monthly
 7. Yearly
 8. Custom

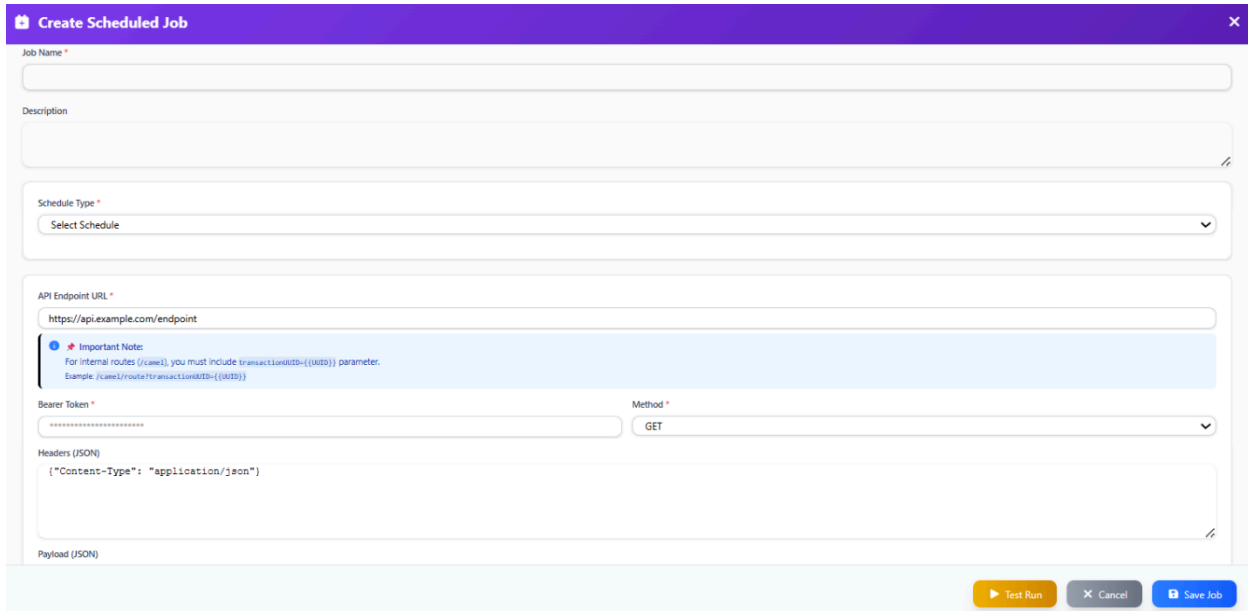
The image shows a 'Create Scheduled Job' dialog box with a purple header. It contains several input fields: 'Job Name', 'Description', 'Schedule Type' (a dropdown menu), 'API Endpoint URL', 'Bearer Token', and 'Method' (a dropdown menu). There is also a section for 'Headers (JSON)' and 'Payload (JSON)'. An 'Important Note' is displayed, stating that for internal routes, a 'transactionId' parameter must be included. At the bottom right, there are three buttons: 'Test Run' (orange), 'Cancel' (grey), and 'Save Job' (blue).

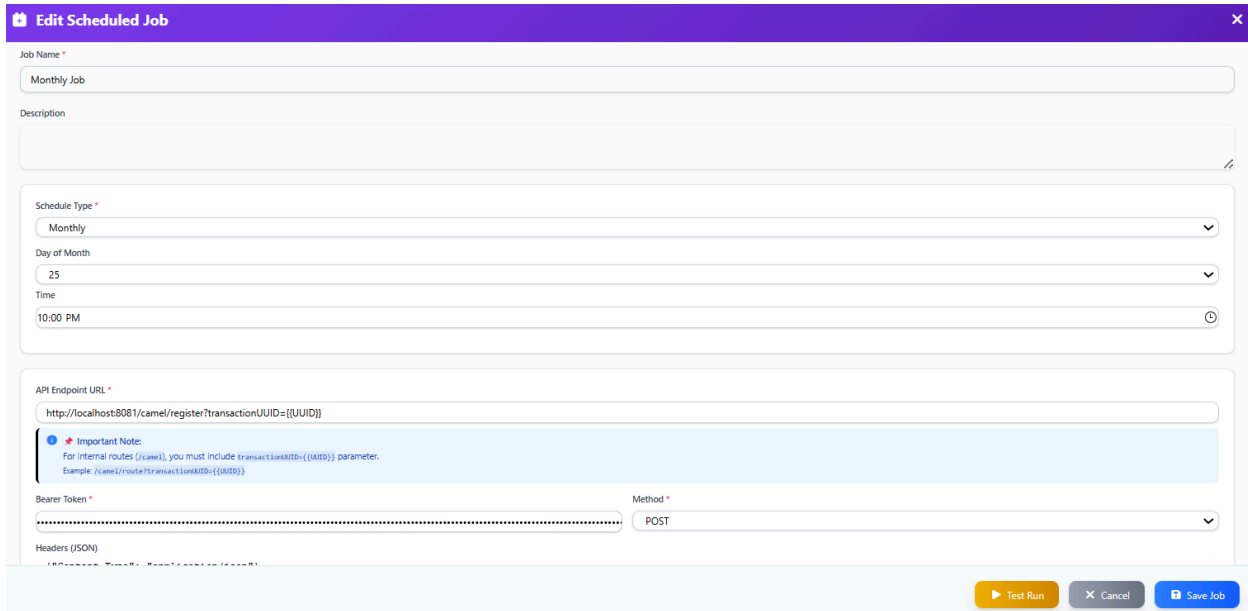
Figure 59 - Create Scheduled Job Dialog

Once all the necessary information is entered, users can either 'Test Run' the job to verify its configuration or 'Save Job' to add it to the list of scheduled tasks. The 'Cancel' button allows users to discard the current entry without saving.

Editing a Scheduled Job

Existing scheduled jobs can be modified to update their name, description, or scheduling parameters.

To edit a job, locate it in the Scheduled Jobs list and click on the corresponding edit icon. This will open the 'Edit Scheduled Job' dialog, which is similar to the creation dialog but pre-populated with the job's current details.



Edit Scheduled Job

Job Name *
Monthly Job

Description

Schedule Type *
Monthly

Day of Month
25

Time
10:00 PM

API Endpoint URL *
http://localhost:8081/camel/register?transactionUUID={{UUID}}

Important Note:
For internal routes (/camel), you must include transactionUUID={{UUID}} parameter.
Example /camel/route?transactionUUID={{UUID}}

Bearer Token *
.....

Method *
POST

Headers (JSON)
{ "Content-Type": "application/json" }

Test Run **Cancel** **Save Job**

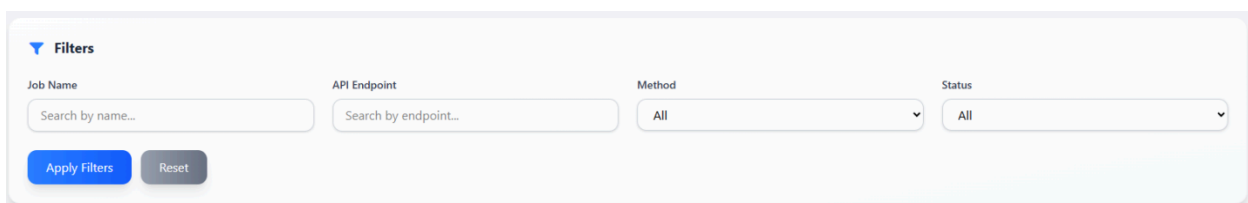
Figure 60 - Edit Scheduled Job Dialog

After making the desired changes, click 'Save Job' to apply them. The system will update the job's configuration, and the changes will take effect from the next scheduled run. Users can also 'Test Run' the modified job or 'Cancel' the editing process.

Search and Filters

Locate jobs using search and filter options:

- Job Name: Search by name.
- API Endpoint: Search by endpoint.
- Method: Filter by HTTP method.
- Status: Filter by status. Click 'Apply Filters' or 'Reset'.



Filters

Job Name
Search by name...

API Endpoint
Search by endpoint...

Method
All

Status
All

Apply Filters **Reset**

Figure 61 - Search and Filters Scheduled Jobs

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Exporting Scheduled Jobs

Export the job list by clicking 'Export', then choose 'Export as Excel' or 'Export as CSV'.

Figure 59 shows export options.

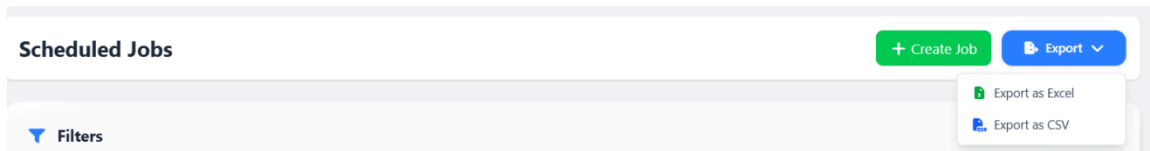


Figure 62 - Export Scheduled Jobs

Activating and Deactivating a Scheduled Job

Scheduled jobs can be activated or deactivated to control their execution without deleting their configuration. This is useful for temporarily pausing a job or enabling it after a period of inactivity. The status of a job is clearly indicated in the 'Status' column of the Scheduled Jobs list.

To change the status, click on the appropriate action icon (e.g., a pause or play button, or a toggle switch) in the 'Actions' column. Before the status change is applied, a 'Confirm Action' dialog will appear, as shown in Figure 36, asking for confirmation.

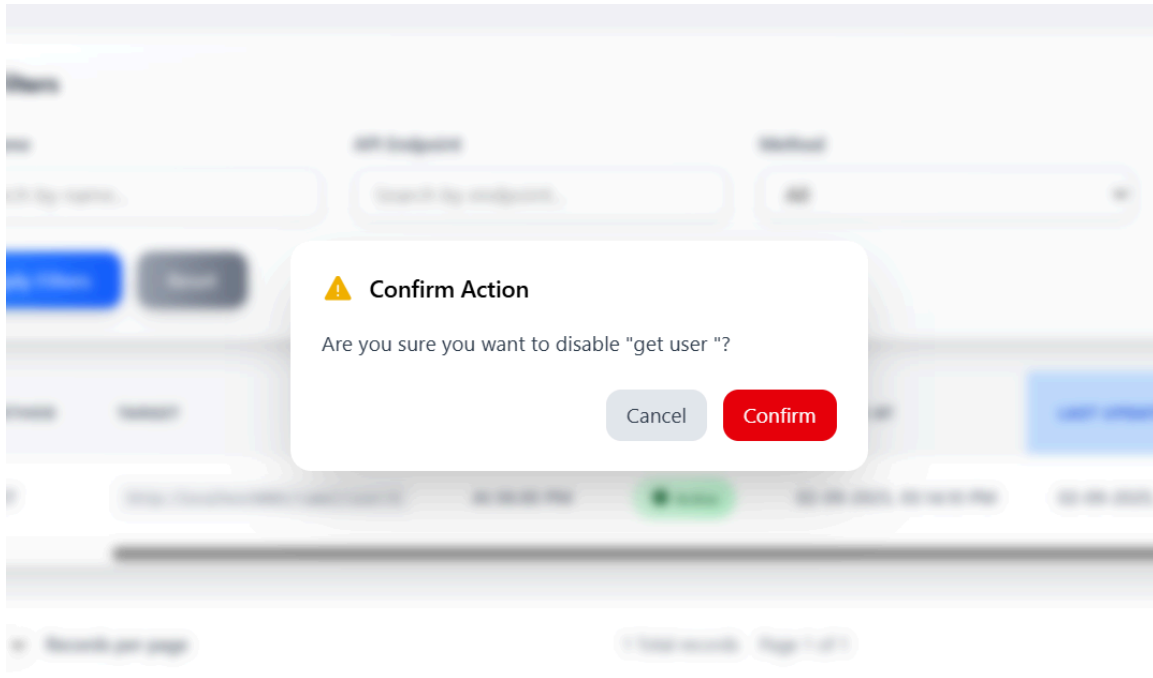


Figure 63 - Confirm Action Dialog for Disabling a Scheduled Job

Confirming the action will update the job's status, and it will either start running according to its schedule (if activated) or cease further executions (if deactivated).

Deleting a Scheduled Job

To permanently remove a scheduled job from the system, click on the delete icon (typically a trash can icon) in the 'Actions' column of the Scheduled Jobs list.

Similar to activating/deactivating, a 'Confirm Action' dialog will appear, as shown in Figure 60, to prevent accidental deletions.

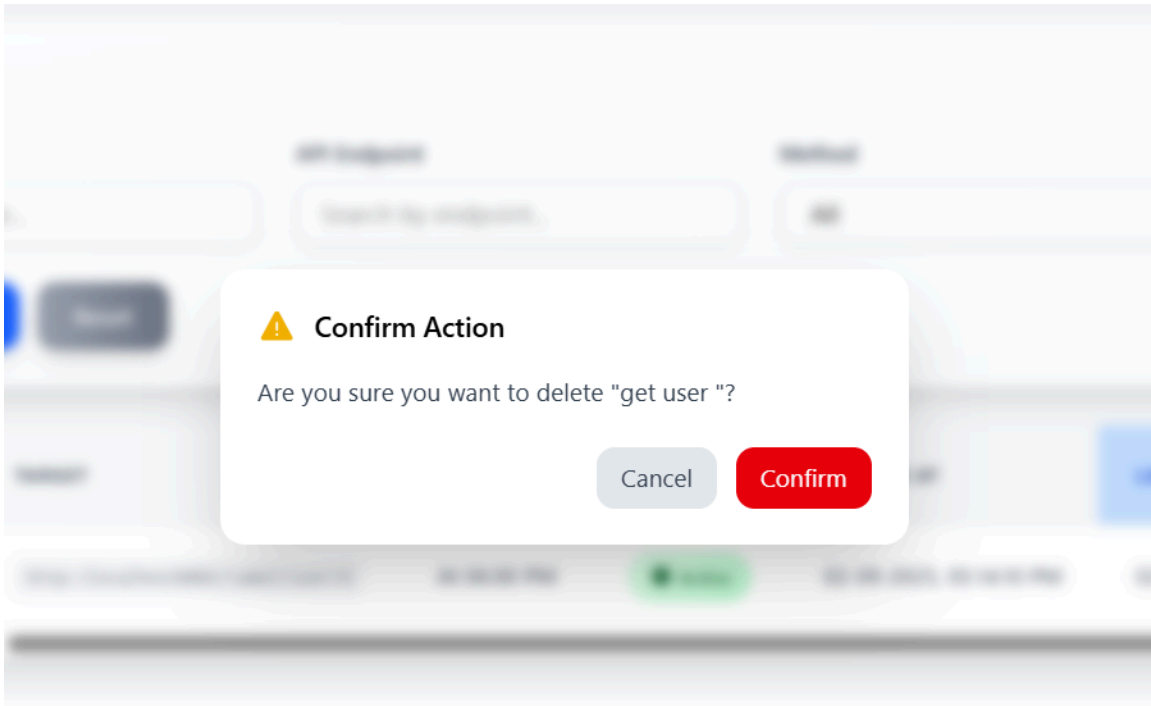


Figure 64 - Confirm Action Dialog for Deleting a Scheduled Job

Upon confirmation, the scheduled job will be removed from the system and will no longer execute.

2.5 Observability

Observability encompasses various logging and monitoring functionalities designed to help administrators and developers understand what is happening within the system in real-time and to troubleshoot issues effectively. The Observability section is for maintaining system stability, identifying bottlenecks, and ensuring smooth operation of all integrated services.

This section is divided into several key menus, each focusing on a specific aspect of system monitoring and logging:

- **Dynamic Routes Logs:** Tracks activities related to the dynamic API routes.
- **Audit Logs:** Records all user and system actions for security and compliance.
- **API Transaction Logs:** Details inbound and outbound API calls and their performance.
- **Scheduled Jobs Logs:** Monitors the execution and status of scheduled tasks.
- **Scheduled Execution Logs:** Provides logs for the actual execution instances of scheduled jobs.
- **Notification Logs:** Keeps a record of all system-generated notifications.

These sub-sections collectively offer a holistic view of the middleware's operational state, enabling proactive management and rapid problem resolution.

2.5.1 Dynamic Routes Logs

The Dynamic Routes Logs provide a detailed record of all activities and events related to the dynamic API routes configured within the Middleware System.

Overview

This module captures critical information about how dynamic routes are being used and modified. It records actions such as route creation, updates, activations, deactivations, and executions. By providing a centralized view of these events, the Dynamic Routes Logs enable quick identification of operational anomalies, performance issues, or unauthorized access attempts related to your API endpoints.

Key Features and Functionality

The Dynamic Routes Logs interface, as depicted in the provided screenshot, offers several features to facilitate effective monitoring and analysis:

- **Quick Search and Filters:** Users can efficiently search and filter log entries based on various criteria:

1. **User:** Filter logs by the user who performed the action or initiated the route execution.
2. **Action Type:** Specify the type of action recorded (e.g., "Re-added and started route in context", "start").
3. **Date Range:** Filter logs within a specific time period to focus on recent activities or historical data.
4. **Route ID:** Search for logs related to a particular dynamic route using its unique identifier.

- **Log Entry Details:** Each log entry provides comprehensive information, including:

1. **ID:** A unique identifier for the log entry.
2. **Actions:** The specific action performed (e.g., sync, start).

- 3. Details: A descriptive message about the event (e.g., "Re-added and started route in context", "Route started").
- 4. Route ID: The identifier of the dynamic route associated with the log entry.
- 5. Event Time: The timestamp when the event occurred, providing a chronological record.
- 6. Route Version: The version of the dynamic route involved in the event.
- 7. User: The user or system component that initiated the action (e.g., System, admin@example.com).

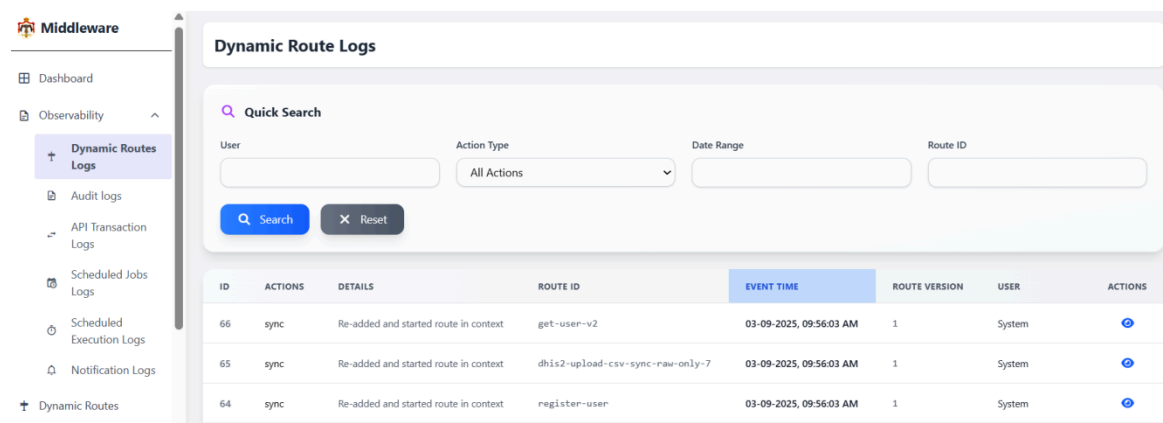


Figure 65 - Dynamic Routes Logs

2.5.2 Audit Logs

Overview

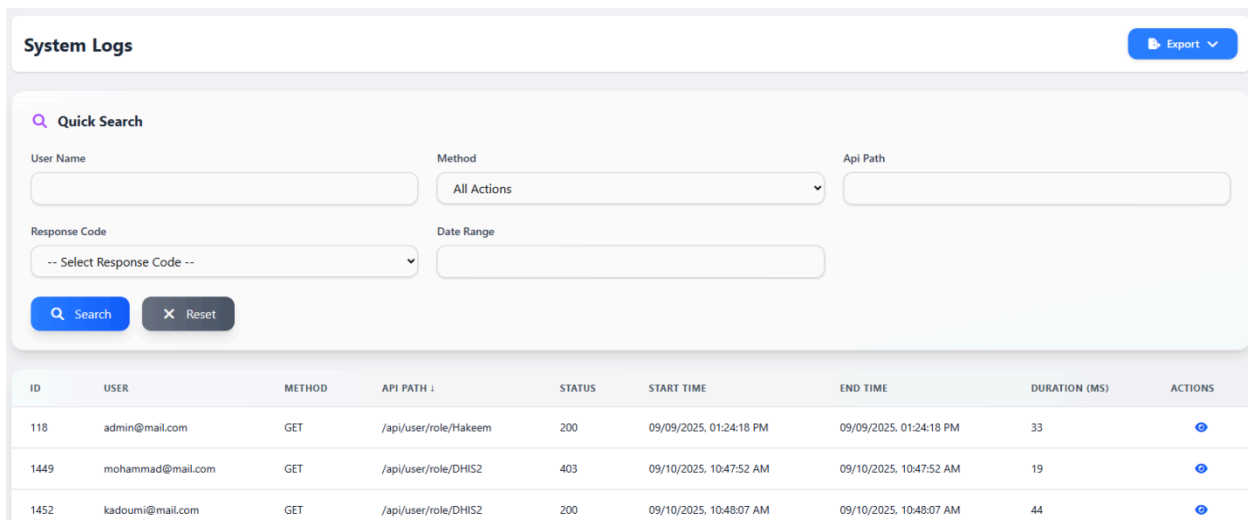
The Audit Logs module provides a comprehensive record of all user and system actions performed within the Middleware System. This functionality is crucial for security, compliance, and troubleshooting, offering an immutable trail of events that can be reviewed to understand system behavior and user activities. Every significant interaction, from configuration changes to data access, is logged, ensuring transparency and accountability.

Key Features and Functionality

The Audit Logs interface, labeled as "System Logs" in the provided screenshots, offers robust search, filtering, and export capabilities to manage and analyze the logged data effectively:

- **Quick Search and Filters:** The interface provides multiple criteria to efficiently search and narrow down log entries:
 1. **User Name:** Filter logs by the specific user who performed the action (e.g., admin@mail.com, viewer@mail.com).
 2. **Method:** Search by the HTTP method of the API call (e.g., GET, POST, PUT, DELETE).
 3. **API Path:** Filter by the specific API endpoint that was accessed (e.g., /routes/audits/66, /permission, /integration-mappings).
 4. **Response Code:** Select a specific HTTP response code (e.g., 200 OK, 403 Forbidden, 404 Not Found) to view successful or failed operations.
 5. **Date Range:** Specify a time period to view logs from a particular day, week, or month.
- **Export Functionality:** The system allows for easy export of audit log data, which is essential for offline analysis, reporting, or archiving. Logs can be exported in common formats:
 1. **Export as Excel:** Downloads the log data in .xlsx format for detailed spreadsheet analysis.
 2. **Export as CSV:** Downloads the log data in a comma-separated values format, suitable for various data processing tools.
- **Log Entry Details:** The table view presents comprehensive details for each audit log entry:
 1. **ID:** A unique identifier for each audit record.
 2. **USER:** The user account associated with the action.
 3. **METHOD:** The HTTP method used for the action.

4. **API PATH:** The specific API endpoint involved.
5. **STATUS:** The HTTP status code of the response, indicating success or failure (e.g., 200, 403, 404).
6. **START TIME:** The timestamp when the action began.
7. **END TIME:** The timestamp when the action completed.
8. **DURATION (MS):** The time taken for the action to complete, in milliseconds.
9. **ACTIONS:** Provides options to view more details or perform related actions for the specific log entry.



The screenshot shows the 'System Logs' interface. At the top right is an 'Export' button. Below it is a 'Quick Search' section with filters for 'User Name', 'Method' (set to 'All Actions'), 'Api Path', 'Response Code' (set to '-- Select Response Code --'), and 'Date Range'. There are 'Search' and 'Reset' buttons. Below the search section is a table with the following data:

ID	USER	METHOD	API PATH	STATUS	START TIME	END TIME	DURATION (MS)	ACTIONS
118	admin@mail.com	GET	/api/user/role/Hakeem	200	09/09/2025, 01:24:18 PM	09/09/2025, 01:24:18 PM	33	View
1449	mohammad@mail.com	GET	/api/user/role/DHIS2	403	09/10/2025, 10:47:52 AM	09/10/2025, 10:47:52 AM	19	View
1452	kadoumi@mail.com	GET	/api/user/role/DHIS2	200	09/10/2025, 10:48:07 AM	09/10/2025, 10:48:07 AM	44	View

Figure 66 - Audit Logs

2.5.3 API Transaction Logs

Overview

The API Transaction Logs module provides a detailed record of all inbound and outbound API calls processed by the Middleware System. This log is crucial for monitoring API performance, ensuring data flow integrity, and troubleshooting any issues that may arise during API interactions. It offers a comprehensive view of each transaction, from the moment a request is received to the completion of its processing.

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This module captures extensive information about every API call, including request and response details, performance metrics, and any errors encountered.

Key Features

The API Transaction Logs interface, also presented under "System / Audit Logs" in the provided screenshots, offers robust capabilities for searching, filtering, and exporting transaction data:

- **Quick Search and Filters:** The interface allows users to efficiently search and filter log entries based on various criteria:
 1. User Name: Filter logs by the user associated with the API call.
 2. Method: Search by the HTTP method of the API call (e.g., GET, POST, PUT, DELETE).
 3. API Path: Filter by the specific API endpoint that was accessed.
 4. Response Code: Select a specific HTTP response code (e.g., 200 OK, 403 Forbidden, 404 Not Found) to view successful or failed transactions.
 5. Date Range: Specify a time period to view logs from a particular day, week, or month.
 6. Transaction ID.
 7. Source Transaction UUID.
- **Export Functionality:** Similar to Audit Logs, API Transaction Logs can be exported for offline analysis, reporting, or archiving:
 1. Export as Excel: Downloads the log data in .xlsx format.
 2. Export as CSV: Downloads the log data in a comma-separated values format.
- **Log Entry Details:** The table view provides comprehensive details for each API transaction log entry:
 1. ID: A unique identifier for each log entry.

2. USER: The user or system that initiated the API call.
3. METHOD: The HTTP method used for the API call.
4. API PATH: The specific API endpoint involved.
5. STATUS: The HTTP status code of the response, indicating success or failure (e.g., 200, 403, 404).
6. START TIME: The timestamp when the API call began.
7. END TIME: The timestamp when the API call completed.
8. DURATION (MS): The total processing duration of the API call, in milliseconds.
9. ACTIONS: Provides options to view more details or perform related actions for the specific log entry.  

- **Failed Transactions & Reprocess :**

Lets operators inspect failed API transactions (validation errors, downstream 5xx, timeouts) and retry safely.

Workflow

1. Open Observability : API Transaction Logs.
2. Filter Status = Failed and inspect the standardized error and raw details.
3. Fix the root cause (payload/mapping/partner outage).
4. Click Retry (single).
5. Watch re-attempts and final status in the same log table.

Notes

- Retries preserve the original UUID and increment retry count.
- Idempotency is enforced on the route to avoid duplicates.

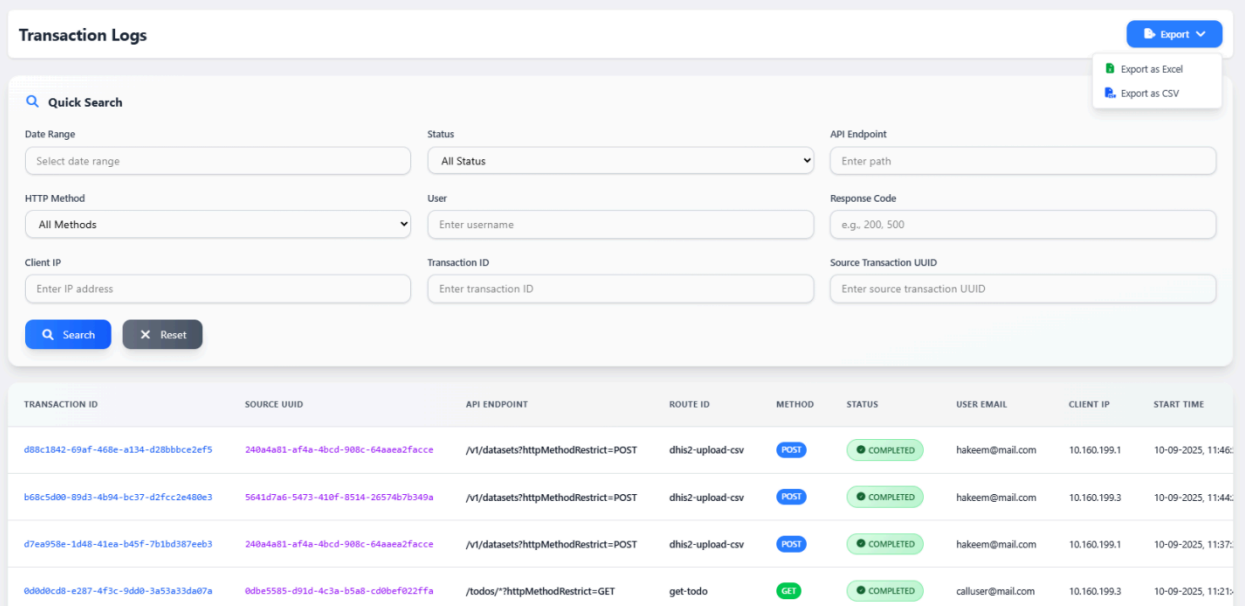


Figure 67 - API Transactions Logs

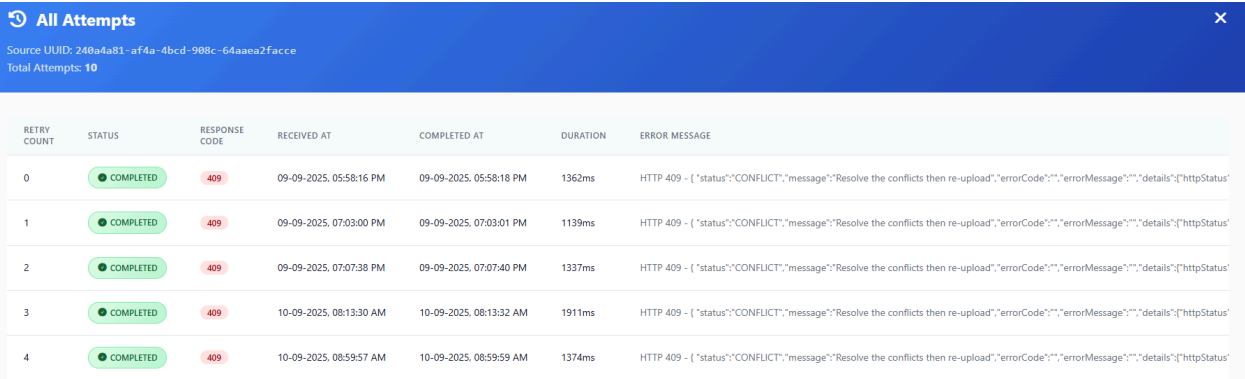


Figure 68 - API Transactions / All Attempts

2.5.4 Scheduled Jobs Logs

Overview

The Scheduled Jobs Logs module provides detailed records of all scheduled jobs, including their status (success/failed) and any associated messages. This is essential for monitoring the health and performance of automated tasks.

Key Features

The Scheduled Jobs Logs interface offers robust capabilities for searching, filtering, and exporting scheduled job logs data:

Execution Filters: The interface allows users to efficiently search and filter log entries based on various criteria:

1. Job Name: Search by the name of the scheduled job.
2. User: Search by the email of the user associated with the job.
3. Status: Filter by the execution status (e.g., All , Success , Failed).
4. Date Range: Select a specific time period to view logs.
5. API Path: Filter by the API path associated with the job (e.g., <http://example/api>).

Export Functionality: The system allows for easy export of scheduled job log data, which is essential for offline analysis, reporting, or archiving:

1. Export as Excel: Downloads the log data in .xlsx format.
2. Export as CSV: Downloads the log data in a comma-separated values format.

Log Entry Details: The table view provides comprehensive details for each scheduled job log entry:

1. EXECUTION ID: A unique identifier for each job execution instance.

2. **JOB NAME:** The name of the scheduled job.
3. **JOB METHOD:** The HTTP method used by the job (e.g., GET).
4. **JOB API:** The API endpoint called by the job (e.g., <http://localhost:8081/camel/user/6>).
5. **EVENT TIME:** The timestamp when the job event occurred.
6. **STATUS:** The outcome of the job execution (e.g., Success , Failed).
7. **ACTION:** The action performed (e.g., edit , test).
8. **USER:** The user associated with the job execution (e.g., admin@mail.com).
9. **ACTIONS:** Provides options to view more details or perform related actions for the specific log entry.

Scheduled Jobs Logs

Export

Execution Filters

Job Name

User

Search by job name...

Search by User Email...

Status

Date Range

Api Path

All

Select date range

http://example/api

Apply Filters

Reset

EXECUTION ID	JOB NAME	JOB METHOD	JOB API	EVENT TIME	STATUS	ACTION	USER	ACTIONS
29	DHIS2	GET	https://publicapsesw1.dev/selgate-api	9/10/2025 12:45:51 PM	Failed	test	kadoumi@mail.com	
3	Add new User	POST	https://jsonplaceholder.typicode.com/posts	9/9/2025 5:57:36 PM	Success	pause	kadoumi@mail.com	
8	get User	GET	https://jsonplaceholder.typicode.com/posts/1	9/9/2025 6:01:38 PM	Success	pause	kadoumi@mail.com	

Figure 69 - Scheduled Jobs Logs

2.5.5 Scheduled Execution Logs

Overview

The Scheduled Execution Logs module offers details about each instance of a scheduled job's execution. This includes specific execution details and allows for filtering to pinpoint issues or analyze performance.

Key Features

The Scheduled Execution Logs interface, labeled as "Execution Log" in the provided screenshots, offers robust capabilities for searching, filtering, and exporting execution data:

Execution Filters: The interface allows users to efficiently search and filter log entries based on various criteria:

1. Job Name: Search by the name of the scheduled job.
2. User: Search by the email of the user associated with the job.
3. Status: Filter by the execution status (e.g., All , Success , Failed).
4. Date Range: Select a specific time period to view logs.
5. API Path: Filter by the API path associated with the job (e.g., <http://example/api>).

Export Functionality: The system allows for easy export of scheduled execution log

data, which is essential for offline analysis, reporting, or archiving:

1. Export as Excel: Downloads the log data in .xlsx format.
2. Export as CSV: Downloads the log data in a comma-separated values format.

Log Entry Details: The table view provides comprehensive details for each scheduled execution log entry:

1. EXECUTION ID: A unique identifier for each job execution instance.
2. JOB NAME: The name of the scheduled job.

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3. **ENDPOINT:** The API endpoint called by the job (e.g., `http://localhost:8081/camel/user/6`).
4. **START TIME:** The timestamp when the job execution began.
5. **END TIME:** The timestamp when the job execution completed.
6. **DURATION (MS):** The total processing duration of the job execution in milliseconds.
7. **STATUS:** The outcome of the job execution (e.g., `Success` , `Failed`).
8. **ACTIONS:** Provides options to view more details or perform related actions for the specific log entry.

EXECUTION ID	JOB NAME	ENDPOINT	START TIME	END TIME	DURATION (MS)	STATUS	ACTIONS
62	get User	https://jsonplaceholder.typicode.com/posts/1	9/10/2025, 4:25:40 PM	9/10/2025, 4:25:40 PM	245	SUCCESS	🔍
61	get User	https://jsonplaceholder.typicode.com/posts/1	9/10/2025, 4:25:30 PM	9/10/2025, 4:25:30 PM	229	SUCCESS	🔍
60	get User	https://jsonplaceholder.typicode.com/posts/1	9/10/2025, 4:25:20 PM	9/10/2025, 4:25:20 PM	244	SUCCESS	🔍
59	get User	https://jsonplaceholder.typicode.com/posts/1	9/10/2025, 4:25:10 PM	9/10/2025, 4:25:10 PM	232	SUCCESS	🔍

Figure 70 - Scheduled Execution Logs

2.5.6 Notification Logs

Overview

The Notification Logs module keeps a record of all system-generated notifications, such as alerts or informational messages. This allows administrators to review notification history and ensure critical information is being disseminated effectively.

Key Features

Search & Filters

- **Action:** Filter by operation type (CREATE/PUT/DELETE/SEND, etc.).

- Email: The user email that executed the action.
- Description: Keywords contained in the event description.
- Creation Date: Filter by date or date range.
- Apply Filters / Reset: Apply or clear filters.

Table Columns

- ID: Log entry identifier.
- Action: Operation (e.g., PUT, DELETE, SEND).
- Email: User who performed the action (e.g., admin@mail.com).
- Event Time: Timestamp of the action.
- Actions: View icon opens a detail modal (full description and, when available, request/response snippets to aid troubleshooting).

ID	ACTION	EMAIL	EVENT TIME	ACTIONS
82	PUT	admin@mail.com	10/1/2025, 5:52:07 PM	
81	DELETE	user14@mail.com	10/1/2025, 4:52:44 PM	

Figure 71 - Notification Action Logs

Export

- Export as Excel / Export as CSV: Download the filtered or full dataset for audit or archival.

Typical Workflows

- Investigate failures: When a send shows FAILED in Messages Config, open Notification Logs to find the SMTP/API error message, template/channel used, and the initiator.
- Security audits: Track who created/edited/deleted templates or channel configurations, and when.

2.6 Access Control

2.6.1 Overview

The Access Control area centralizes identity and authorization management. It contains two menus:

- Users: manage individual accounts and API tokens.
- Roles & Permissions: define reusable permission sets and route access.

All changes are tracked in the audit log.

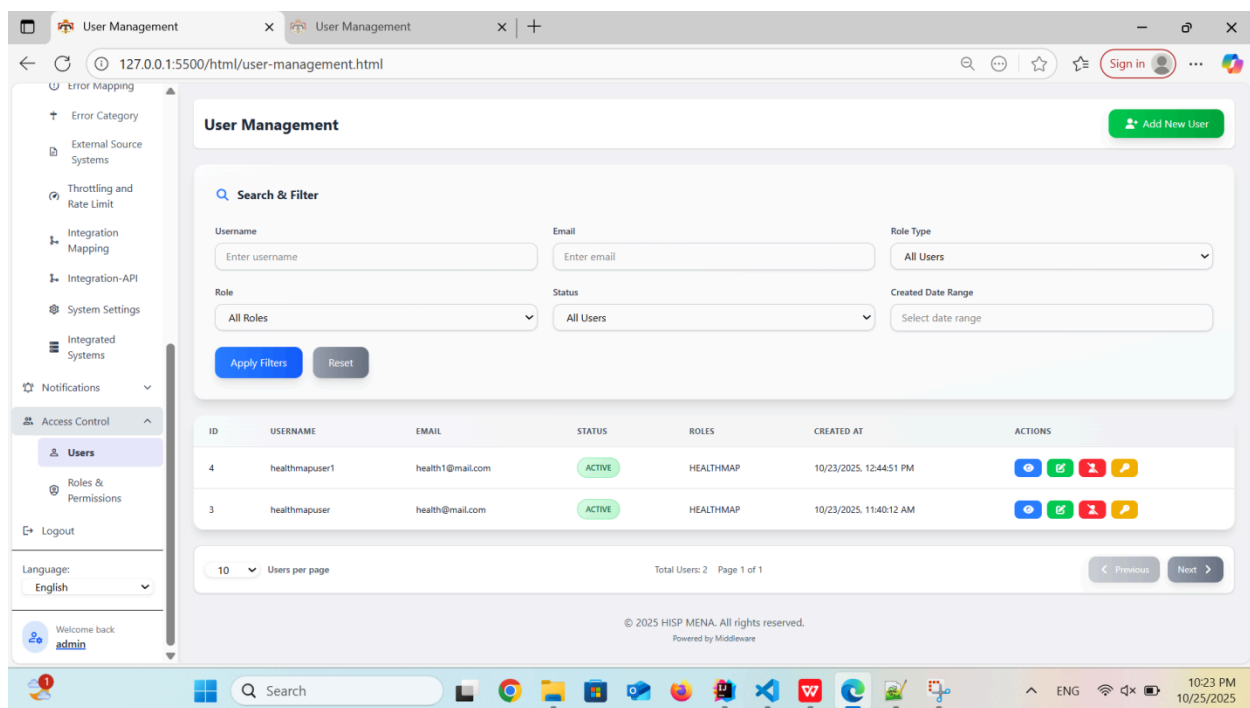


Figure 72 - Access Control

2.6.2 Users

Overview

The Users page lets administrators create, edit, enable/disable, and delete user accounts, assign roles, and manage API tokens used for programmatic access.

Users List

A paginated table shows the registered users with quick actions per row.

Default columns:

1. ID: internal identifier
2. Username
3. Email
4. Full Name (if configured)
5. Roles (comma-separated)
6. Status: ACTIVE / INACTIVE
7. Last Login (if available)
8. Actions: View, Edit, Deactivate/Activate, Delete, Generate/Revoke Token.

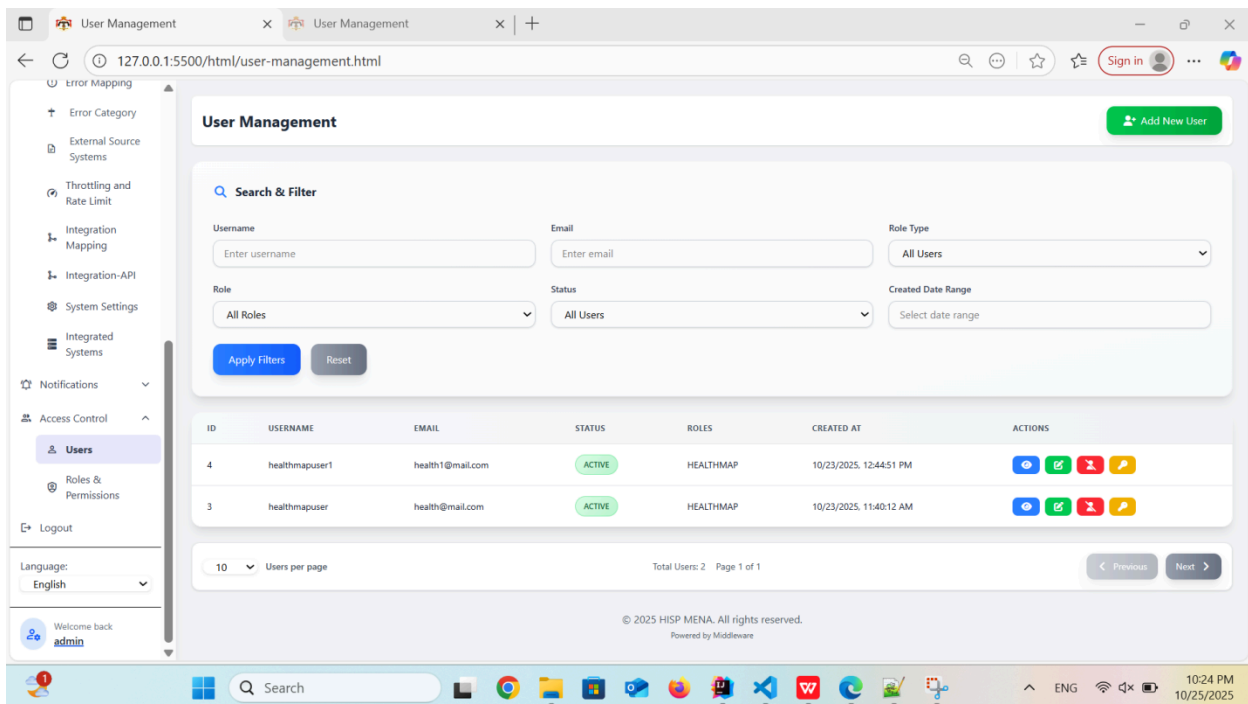


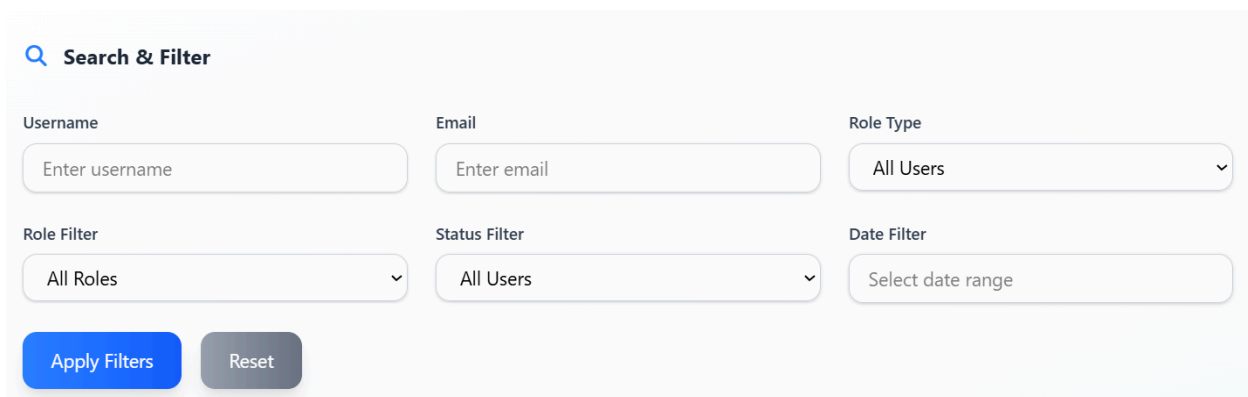
Figure 73 - Users List

Search & Filters

Use the filter panel above the table:

1. Search: by Username, Email, or Full Name
2. Role: show only users with selected roles
3. Status: ACTIVE / INACTIVE
4. Date: by Created At or Last Login (depending on the field in your deployment)
5. Apply Filters / Reset: run or clear filters

Tip: After Reset, the table returns to the default view; re-apply filters to start a fresh search.



The screenshot shows a 'Search & Filter' panel with a search icon and title. It contains several input fields and dropdown menus for filtering users. The fields are arranged in two rows. The first row has 'Username' (text input), 'Email' (text input), and 'Role Type' (dropdown menu). The second row has 'Role Filter' (dropdown menu), 'Status Filter' (dropdown menu), and 'Date Filter' (text input). Below these fields are two buttons: 'Apply Filters' (blue) and 'Reset' (grey).

Username	Email	Role Type
Enter username	Enter email	All Users

Role Filter	Status Filter	Date Filter
All Roles	All Users	Select date range

Apply Filters Reset

Figure 74 - Search / Filter Users

Add a New User

1. Click Add User.
2. Fill in Username, Email, Password (and confirm), Full Name, Phone (optional), Status.
3. Assign one or more Roles.
4. Save.

Validation & rules

- Username and email must be unique.

- Passwords must meet the configured policy (length/complexity).
- At least one role is recommended for system access.

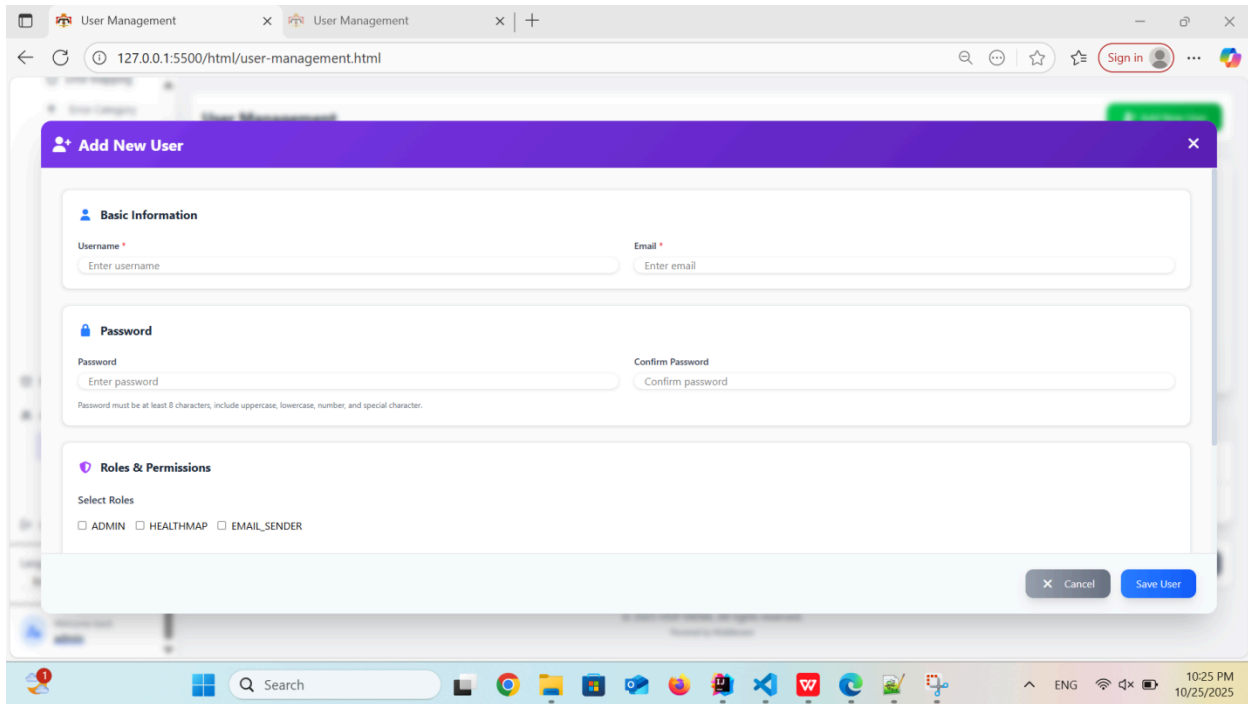
The screenshot shows a web browser window with two tabs labeled 'User Management'. The address bar shows the URL '127.0.0.1:5500/html/user-management.html'. A 'Sign in' button is visible in the top right corner. The main content area displays a modal form titled 'Add New User'. The form is divided into three sections: 'Basic Information' with fields for 'Username' and 'Email'; 'Password' with fields for 'Password' and 'Confirm Password', accompanied by a note that passwords must be at least 8 characters long and include uppercase, lowercase, numbers, and special characters; and 'Roles & Permissions' with a 'Select Roles' section containing three checkboxes: 'ADMIN', 'HEALTHMAP', and 'EMAIL_SENDER'. At the bottom right of the form are 'Cancel' and 'Save User' buttons. The Windows taskbar is visible at the bottom of the screen, showing the search bar and various application icons.

Figure 75 - Add New User

Edit an Existing User

1. Locate the user (search or filter).
2. Click Edit.
3. Update fields (profile data, roles, status).
4. Save.

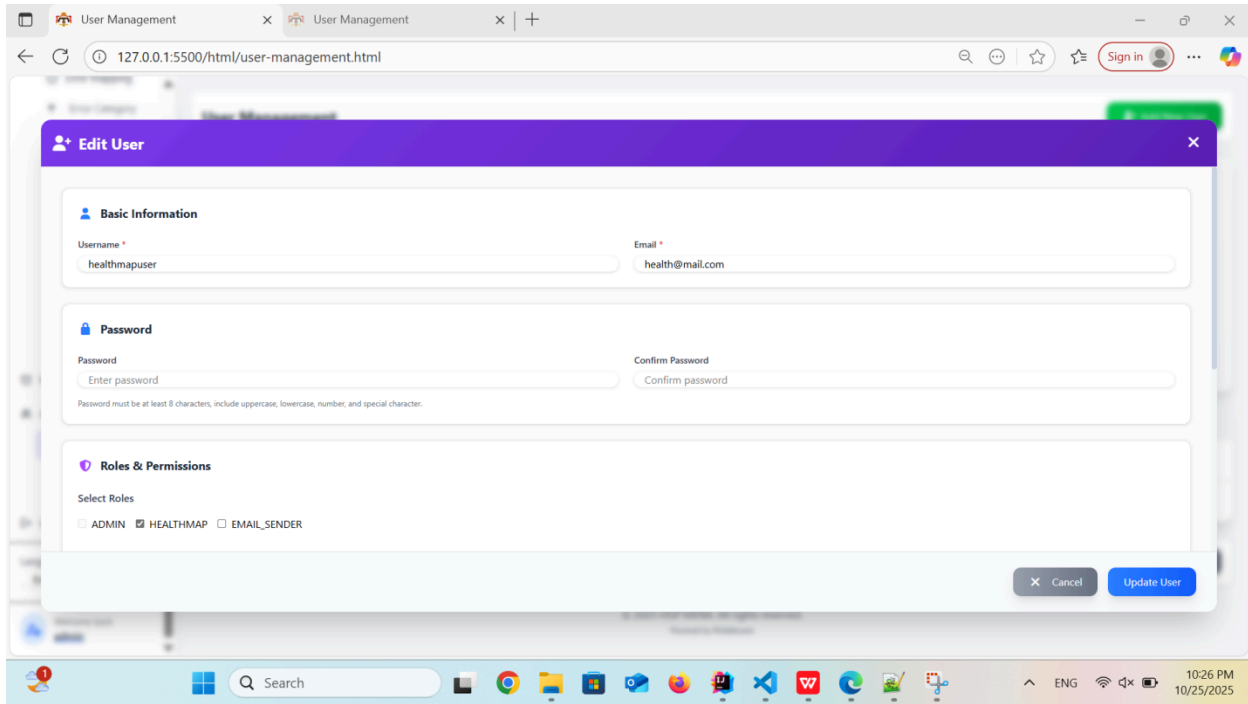


Figure 76 - Edit User

Deactivate / Reactivate

1. Use the Deactivate (or Activate) action on the user row.
2. Confirm the action.
3. Inactive users cannot sign in or use APIs via tokens.

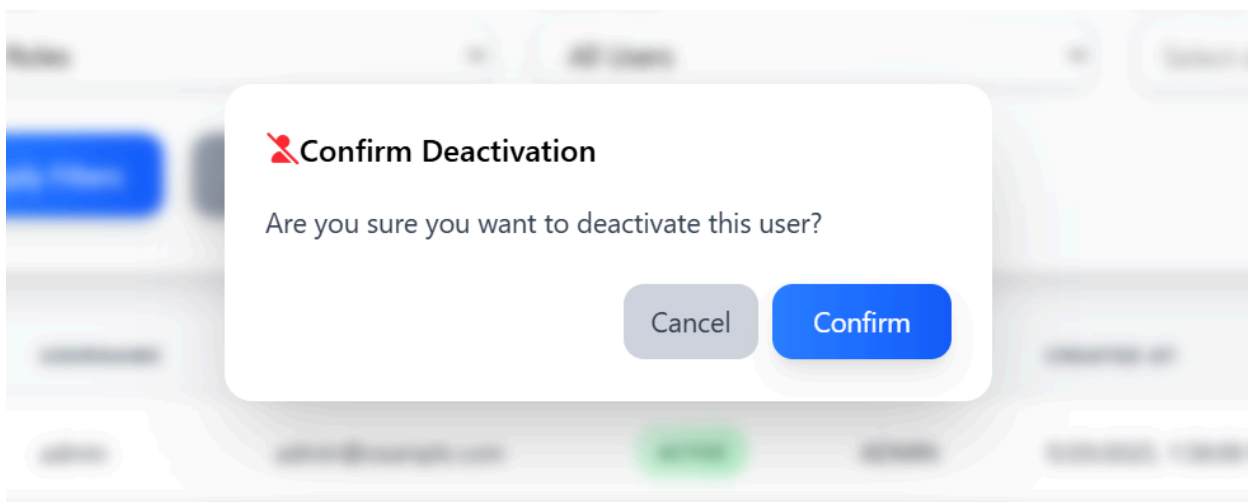


Figure 77 - Deactivate / Activate User

Reset Password

1. Click Edit on the target user.
2. Enter a New Password and confirm.
3. Save.

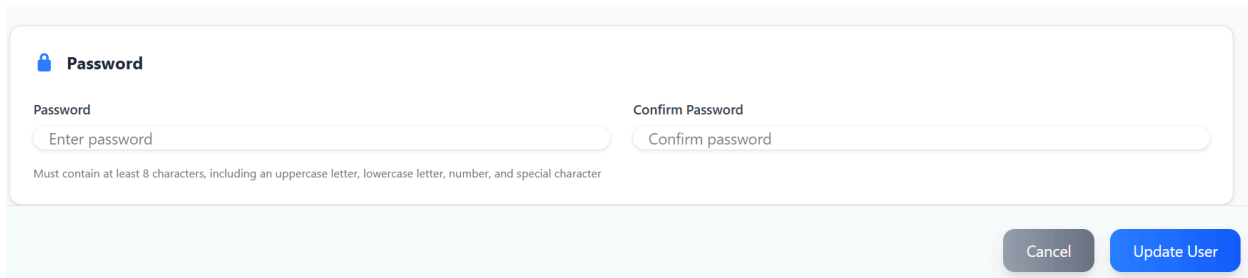
The screenshot shows a web form titled "Password" with a lock icon. It contains two input fields: "Password" with the placeholder text "Enter password" and "Confirm Password" with the placeholder text "Confirm password". Below the "Password" field, there is a small text note: "Must contain at least 8 characters, including an uppercase letter, lowercase letter, number, and special character". At the bottom right of the form, there are two buttons: a grey "Cancel" button and a blue "Update User" button.

Figure 78 - Reset Password

API Tokens (Programmatic Access)

Tokens authenticate background jobs and integrations.

Generate a token

1. Open the user's row menu : Generate Token.
2. Choose Expiration (e.g., 30/90 days or Never).
3. Generate and copy the token (it is shown only once).
4. Store it securely (password vault).

Revoke a token

- From the same user, choose Revoke Token to immediately invalidate it.

Good practice

- Prefer short-lived tokens.
- Use dedicated “service users” for integrations, not personal accounts.

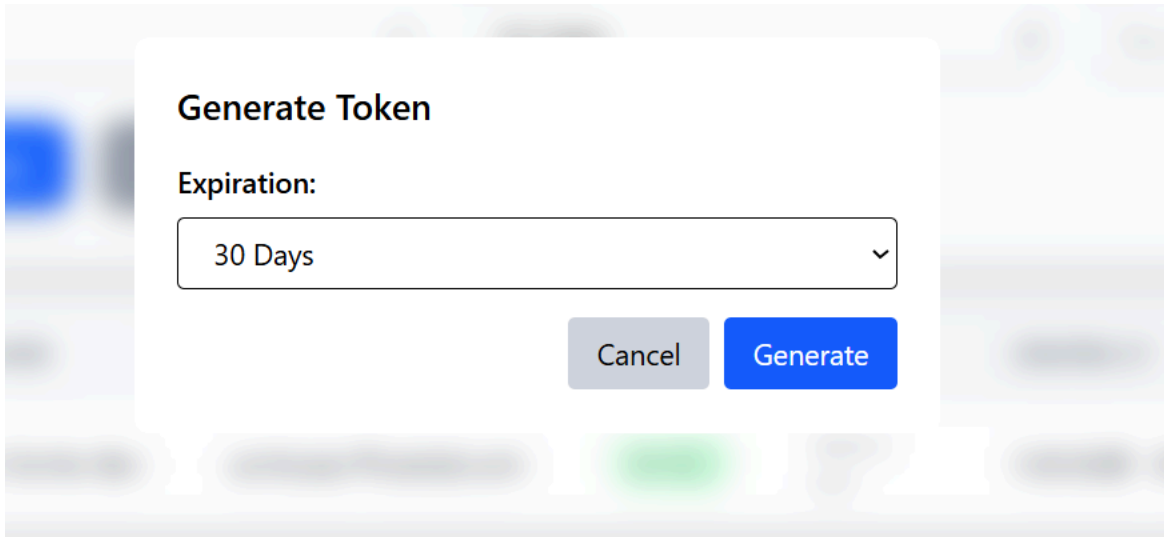


Figure 79 - Generate Token

2.6.3 Roles & Permissions

Overview

Roles are reusable permission bundles. Users inherit all permissions of their assigned roles. This implements RBAC (Role-Based Access Control).

Roles List

A table lists all roles with:

1. Role Name
2. Type (if used: e.g., System, Custom)
3. Description (short purpose)
4. Actions: View, Edit, Clone, Delete (or Archive)

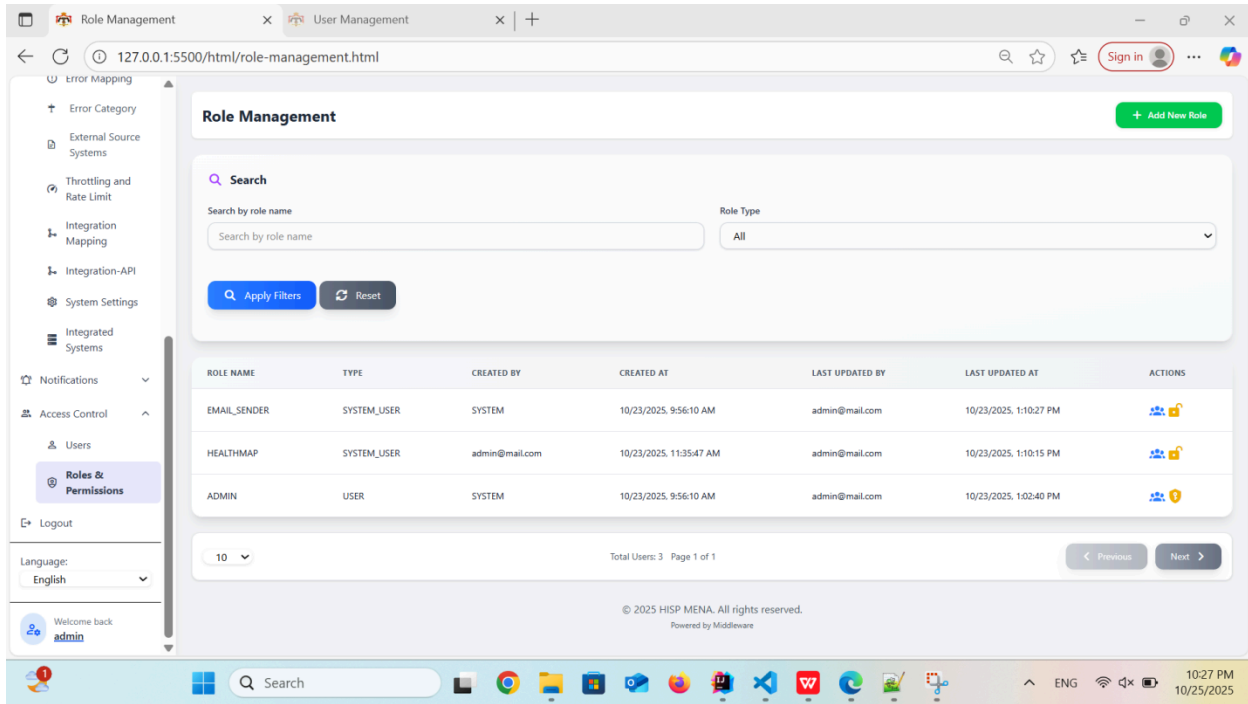


Figure 80 - Roles List

Create a Role

1. Click Create Role.
2. Enter Role Name and optional Description.
3. Click Save to create the shell.
4. Proceed to Permissions.

The 'Create New Role' form is shown with a purple header bar containing a plus icon and the text 'Create New Role'. The form has two main input fields:

- Role Name ***: A text input field with the placeholder text 'Enter role name'.
- Role Type ***: A dropdown menu with 'User' selected.

At the bottom right of the form, there are two buttons: 'Cancel' (with a close icon) and 'Create Role' (with a save icon).

Figure 81 - Create New Role

Edit Permissions

Open the role : Permissions.

There are two panels:

A) Module Permissions (UI/Features)

Grant access to screens and actions, e.g.:

- Configuration: View / Create / Update / Delete
- Dynamic Routes: View / Publish / Retire
- Notifications: Templates, Channels, Groups, Receivers, Send & Logs
- Observability: Audit Logs, API Transactions, Scheduled Jobs, Notification Logs
- Access Control: Users, Roles & Permissions

Each item typically supports granular actions: View, Create, Update, Delete, Export, etc.

B) Route Permissions (API Endpoints)

Control which HTTP routes a role can access:

- Route pattern (e.g., /api/templates/**)
- Allowed methods (GET/POST/PUT/DELETE)
- Enable/disable a route entry

Use route permissions for service users or API-only roles. UI roles commonly rely on Module permissions.

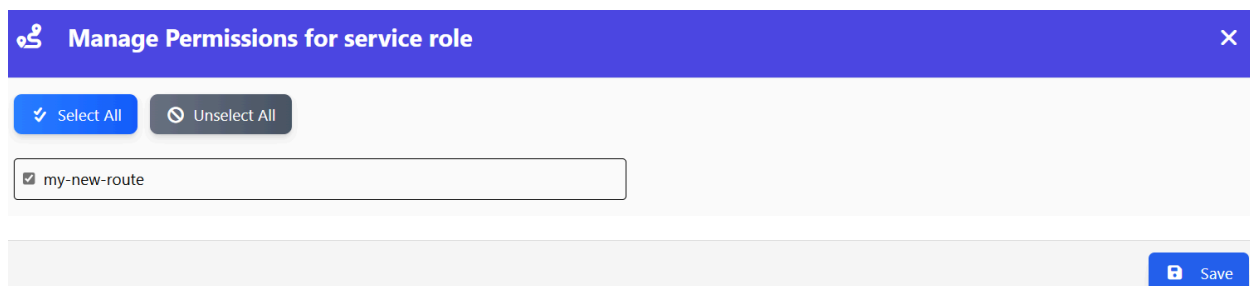


Figure 82 - Manage Route Permissions

 Manage Permissions for Editor

Select All

Unselect All

☒ auditLogs:view	☐ auditLogs:export
☒ errorCategories:view	☐ errorCategories:export
☐ errorCategories:create	☒ errorCategories:edit
☒ errorMappings:view	☐ errorMappings:export
☐ errorMappings:create	☒ errorMappings:edit
☒ sourceSystems:view	☐ sourceSystems:export
☐ sourceSystems:create	☒ sourceSystems:edit
☐ dashboard:view	☐ dashboard:export

Save

Figure 83 - Manage Module Permissions

Assign Roles to Users

- In Users : Edit, add the target role to Roles.
- A user can have multiple roles; effective permissions are the union.

 Edit User

Password

Enter password

Confirm Password

Confirm password

Must contain at least 8 characters, including an uppercase letter, lowercase letter, number, and special character

Roles & Permissions

Select Roles

☐ ADMIN ☐ Viewer ☐ Editor ☐ Notification ☒ service role

User Settings

Cancel

Update User

Figure 84 - Assign Roles To Users

2.7 Notifications

2.7.1 Overview

The Notifications module centralizes everything needed to define message templates, manage recipients, organize them into groups, configure delivery channels (e.g., Email/SMS), and send messages immediately or on a schedule. It also integrates with Observability so every send attempt is traceable.

Sub-menus

- Notification Templates: Create and manage reusable message templates.
- Receivers: Define recipients and their contact details.
- Notification Groups: Create recipient groups (e.g., “NOC: Night Shift”).
- Notification Groups Receivers: Assign receivers to groups.
- Notification Channel’s Config: Configure delivery channels (e.g., SMTP over SSL).
- Messages Config: Trigger immediate sends or generate a request to be used by scheduled jobs/integrations.

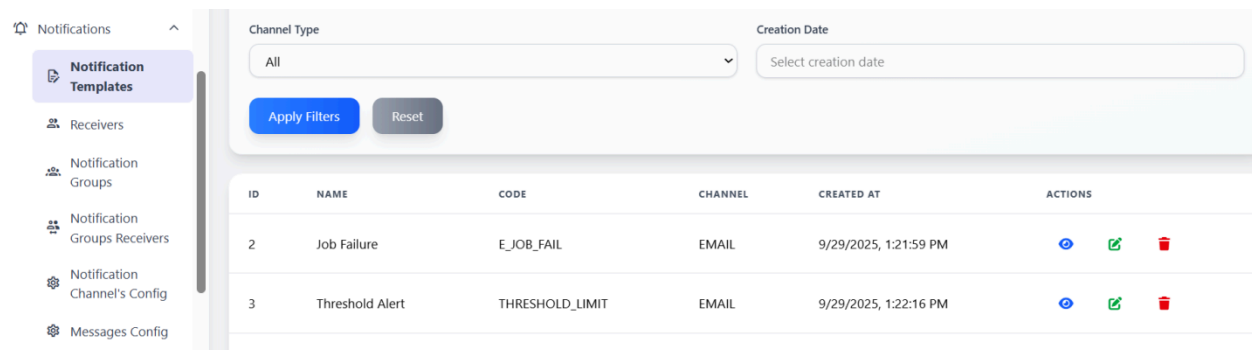


Figure 85 - Notifications Menu / Sub-Menus

2.7.2 Notification Templates

Purpose

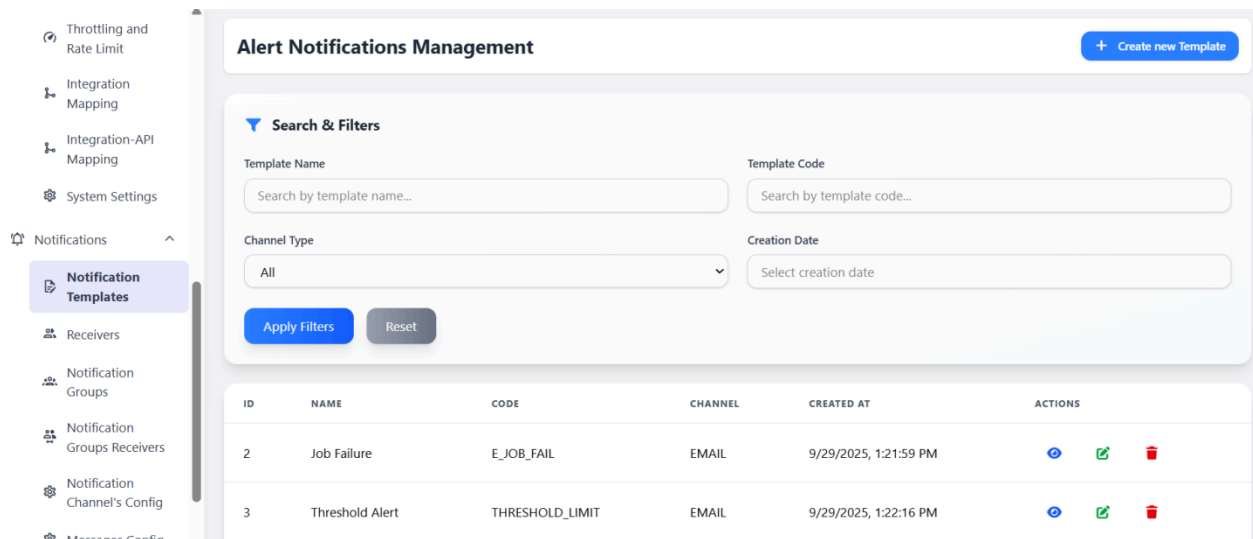
Define reusable message content for different channels. Templates can include placeholders (variables) that are replaced at send time.

Key actions

- Search & Filters: by Name, Code, Type, Creation Date.
- Create: set Name, unique Code, Type (Email/SMS), Subject (for email), and Body (supports placeholders like {jobName}, {details}).
- Edit / Delete / View details with confirmation dialogs for destructive actions.

Data integrity

- If a template is referenced by sent notifications, deleting it will be blocked by database constraints; the UI returns a clear error message (e.g., “Cannot delete, referenced by notification_logs”).



The screenshot displays the 'Alert Notifications Management' interface. On the left is a sidebar with navigation items: Throttling and Rate Limit, Integration Mapping, Integration-API Mapping, System Settings, Notifications (expanded), Notification Templates (selected), Receivers, Notification Groups, Notification Groups Receivers, Notification Channel's Config, and Message Config. The main panel has a title 'Alert Notifications Management' and a '+ Create new Template' button. Below the title is a 'Search & Filters' section with four input fields: 'Template Name' (placeholder: Search by template name...), 'Template Code' (placeholder: Search by template code...), 'Channel Type' (dropdown menu with 'All' selected), and 'Creation Date' (placeholder: Select creation date). There are 'Apply Filters' and 'Reset' buttons. Below the filters is a table with the following data:

ID	NAME	CODE	CHANNEL	CREATED AT	ACTIONS
2	Job Failure	E_JOB_FAIL	EMAIL	9/29/2025, 1:21:59 PM	
3	Threshold Alert	THRESHOLD_LIMIT	EMAIL	9/29/2025, 1:22:16 PM	

Figure 86 - Notification Template

Create Notification Template

Name *

Greetings

Code *

G_SMS

Channel *

SMS

Message Template (JSONB) *

```
{
  "subject": "Welcome { name } !",
  "body": "Hello {name},Thanks for joining , We will contact you at phone"
}
```

Cancel

Validate

Save

Figure 87 - Create Notification Template

View Notification Template

ID: 5

Name: Greetings

Channel: SMS

Code: G_SMS

Message Template:

```
{
  "subject": "Welcome { name } !",
  "body": "Hello {name},Thanks for joining , We will contact you at phone"
}
```

Created By: admin@mail.com

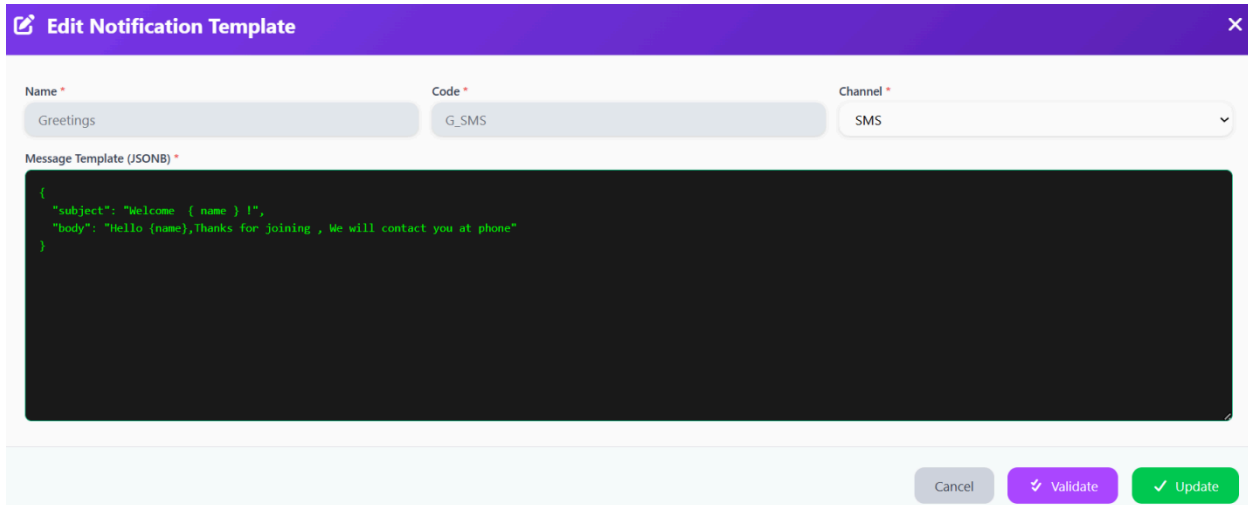
Created At: 10/6/2025, 12:43:00 PM

Updated By: admin@mail.com

Updated At: 10/6/2025, 12:43:00 PM

Close

Figure 88 - View Notification Template



The screenshot shows a web interface for editing a notification template. At the top is a purple header bar with the title 'Edit Notification Template' and a close button. Below the header, there are three input fields: 'Name' with the value 'Greetings', 'Code' with the value 'G_SMS', and 'Channel' with a dropdown menu showing 'SMS'. Below these fields is a large text area labeled 'Message Template (JSONB)' containing a JSON object:

```
{  "subject": "Welcome { name }!",  "body": "Hello {name}, thanks for joining , We will contact you at phone"}
```

. At the bottom right of the form are three buttons: 'Cancel', 'Validate' (with a checkmark icon), and 'Update' (with a checkmark icon).

Figure 89 - Edit Notification Template

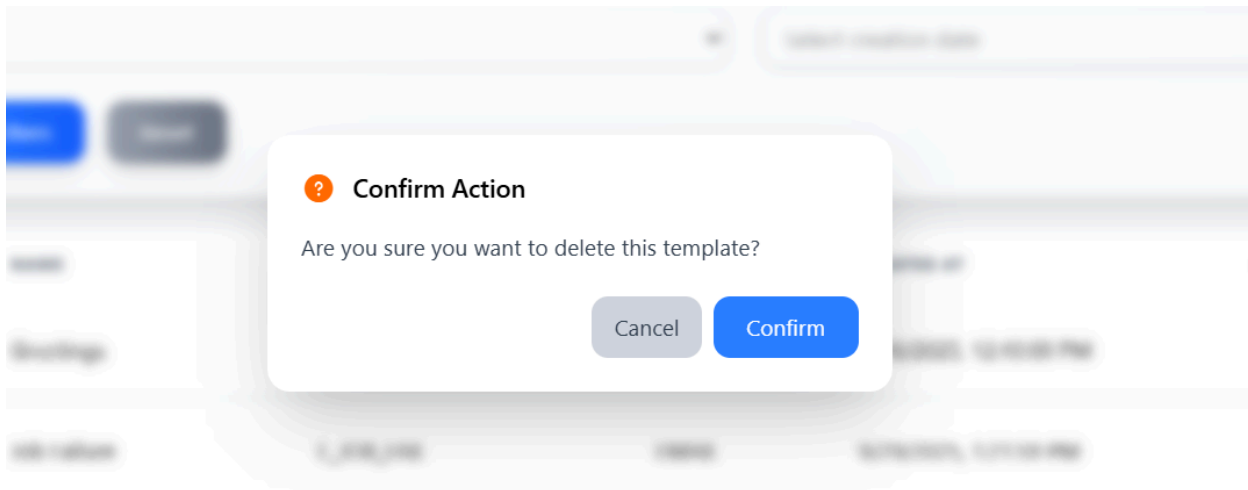


Figure 90 - Delete Notification Template

2.7.3 Receivers

Purpose

Maintain recipient profiles and their contact channels.

Confidential: For authorized use only

Key actions

- Search & Filters: by name/email/phone/status.
- Create/Edit: Name, Email, optional Phone/Mobile, Status (Active/Inactive).
- Activate/Deactivate/Delete with confirmation.
- Export: Excel/CSV.

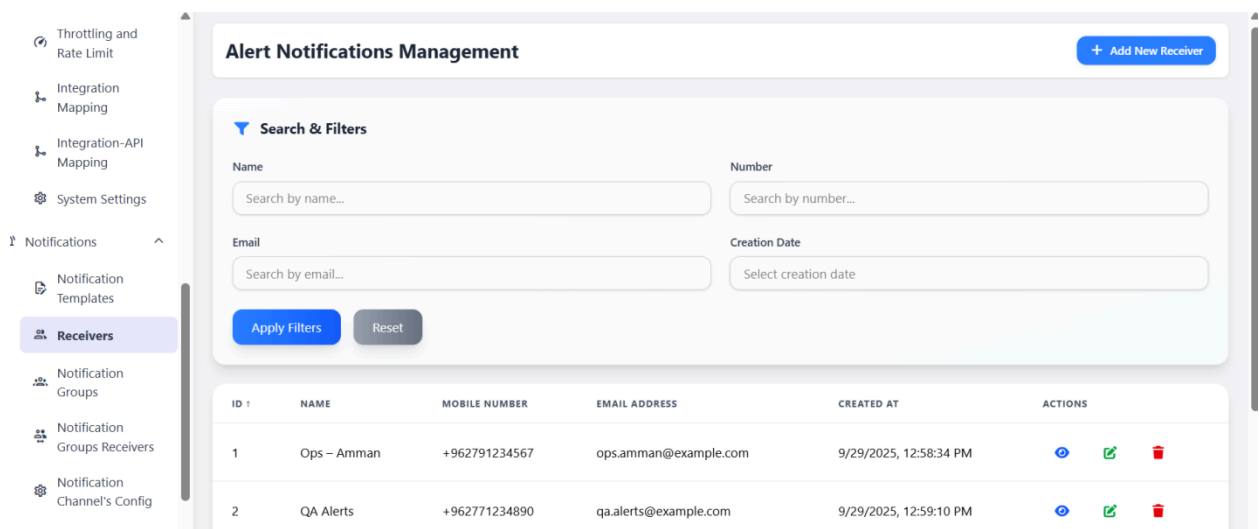


Figure 91 - Receivers

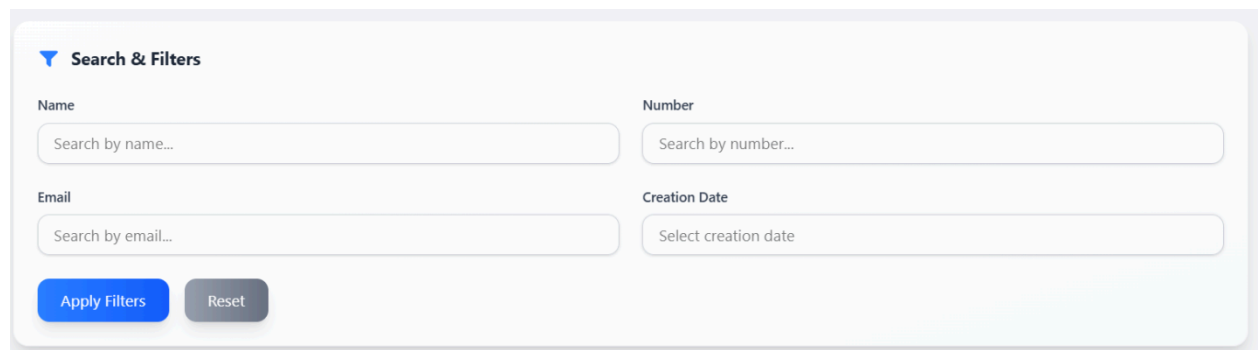


Figure 92 - Search Receiver

Create Receiver

Name

Enter receiver name

Mobile Number

Enter mobile number

Email Address

Enter email address

Cancel

Save

Figure 93 - Create Receiver

Edit Receiver

Name

Ops – Amman

Mobile Number *

+962791234567

Email Address *

ops.amman@example.com

Cancel

Update

Figure 94 - Edit Receiver

?

Confirm Action

Are you sure you want to delete this receiver?

Cancel

Confirm

Figure 95 - Delete Receiver

View Receiver

ID: 1

Mobile: +962791234567

Created By: admin@mail.com

Updated By: admin@mail.com

Name: Ops – Amman

Email: ops.amman@example.com

Created At: 9/29/2025, 12:58:34 PM

Updated At: 9/29/2025, 12:58:34 PM

Close

Figure 96 - View Receiver

2.7.4 Notification Groups

Purpose

Organize receivers into logical groups to simplify sending (e.g., shifts, departments).

Key actions

- Search & Filters: by group name/description/status/creation date.
- Create/Edit/Activate/Deactivate/Delete.
- Export: Excel/CSV.

Notifications

Notification Templates

Receivers

Notification Groups

Notification Groups Receivers

Notification Channel's Config

Messages Config

Access Control

Users

Roles & Permissions

Logout

Alert Notifications Management

Create New Group

Search & Filters

Name

Search by name...

Code

Search by code...

Description

Search by description...

Creation Date

Select creation date

Apply Filters

Reset

ID	NAME	CODE	DESCRIPTION	CREATED AT	ACTIONS
2	NOC – Night Shift	NOC_NIGHT	Network operations center night shift (20:00–08:00).	9/29/2025, 4:55:30 PM	

10

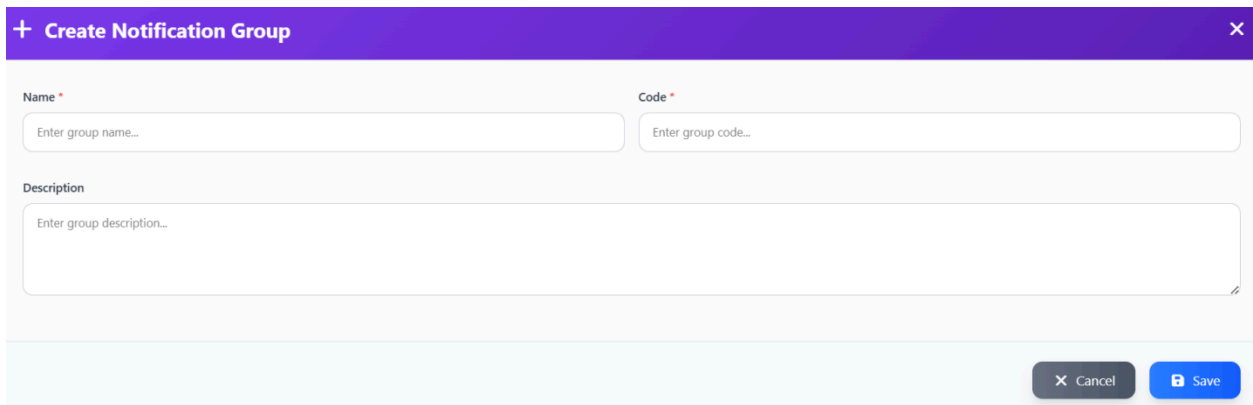
Records per page

Total Records: 1 Page 1 of 1

Previous

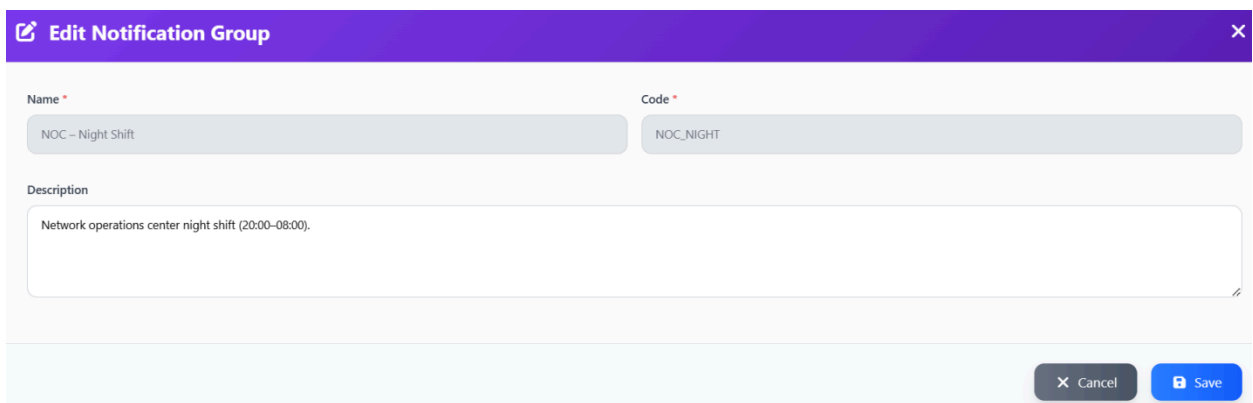
Next

Figure 97 - Notification Groups



The 'Create Notification Group' dialog features a purple header with a plus icon and a close button. It contains three input fields: 'Name' (placeholder: 'Enter group name...'), 'Code' (placeholder: 'Enter group code...'), and 'Description' (placeholder: 'Enter group description...'). At the bottom right, there are 'Cancel' and 'Save' buttons.

Figure 98 - Create Notification Group



The 'Edit Notification Group' dialog has a purple header with an edit icon and a close button. The 'Name' field contains 'NOC – Night Shift' and the 'Code' field contains 'NOC_NIGHT'. The 'Description' field contains 'Network operations center night shift (20:00-08:00)'. 'Cancel' and 'Save' buttons are at the bottom right.

Figure 99 - Edit Notification Group

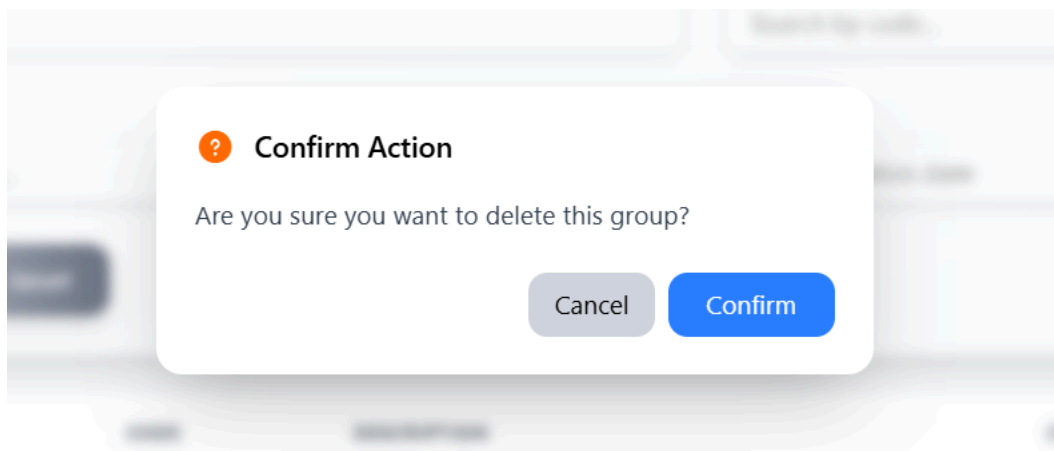


Figure 100 - Delete Notification Group

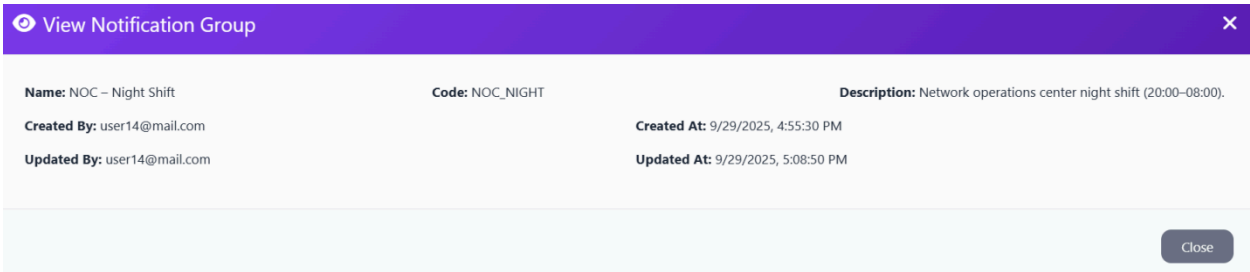


Figure 101 - View Notification Group

2.7.5 Notification Groups Receivers

Purpose

Assign receivers to groups.

Key actions

- Add/Remove receivers to/from a selected group.
- Export: Excel/CSV.

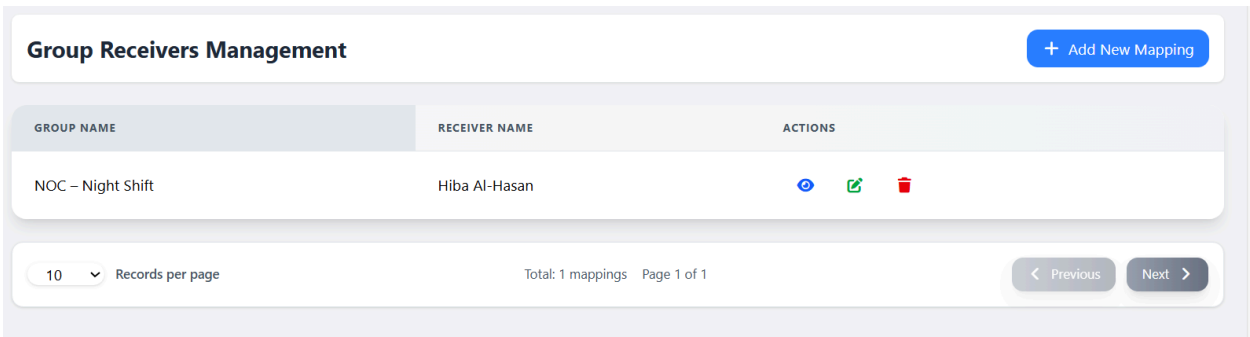


Figure 102 - Group Receivers

Create Group Receiver Mapping

Group *

Select Group...

Receivers *

Select Receivers...

Cancel

Save

Figure 103 - Create Group Receiver Mapping

Edit Group Receiver Mapping

Group *

NOC – Night Shift

Receivers *

Hiba Al-Hasan

Cancel

Save

Figure 104 - Edit Group Receiver Mapping

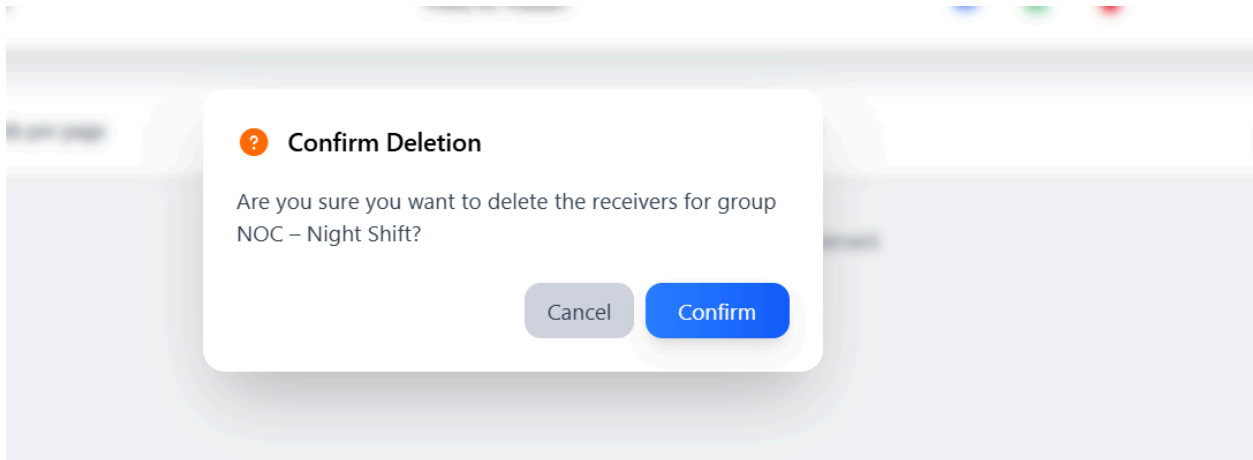


Figure 105 - Delete Group Receivers

2.7.6 Notification Channel's Config

Purpose

Store delivery channel settings (e.g., SMTP for email, provider API for SMS).

Key actions

- **Search & Filters:** by name/code/channel type/creation date.
- **Create:** choose Channel (EMAIL/SMS/...), then provide the JSON configuration.

Example:

Gmail over (port 587)

```
{  
  "smtpHost": "smtp.gmail.com",  
  "smtpPort": 587,  
  "username": "example@gmail.com",  
  "password": "secret",  
  "fromAddress": "example@gmail.com",
```

```
"displayName": "Your App Name"
```

```
}
```

- View / Edit / Delete with confirmation.

Security note

Use a provider-approved method (e.g., Gmail App Password).

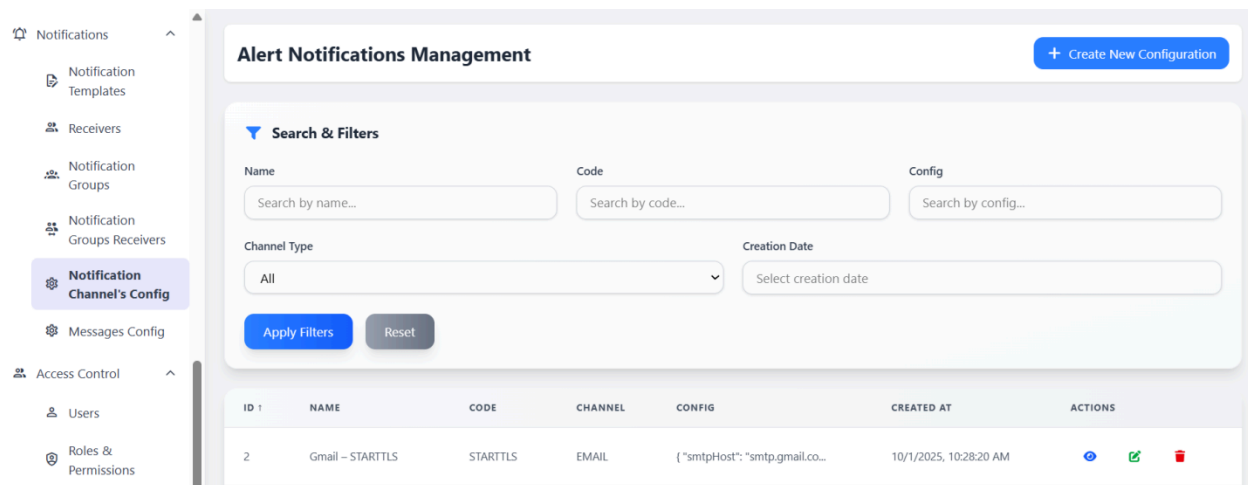


Figure 106 - Notification Channel's Config

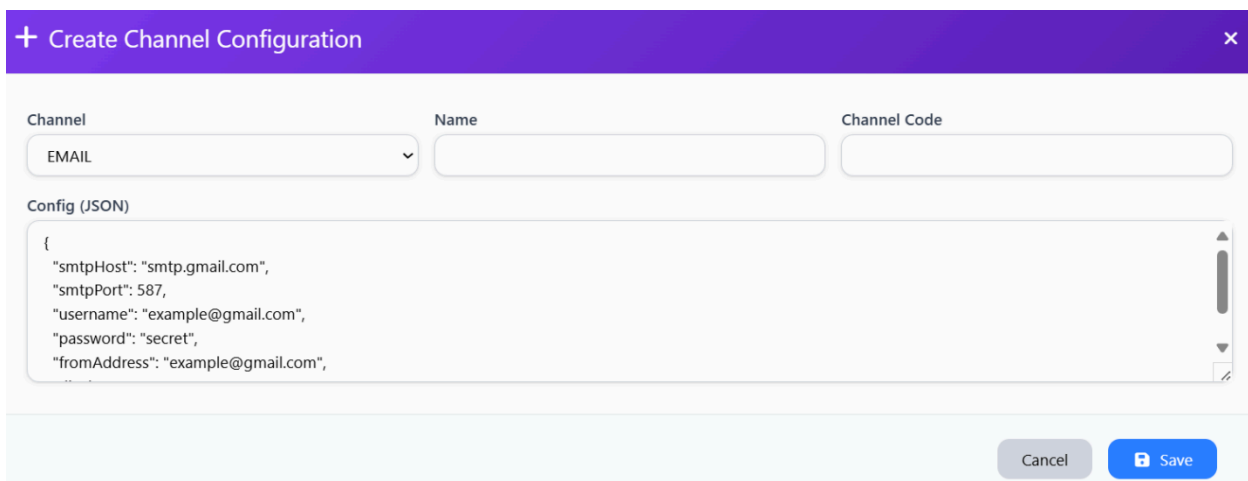


Figure 107 - Create Channel Configuration

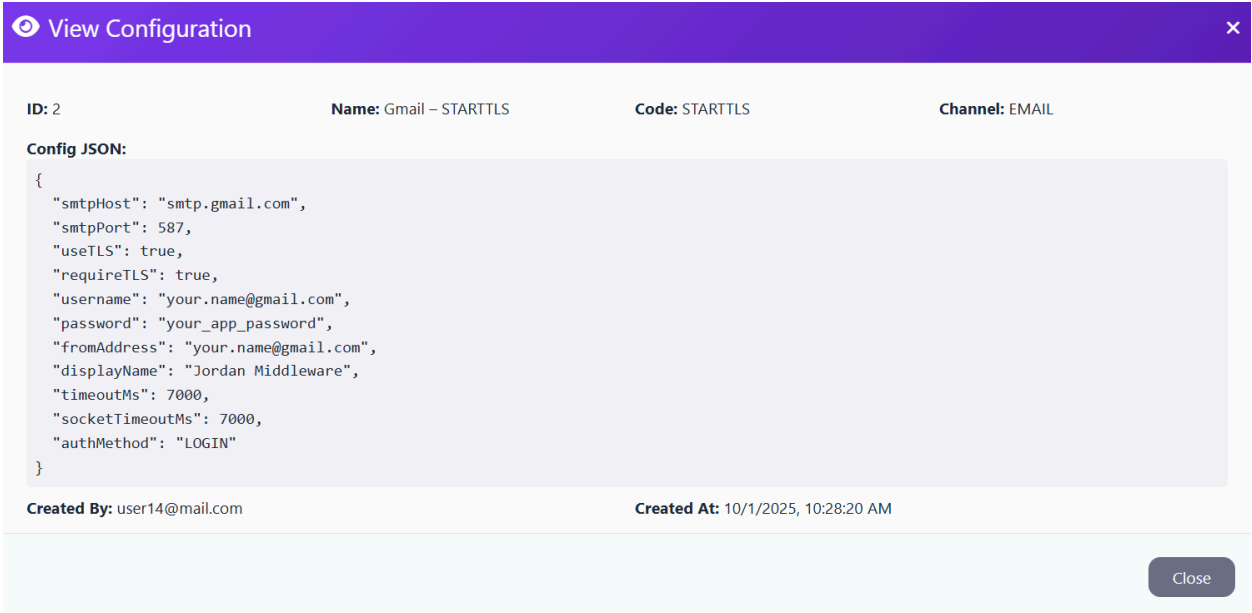


Figure 108 - View Notification Channel's Configuration

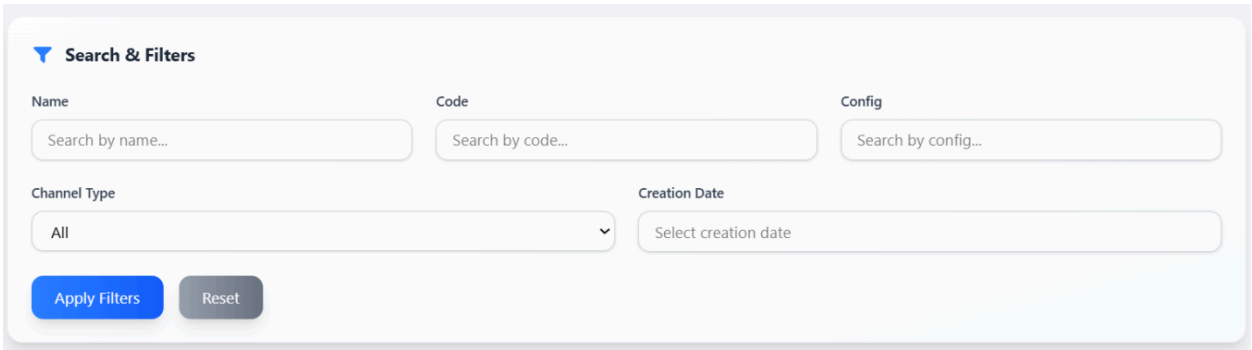
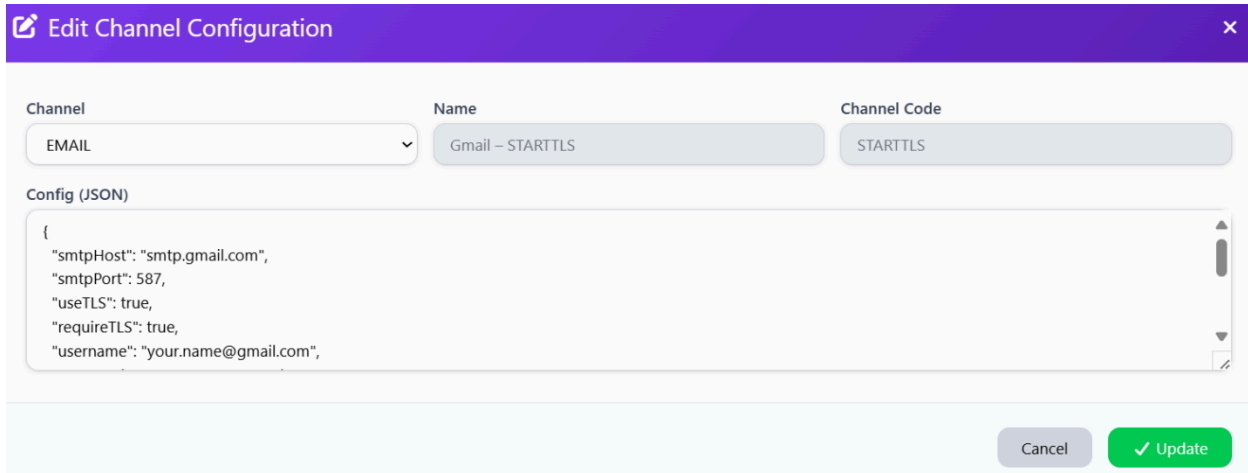


Figure 109 - Search Notification Channel's Configuration



The 'Edit Channel Configuration' dialog features a purple header with a close button. It contains three input fields: 'Channel' (a dropdown menu showing 'EMAIL'), 'Name' (a text field with 'Gmail - STARTTLS'), and 'Channel Code' (a text field with 'STARTTLS'). Below these is a 'Config (JSON)' text area containing a JSON object with SMTP settings. At the bottom right are 'Cancel' and 'Update' buttons.

Edit Channel Configuration

Channel: EMAIL

Name: Gmail - STARTTLS

Channel Code: STARTTLS

Config (JSON)

```
{  
  "smtpHost": "smtp.gmail.com",  
  "smtpPort": 587,  
  "useTLS": true,  
  "requireTLS": true,  
  "username": "your.name@gmail.com",  
}
```

Cancel Update

Figure 110 - Edit Channel's Configuration

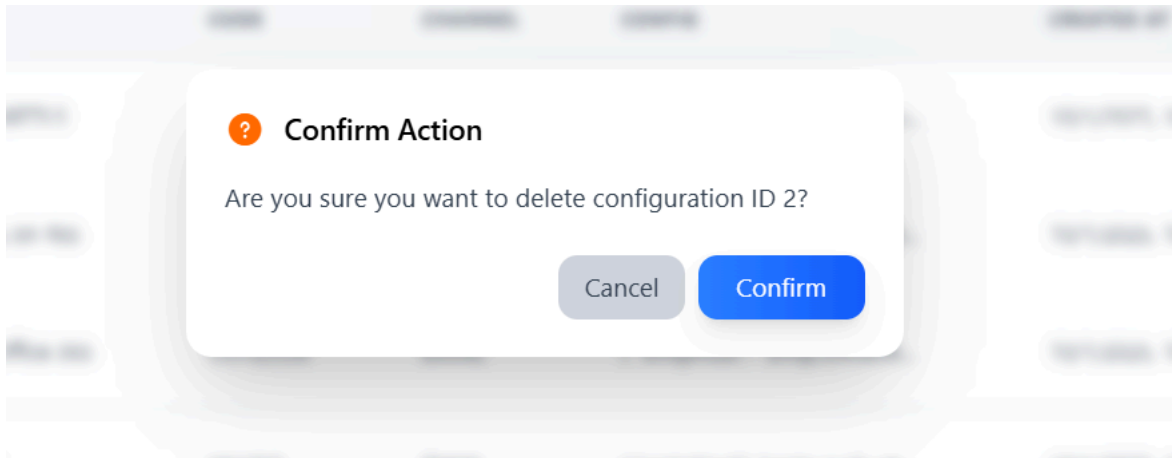


Figure 111 - Delete Channel's Configuration

2.7.7 Messages Config (Send and Schedule Messages)

Purpose

Send a message now to one or more groups using a selected template and channel, or copy the fully-formed API request (endpoint + JSON body) to use in schedulers or external tools.

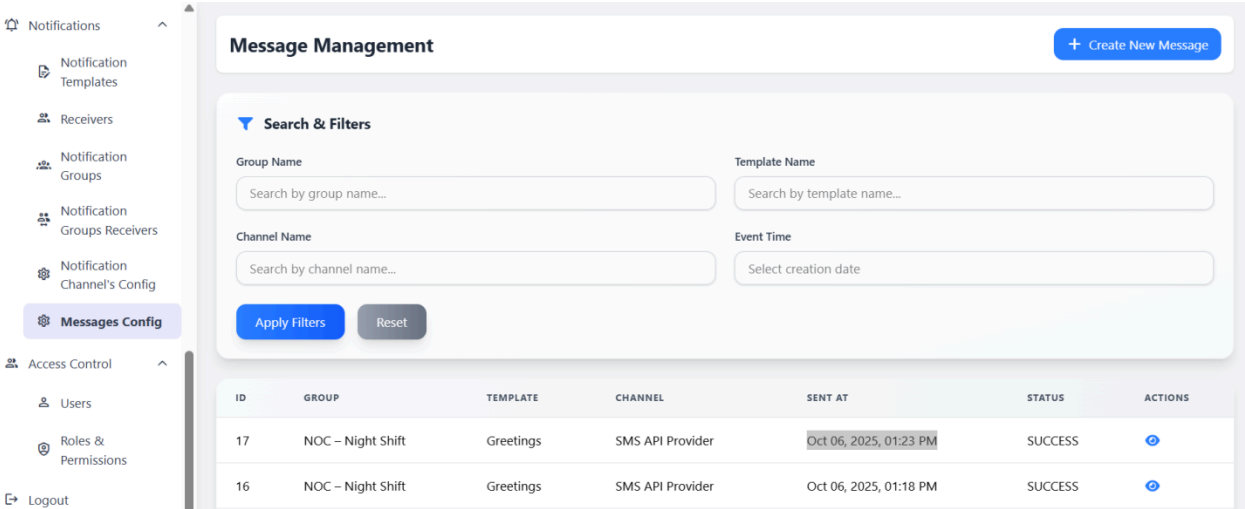


Figure 112 - Messages Config

Search & Filters (log list)

Filter historical sends by Group, Template, Channel, Event Time; Reset returns the grid to default view.

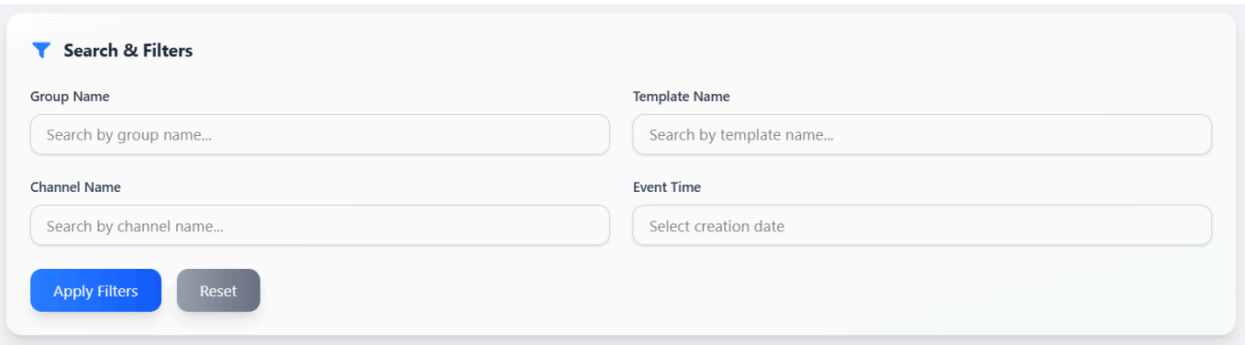


Figure 113 - Search Message's Config

Create Message

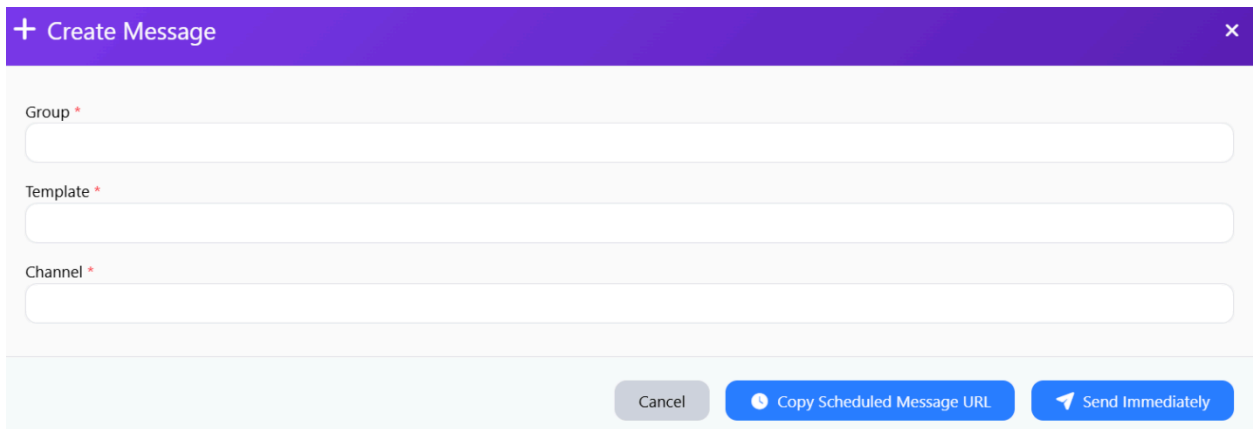
1. Choose Group (e.g., NOC: Night Shift).
2. Choose Template (e.g., Job Failure).

3. Choose Channel (e.g., Gmail: on 587).
4. Click Send Immediately to dispatch right away.
 - If the template contains placeholders, they are filled from the params provided by API calls or default values if defined.
5. Click Copy Scheduled Message URL to open a dialog containing:

- Endpoint URL: /api/notifications/send
- Request Body (example):

```
{  
  "groupCodes": ["NOC_NIGHT"],  
  "templateCode": "E_JOB_FAIL",  
  "channelCode": "GMAIL_SSL",  
  "params": {  
    "jobName": "Nightly Import",  
    "details": "2025-10-01 23:31"  
  }  
}
```

Use this payload in Scheduled Jobs or any external scheduler.



+ Create Message ×

Group *

Template *

Channel *

Cancel Copy Scheduled Message URL Send Immediately

Figure 114 - Create Message's Config

Scheduled Message Details

Endpoint URL
http://localhost:8081/api/notifications/send

Request Body
{
 "groupCodes": [
 "NOC_NIGHT"
],
 "templateCode": "G_SMS",
 "channelCode": "SMS"
}

Close

Figure 115 - Copy Scheduled Message Details

Result tracking

Each send attempt appears in the list with SUCCESS/FAILED status and can be audited further in Observability > Notification Logs.

2.7.8 Quick Start (End-to-End)

1. Create Templates (with placeholders you need).
2. Add Receivers (emails/phones).
3. Create Group and link receivers in Groups Receivers.
4. Configure Channel (e.g., SMTP/SSL).
5. Send from Messages Config or copy the request and schedule it.
6. Verify outcome in Notification Logs (Observability).

3. Conclusion

This manual provides comprehensive guidance on all modules of the Middleware System. It enables users and administrators to effectively manage and operate the system. For best results, ensure that each operation is performed according to the documented steps.